

Unit-1

1. Compare operational data stores and data warehouse.

OR

What is ODS? Provide its structure.

OR

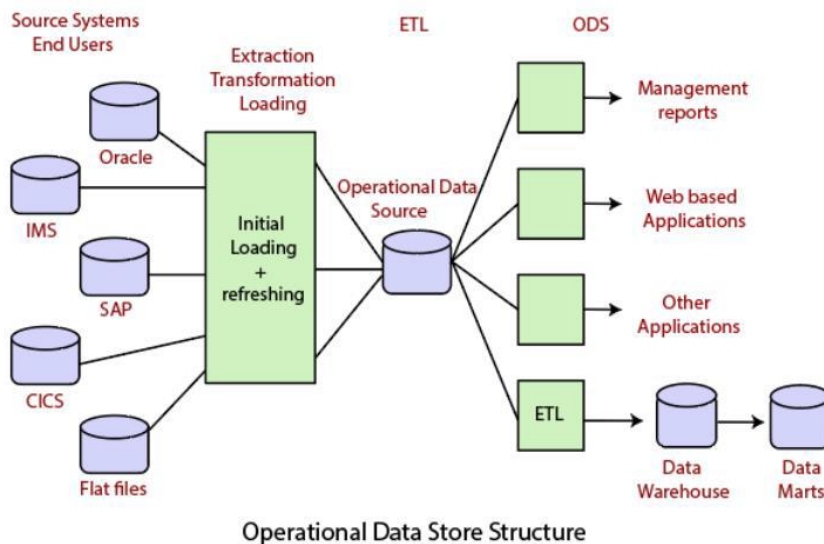
Explain relationship between OLTP, OLAP and ODS.

Ans: A subject oriented, integrated, volatile, current values data store, **containing only corporate detailed data.**

Volatile: data in ODS changes frequently as new information refreshes the ODS.

Current valued: an ODS is up to date and reflects current status of the information.

Detailed: ODS is detailed enough to serve the needs of the operational management staff in the enterprise.



Relationship between ODS, OLAP and OLTP.

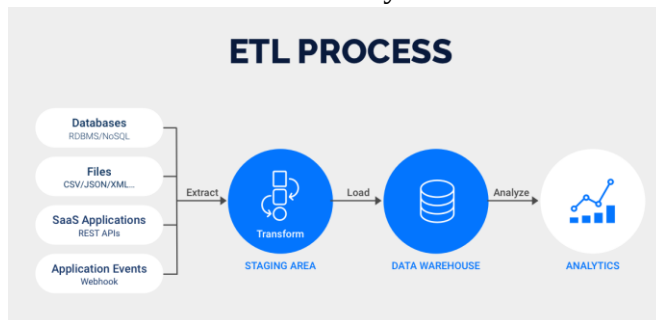
ODS	OLAP	OLTP
ODS means for operational reporting and supports current or near real time reporting requirements.	OLAP is intended for historical and trend analysis, usually reporting on a large volume of data.	Maintains the data only for transactions these are designed for recording daily operations and transactions of a business.
It consist of only a short window of data.	It includes the entire history of data.	It includes current data and guaranteed up to date.
It is typically detailed data only.	It contains summarized and detailed data.	It contains primitive and highly detailed data.
It is used for detailed decision making and operational reporting.	It is used for long term decision making and management reporting.	It is used to perform data to day operations.
It is used at the operational level.	It is used at management level.	It is used at operational and surface level transactions.

It is updated often as the transactions system generates new data.	It is usually updated in batch processing mode on a set schedule.	It is updated fast initiated by user in short terms.
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2. Describe different steps in data cleaning with respect to ETL.

Describe ETL process.

Ans: ETL is a process that extracts the data from different source systems, then transforms the data and finally loads the data into the data warehouse system.



Extraction:

- Process of extracting the data from the source system.
- Extracting data correctly sets the stage for the success of subsequent processes.
Most data warehousing projects combine data from different source systems- relational databases, XML, JSON and flat files.
- Data warehouse needs to integrate systems that have different DBMS, hardware, operating systems and communication protocols.
- Source could include legacy applications like mainframes, Call switches, text files, spreadsheets, customized applications, point of contact devices like ATM, data from vendors etc.,
- Examples for validations done during extraction:
 - Data type check.
 - Remove all duplicate/redundant data.
 - Make sure that no unwanted/spam data load.

Transform:

- In the data transformation stage, a series of rules or functions are applied to the extracted data in order to prepare it for loading into the end target.
- Data extracted from source server is raw and not usable in its original form. Therefore it needs to be cleaned, mapped and transformed. This steps adds value and changes data.
- Transformations if they are done in staging area so that performance of source system is not degraded. Also, if corrupted data is copied directly from the source into data warehouse database, rollback will be a challenge and validate extracted data before it moves into the data warehouse.
- Examples:
 - Selecting certain columns to load.

- Translating coded value (source system code: 1 as male and 2 as female, but the warehouse codes male as “M” and female as “F”).
- Encoding free-form values (ex: mapping “Male” to “M”).
- Deriving a new calculated value (ex: sale_amt=qty*unit_price).
- Sorting the data based on list of columns to improve search performance.
- Aggregating.
- Splitting a column into multiple columns.
- Applying any form of data validation.

Loading:

- Process of loading data into the target data warehouse database.
- In a typical data warehouse, huge volume of data needs to be loaded in a relatively short period. Hence load process must be optimized for performance.
- In case of load failure, recover mechanisms should be configured to restart from the point of failure without data integrity loss.
- Updating extracted data is frequently done on a daily, weekly, or monthly basis.
- Some data warehouse overwrite existing information where as other data warehouses may add new data in a historical form.
- Data cleaning: detects errors in the data and rectifies them when possible.
- Refresh: propagates the updates from the data sources to the warehouse.

Examples for ETL tools: Informatica PowerCenter, Oracle Data Integrator, Talend, Stitch, Skyvia etc

3. Define a data warehouse and explain with example.

OR

Explain the key features of data warehouse.

OR

Explain the characteristics of data warehouse.

Ans: According to William H. Inmon, “A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management’s decision making process”

- Data warehouse provides architectures and tools for business executives to systematically organize, understand, and use their data to make strategic decisions.
- Subject oriented:
 - A data warehouse is organized around major subjects (ex: customer, supplier, product, and sales).
 - Focuses on the modeling and analysis of data for decision makers.
 - Hence data warehouse typically provide a simple and concise view of particular subject issues by excluding data that are not useful in the decision support process.
- Integrated:

- A data warehouse is usually constructed by integrating multiple heterogeneous sources, such as relational databases, flat files, and online transaction records.
- Data cleaning and data integration techniques are applied to ensure consistency in naming conventions, attribute measures, and so on.
- Time variant:
 - Data are stored to provide information from an historic perspective (ex: the past 5-10 years).
- Nonvolatile:
 - A data warehouse is always a physically separate store of data transformed from the application data found in the operational environment.
 - Due to this separation, a data warehouse does not require transaction processing, recovery, and concurrency control mechanisms.
 - It stores an information that an enterprise needs to make strategic decisions.

4) How are organizations using the information from data warehouses?

Ans: Organizations use information to support business decision making activities:

- Increasing customer focus, which includes the analysis of customer buying patterns
- Repositioning products and managing product portfolios by comparing the performance of sales by quarter, by year, and by geographic regions in order to fine-tune production strategies
- Analysing operations and looking for sources of profit.
- Managing customer relationships, making environmental corrections, and managing the cost of corporate assets.
- Data warehousing is also very useful from the point of view of heterogeneous database integration. Organizations typically collect diverse kinds of data and maintain large databases from multiple, heterogeneous, autonomous, and distributed information sources.
- The traditional database approach to heterogeneous database integration is to build wrappers and integrators (or mediators) on top of multiple, heterogeneous databases.

5) Explain various data cube operations with suitable examples

OR

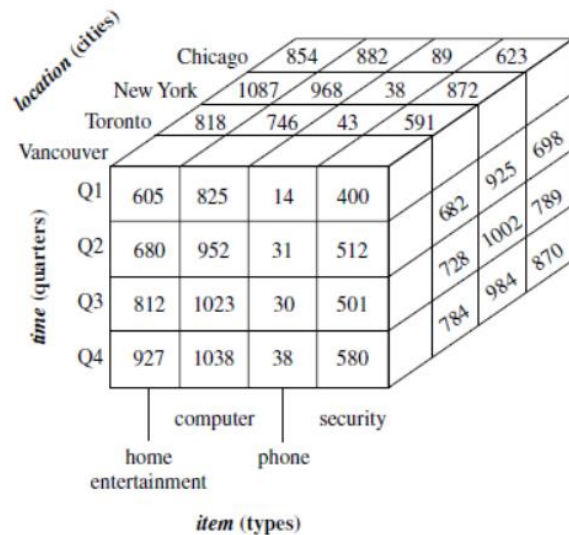
What are Data cubes? Give an example of three-dimensional and multi-dimensional data cube. Briefly explain different data cube implementations.

OR

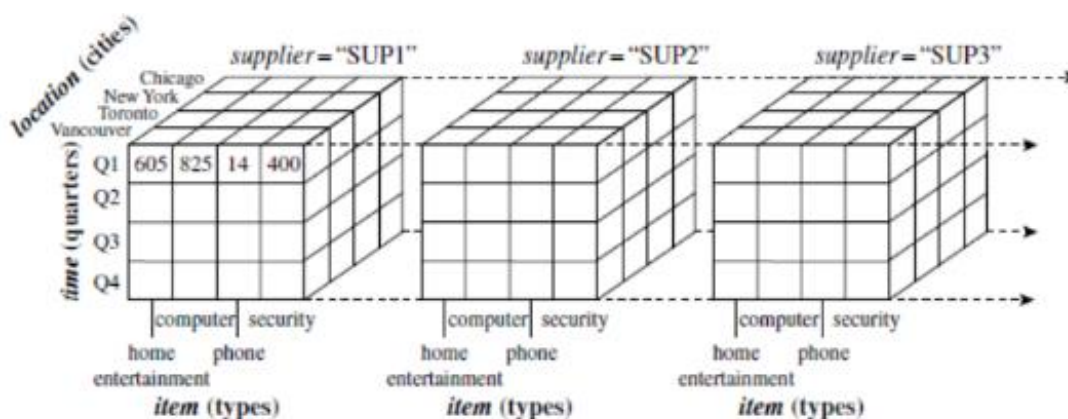
Explain the different operations that can be performed on the data cube with example.

Ans: A data cube allows data to be **modelled and viewed in multiple dimensions**. It is defined by dimensions and facts.

Three – dimensional data cube example:



Multi – dimensional data cube example

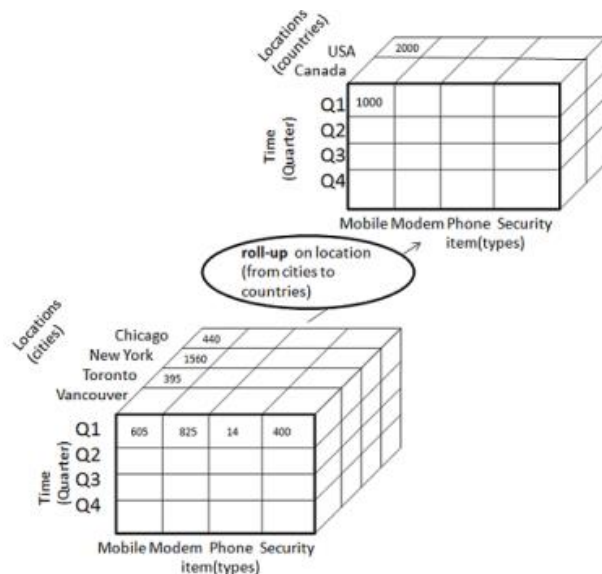


A 4-D data cube representation of sales data, according to *time*, *item*, *location*, and *supplier*. The measure displayed is *dollars_sold* (in thousands). For improved readability, only some of the cube values are shown.

Roll-up/drill-up

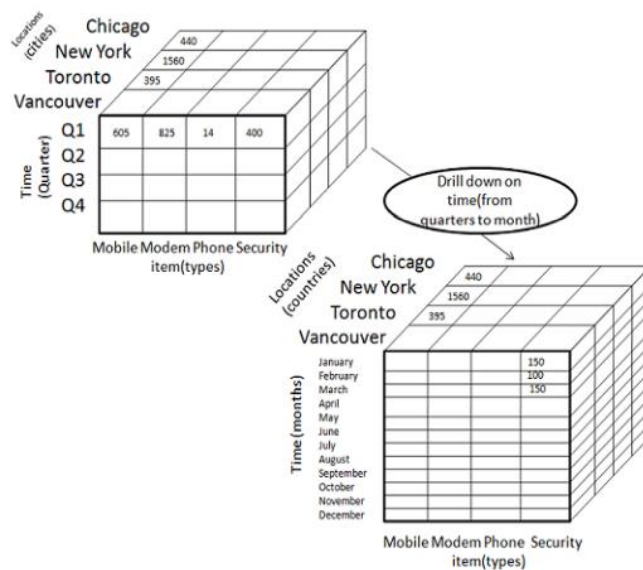
- Performs aggregation on a data cube, either by **climbing up a concept hierarchy for a dimension or by dimension reduction**
- **Concept hierarchy: total order “street < city < province or state < country”**
- Roll-up operation aggregates the data by ascending the location hierarchy from the level of city to the level of country
- Rather than grouping the data by city, the resulting cube groups the data by country
- **When roll-up is performed by dimension reduction, one or more dimensions are removed from the given cube**
- **Example: consider a sales data cube containing only the location and time dimensions**

- Roll-up may be performed by removing, say, the time dimension, resulting in an aggregation of the total sales by location, rather than by location and by time.



Drill-down

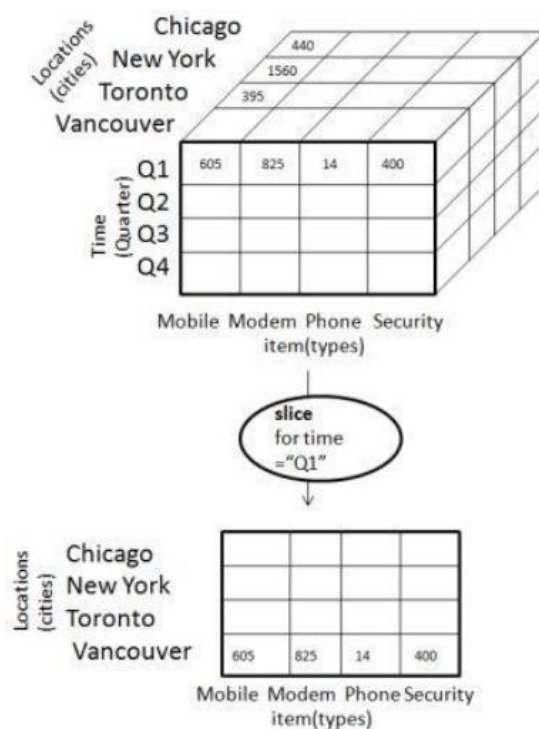
- Reverse of roll-up.
- It navigates from less detailed data to more detailed data.
- Can be realized by either stepping down a concept hierarchy for a dimension or introducing additional dimensions.
- Example: drill-down operation performed on the central cube by stepping down a concept hierarchy for time defined as "day < month < quarter < year."
- Drill-down occurs by descending the time hierarchy from the level of quarter to the more detailed level of month.
- The resulting data cube details the total sales per month rather than summarizing them by quarter.
- Because a drill-down adds more detail to the given data, it can also be performed by adding new dimensions to a cube.



Slice

- This operation performs a selection on one dimension of the given cube, resulting in a sub cube.

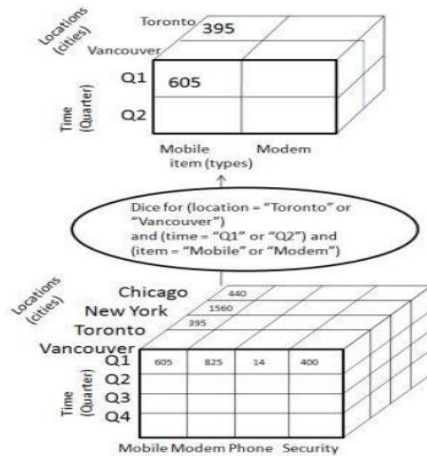
Sales data are selected from the central cube for the dimension time using the criterion time = "Q1".



Dice

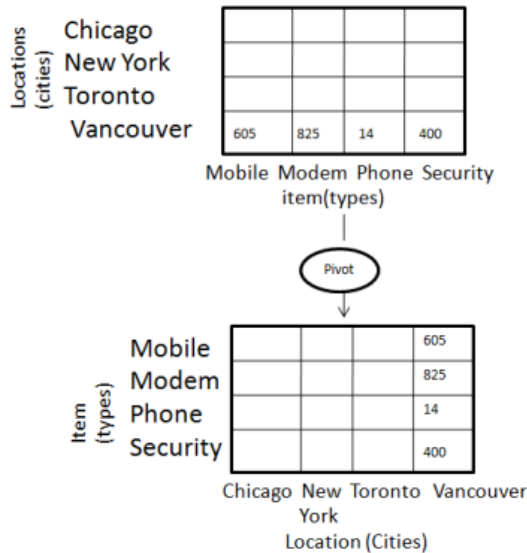
- This operation defines a sub cube by performing a selection on two or more dimensions

- Dice operation on the central cube based on the following selection criteria that involve three dimensions: (location = “Toronto” or “Vancouver”), (time = “Q1” or “Q2”) and (item = “home entertainment” or “computer”)



Pivot (rotate)

- It is a visualization operation that rotates the data axes in view to provide an alternative data presentation
- The item and location axes in a 2-D slice are rotated
- Examples: rotating the axes in a 3- D cube, transforming a 3-D cube into a series of 2- D plane



6) Explain How Star Schema and Snowflake Schema are different from each other.

Star Schema	Snow Flake Schema
The star schema is the simplest data warehouse schema.	Snowflake schema is a more complex data warehouse model than a star schema.
In star schema each of the dimensions is represented in a single table. It should not have any hierarchies between dimensions.	In snow flake schema at least one hierarchy should exists between dimension tables.

It contains a fact table surrounded by dimension tables. If the dimensions are denormalized, then it is a star schema design.	It contains a fact table surrounded by dimension tables. If a dimension is normalized, then it is a snow flaked design.
In star schema only one join establishes the relationship between the fact table and any one of the dimension tables.	In snow flake schema since there is relationship between the dimensions tables it has to do many joins to fetch the data.
A star schema optimizes the performance by keeping queries simple and providing fast response time. All the information about each level is stored in one row.	Snowflake schemas normalize dimensions to eliminated redundancy. The result is more complex queries and reduced query performance.
It is called a star schema because the diagram resembles a star.	It is called a snowflake schema because the diagram resembles a snowflake.

7) Discuss the different schemas for developing the data model for Data Warehousing.

Or

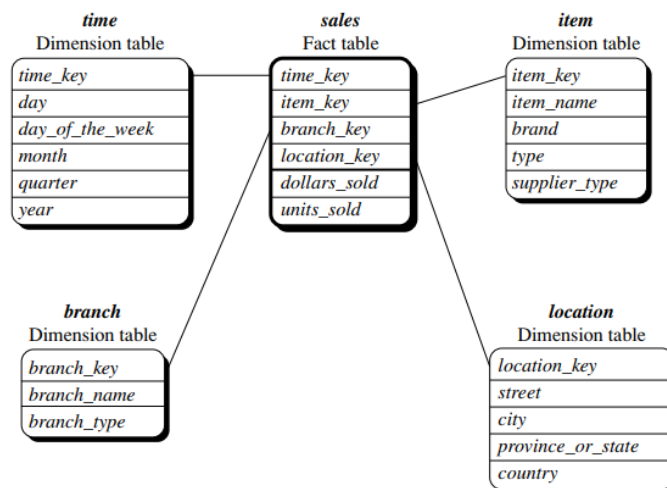
What are the schemas used to construct a data warehousing? Explain.

Explain fact constellation Schema with example.

Or

Present simple Snowflake Schema for the two-dimensional data.

Star schema:

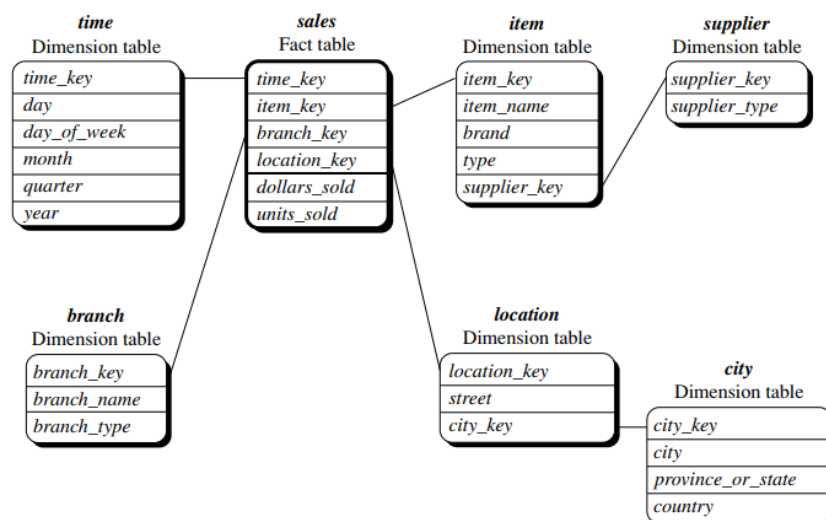


Star schema of sales data warehouse.

- The most common modeling paradigm is the star schema, in which the data warehouse contains a large central table (fact table) containing the bulk of the data, with no redundancy, and a set of smaller attendant tables (dimension tables), one for each dimension.
- The schema graph resembles a starburst, with the dimension tables displayed in a radial pattern around the central fact table
- A star schema for AllElectronics sales is shown in Figure. Sales are considered along four dimensions: time, item, branch, and location.

- The schema contains a central fact table for sales that contains keys to each of the four dimensions, along with two measures: dollars sold and units sold.
- To minimize the size of the fact table, dimension identifiers (e.g., time key and item key) are system-generated identifiers.
- In the star schema, each dimension is represented by only one table, and each table contains a set of attributes.
- For example, the location dimension table contains the attribute set {location key, street, city, province or state, country}.
- This constraint may introduce some redundancy. For example, “Urbana” and “Chicago” are both cities in the state of Illinois, USA.
- Entries for such cities in the location dimension table will create redundancy among the attribute’s province or state and country; that is, (Urbana, IL, USA) and (Chicago, IL, USA).
- The attributes within a dimension table may form either a hierarchy (total order) or a lattice (partial order).

Snowflake schema:

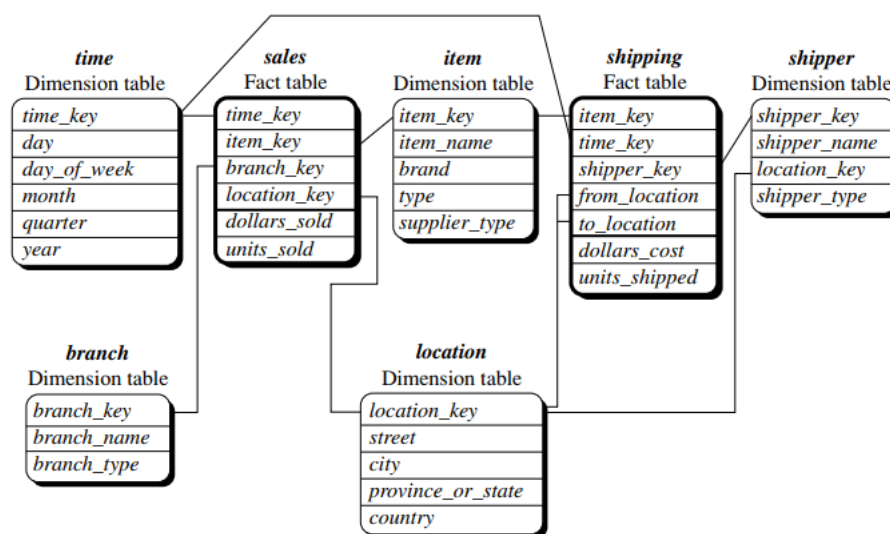


Snowflake schema of a sales data warehouse.

- The snowflake schema is a variant of the star schema model, where some dimension tables are normalized, thereby further splitting the data into additional tables.
- The resulting schema graph forms a shape similar to a snowflake.
- The major difference between the snowflake and star schema models is that the dimension tables of the snowflake model may be kept in normalized form to reduce redundancies.
- Such a table is easy to maintain and saves storage space. This space savings is negligible in comparison to the typical magnitude of the fact table.
- The snowflake structure can reduce the effectiveness of browsing, since more joins will be needed to execute a query.

- The system performance may be adversely impacted. Hence, although the snowflake schema reduces redundancy, it is not as popular as the star schema in data warehouse design
- A snowflake schema for AllElectronics sales is given in Figure. The sales fact table is identical to that of the star schema Figure.
- The main difference between the two schemas is in the definition of dimension tables. The single dimension table for item in the star schema is normalized in the snowflake schema, resulting in new item and supplier tables.
- For example, the item dimension table now contains the attributes item key, item name, brand, type, and supplier key, where supplier key is linked to the supplier dimension table, containing supplier key and supplier type information.
- The single dimension table for location in the star schema can be normalized into two new tables: location and city. The city key in the new location table links to the city dimension.
- Normalization can be performed on province or state and country in the snowflake schema

Fact constellation:



Fact constellation schema of a sales and shipping data warehouse.

- Sophisticated applications may require multiple fact tables to share dimension tables. This kind of schema can be viewed as a collection of stars, and hence is called a galaxy schema or a fact constellation
- A fact constellation schema is shown in Figure. This schema specifies two fact tables, sales and shipping.
- The sales table definition is identical to that of the star schema. The shipping table has five dimensions, or keys—item key, time key, shipper key, from location, and to location—and two measures—dollars cost and units shipped.
- A fact constellation schema allows dimension tables to be shared between fact tables.

- For example, the dimensions tables for time, item, and location are shared between the sales and shipping fact tables.
- In data warehousing, there is a distinction between a data warehouse and a data mart. A data warehouse collects information about subjects that span the entire organization, such as customers, items, sales, assets, and personnel, and thus its scope is enterprise-wide.
- For data warehouses, the fact constellation schema is commonly used, since it can model multiple, interrelated subjects.
- A data mart, on the other hand, is a department subset of the data warehouse that focuses on selected subjects, and thus its scope is departmentwide.
- For data marts, the star or snowflake schema is commonly used, since both are geared toward modeling single subjects, although the star schema is more popular and efficient.

9. Difference between ROLAP and MOLAP ?

ROLAP

- It stands for Relational Online Analytical Processing.
- It is used for large volumes of data.
- The access time in ROLAP is slow.
- It stores data in the form of relation tables.
- The data in ROLAP is fetched from a data warehouse.
- It uses complex SQL queries.
- A static multidimensional view of the data is created in ROLAP.

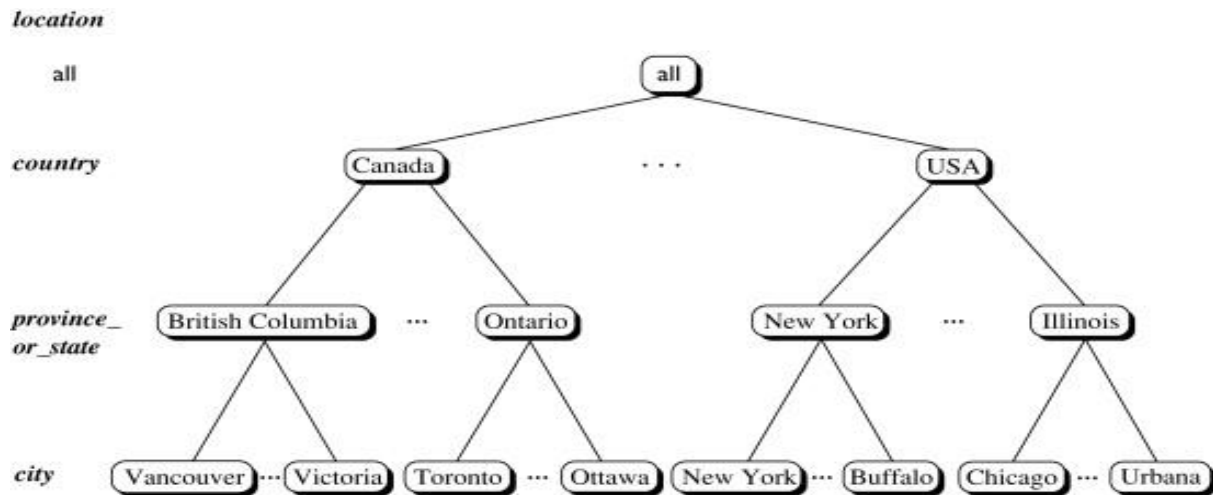
MOLAP

- It stands for Multidimensional Online Analytical Processing.
- It is used for less/limited volumes of data.
- The access time is quick in MOLAP.
- Data is stored in a multidimensional array.
- Data is fetched from the MDDBs database.
- A sparse matrix is used in MOLAP.
- Dynamic multidimensional view of the data is created in MOLAP.

10) Explain the concept of hierarchy in multidimensions data?

- It defines a sequence of mappings from a set of low-level concepts to higher-level, more general concepts .
- A hierarchy is an organizational structure in which items are ranked according to levels of importance .
- Concept hierarchy formation: Concept Hierarchy recursively reduce the data by collecting and replacing low level concepts (such as numeric values for the attribute age) by higher level concepts (such as young, middle-aged, or senior).
- Can be explicitly specified by domain experts and/or datawarehouse designers.

- Example: Consider a concept hierarchy for the dimensions locations.
- City values for location can be mapped to the province or state to which it belongs ▪
The provinces and states can in turn be mapped to the country to which they belong ▪
Set of low-level concepts (i.e., cities) are mapped to higher-level, more general concepts (i.e., countries).

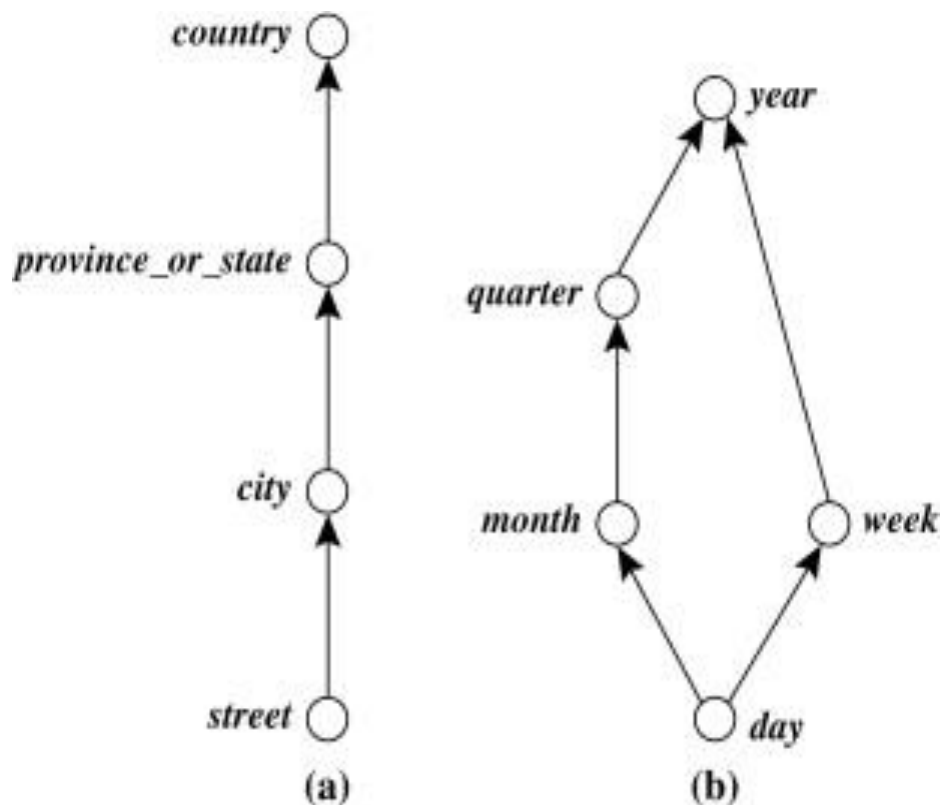


Schema hierarchy

- Many concept hierarchies are implicit within the database schema.
- This hierarchy may formally express semantic relation between attributes .
- Example: dimension location is described by the attributes number, street, city, province or state, zip code, and country .
- These attributes are related by a total order, forming a concept hierarchy such as “street < city < province or state < country” i.e street is conceptually lower than city which is lower than state which is lower than country.

The attributes of a dimension may be organized in a partial order, forming a lattice.

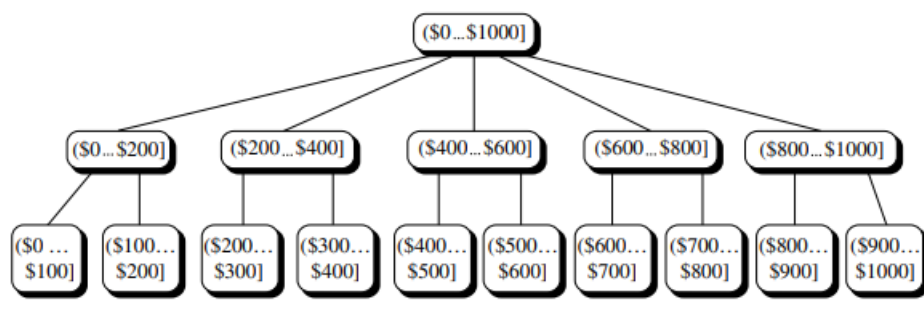
- Partial order for the time dimension is described by the attributes day, week, month, quarter, and year.
- These attributes are related by a Partial order, forming a concept hierarchy as “day < {month < quarter; week} < year”.
- A concept hierarchy that is a total or partial order among attributes in a database schema is called a schema hierarchy.
- Concept hierarchies may be predefined in the data mining system



Hierarchical and lattice structures of attributes in warehouse dimensions:

Set-grouping hierarchy:

- Here, Concept hierarchies are defined by discretizing or grouping values for a given dimension or attribute
- A total or partial order can be defined among groups of values
- Example: for the dimension price, where an interval (\$X ... \$Y] denotes the range from \$X (exclusive) to \$Y (inclusive).



11) Explain different Data warehouse models.

OR

What is Data Mart? Why is it recommended to have data marts in enterprises.

OR

Compare Centralized Data Warehouse and Data Mart.

From the architecture point of view, there are three data warehouse models:

The enterprise warehouse, the data mart, and the virtual warehouse.

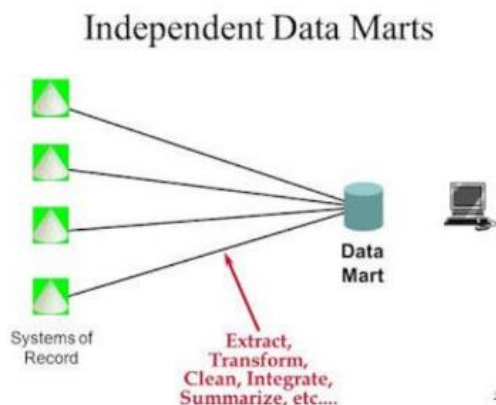
Enterprise warehouse:

- An enterprise warehouse collects all of the information about subjects spanning the entire organization.
- It provides corporate-wide data integration, usually from one or more operational systems or external information providers, and is cross-functional in scope.
- It typically contains detailed data as well as summarized data, and can range in size from a few gigabytes to hundreds of gigabytes, terabytes, or beyond.
- An enterprise data warehouse may be implemented on traditional mainframes, computer superservers, or parallel architecture platforms.
- It requires extensive business modeling and may take years to design and build.

Data mart:

- A data mart contains a subset of corporate-wide data that is of value to a specific group of users.
 - The scope is confined to specific selected subjects. For example, a marketing data mart may confine its subjects to customer, item, and sales.
 - The data contained in data marts tend to be summarized.
 - Data marts are usually implemented on low-cost departmental servers that are Unix/Linux or Windows based.
 - The implementation cycle of a data mart is more likely to be measured in weeks rather than months or years. However, it may involve complex integration in the long run if its design and planning were not enterprise-wide.
- Depending on the source of data, data marts can be categorized as independent or dependent.

Independent data marts are sourced from data captured from one or more operational systems or external information providers, or from data generated locally within a particular department or geographic area.

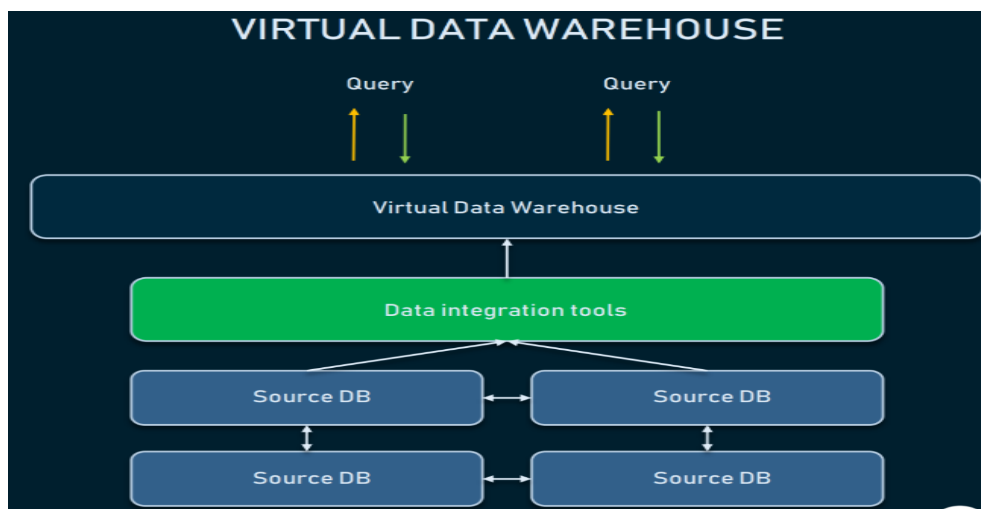


Dependent data marts are sourced directly from enterprise data warehouses.



Virtual warehouse:

- A virtual warehouse is a **set of views over operational databases**
- **For efficient query processing**, only some of the possible summary views may be materialized.
- **It uses middleware to build connections to different data sources**
- **Set of separate databases, which can be queried together, so a user can effectively access all the data as if it was stored in one data warehouse- A layer of virtuality is created**
- They can be fast as they allow users to filter the most important pieces of data from different legacy applications
- **A virtual warehouse is easy to build** but requires excess capacity on operational database servers.



A data mart is a simple form of data warehouse focused on a single subject or line of business.
Or

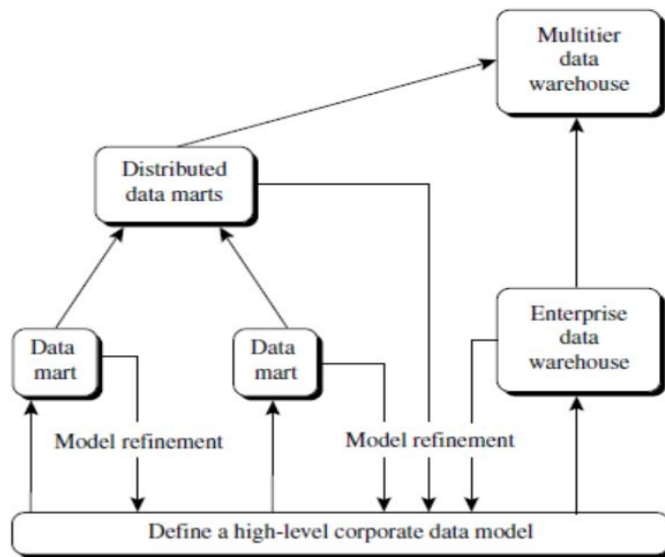
A data mart is a subset of a data warehouse oriented to a specific business line. Data marts contain repositories of summarized data collected for analysis on a specific section or unit within an organization.

A recommended method for the development of data warehouse systems is to implement the warehouse in an incremental and evolutionary manner.

- First, a high-level corporate data model is defined within a reasonably short period (such as one or two months) that provides a corporate-wide, consistent, integrated view of data among different subjects and potential usages. This high-level model, although

it will need to be refined in the further development of enterprise data warehouses and departmental data marts, will greatly reduce future integration problems.

- Second, independent data marts can be implemented in parallel with the enterprise warehouse based on the same corporate data model set noted before.
- Third, distributed data marts can be constructed to integrate different data marts via hub servers. Finally, a multitier data warehouse is constructed where the enterprise warehouse is the sole custodian of all warehouse data, which is then distributed to the various dependent data marts.



Data warehouse	Data Mart
Data warehouse is a centralized system.	While it is decentralized system.
In datawarehouse,lightly denormalization takes place.	In Data mart,highly denormalization takes place.
Data warehouse is top-down model.	While it is a bottom-up model.
To built a warehouse is difficult.	While to build a mart is easy.
Data warehouse is flexible	While it is not flexible.
Data warehouse is the data-oriented in nature	While it is the project-oriented nature.
Data warehouse has long life	While data-mart has short life than warehouse
In Data Warehouse, Data are contained in detail form	While in this, data are contained in summarized form.
Data Warehouse is vast size	While data mart is smaller than warehouse.
It collects data from various data sources	It generally stores data from a data warehouse
Long time for processing the data because of large data	Less time for processing the data because of handling only a small amount of data.

Complicated design process of creating schemas and views	Easy design process of creating schemas and views.
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12) Explain the differences between OLAP and OLTP.

Or

What is OLAP? Explain the different characteristics of OLAP. List the differences between OLAP and OLTP.

Online Analytical Processing Server (OLAP) is based on the **multidimensional data model**. It **allows managers, and analysts to get an insight of the information through fast, consistent, and interactive access to information.**

characteristics of OLAP are as follows:

1. **Multidimensional conceptual view:** OLAP systems let business users have a dimensional and logical view of the data in the data warehouse. It helps in carrying slice and dice operations.
2. **Multi-User Support:** Since the OLAP techniques are shared, the OLAP operation should provide normal database operations, containing retrieval, update, adequacy control, integrity, and security.
3. **Accessibility:** OLAP acts as a mediator between data warehouses and front-end. The OLAP operations should be sitting between data sources (e.g., data warehouses) and an OLAP front-end.
4. **Storing OLAP results:** OLAP results are kept separate from data sources.
5. **Uniform documenting performance:** Increasing the number of dimensions or database size should not significantly degrade the reporting performance of the OLAP system.

Differences between OLTP and OLAP

System orientation and Users:

- OLTP: customer-oriented ; used for transaction and query processing by clerks, clients, and information technology professionals
- OLAP: market-oriented ; used for data analysis by knowledge workers, including managers, executives, and analysts

Data contents:

- OLTP : manages current data that, typically, are too detailed to be easily used for decision making
- OLAP: manages large amounts of historic data, provides facilities for summarization and aggregation, and stores and manages information at different levels of granularity. These features make the data easier to use for informed decision making.

Database design:

- OLTP: usually adopts an entity-relationship (ER) data model and an

application-oriented database design

- OLAP: typically adopts either a star or a snowflake model and a subject-oriented database design

Access patterns:

- OLTP: accesses consist mainly of short, atomic transactions. Such a system requires concurrency control and recovery mechanisms
- OLAP: accesses to OLAP systems are mostly read-only operations

View:

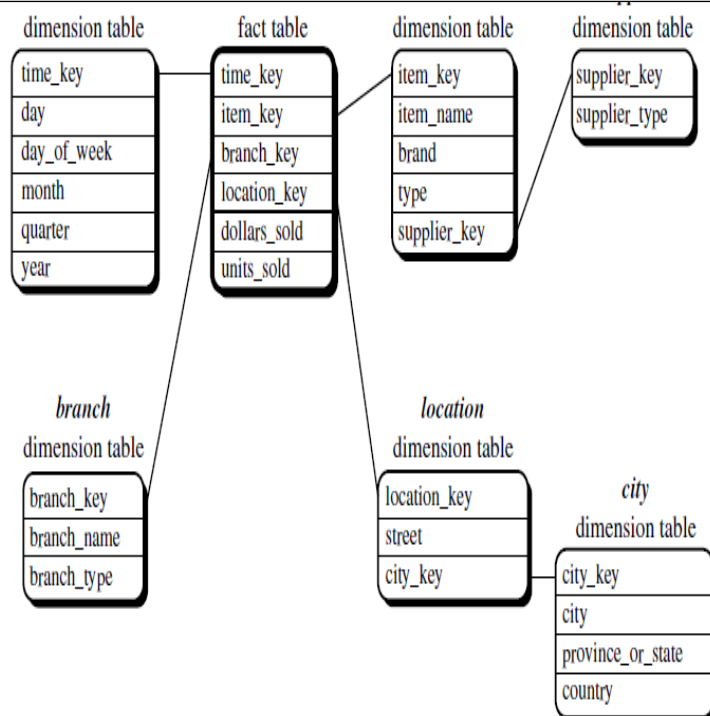
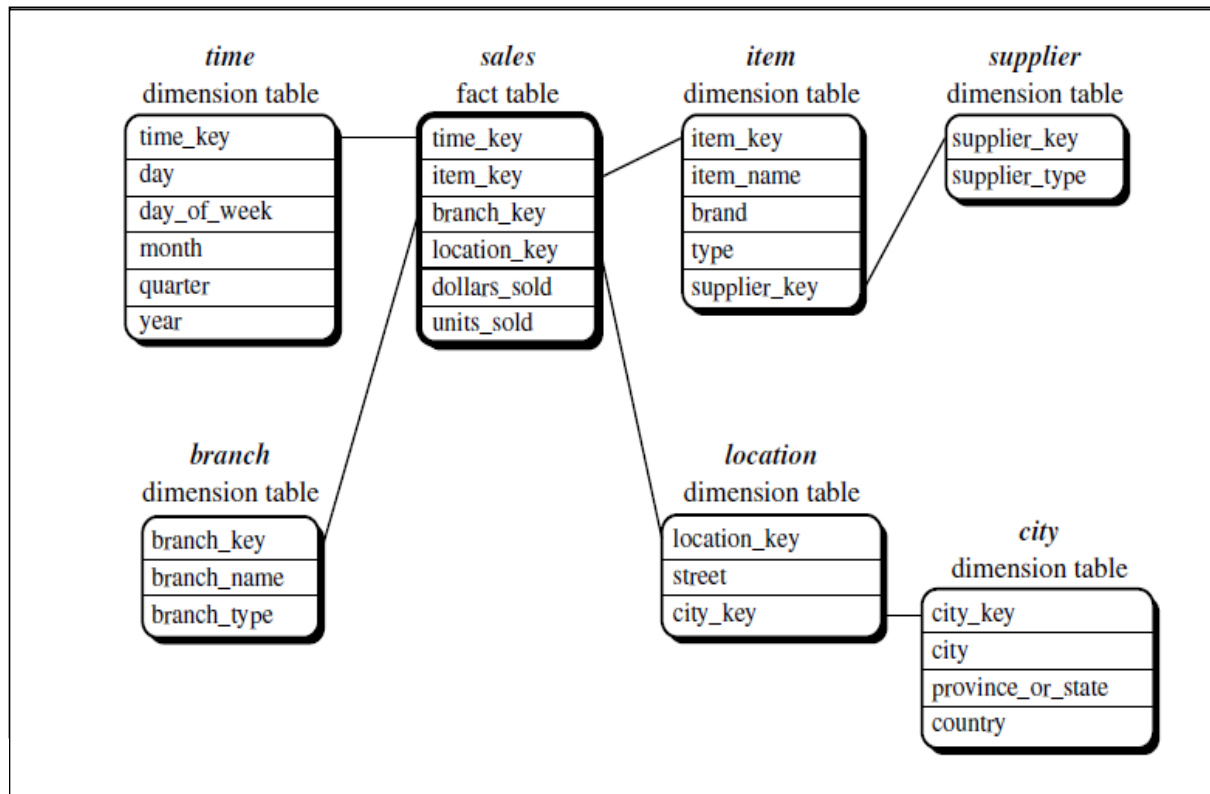
- OLTP: focuses mainly on the current data within an enterprise or department, without referring to historic data or data in different organizations
- OLAP: often spans multiple versions of a database schema, due to the evolutionary process of an organization. OLAP systems also deal with information that originates from different organizations, integrating information from many data stores

13.Explain why data modelling is important in data warehouse. Develop a data model for data warehouses using star schema and snowflake schema for sales assuming sales is given by 4 dimensions-Time, item, location and Branch.

- Data warehouse modelling is the process of designing the schemas of the detailed and summarized information of the data warehouse.
- The goal of data warehouse modelling is to develop a schema describing the reality, or at least a part of the fact, which the data warehouse is needed to support.
- Data warehouse modelling is an essential stage of building a data warehouse for two main reasons. Firstly, through the schema, data warehouse clients can visualize the relationships among the warehouse data, to use them with greater ease.
- Secondly, a well-designed schema allows an effective data warehouse structure to emerge, to help decrease the cost of implementing the warehouse and improve the efficiency of using it.
- Data modelling in data warehouses is different from data modelling in operational database systems. The primary function of data warehouses is to support DSS processes. Thus, the objective of data warehouse modelling is to make the data warehouse efficiently support complex queries on long term information.

Star schema for sales assuming sales is given by 4 dimensions-Time, item, location and Branch.

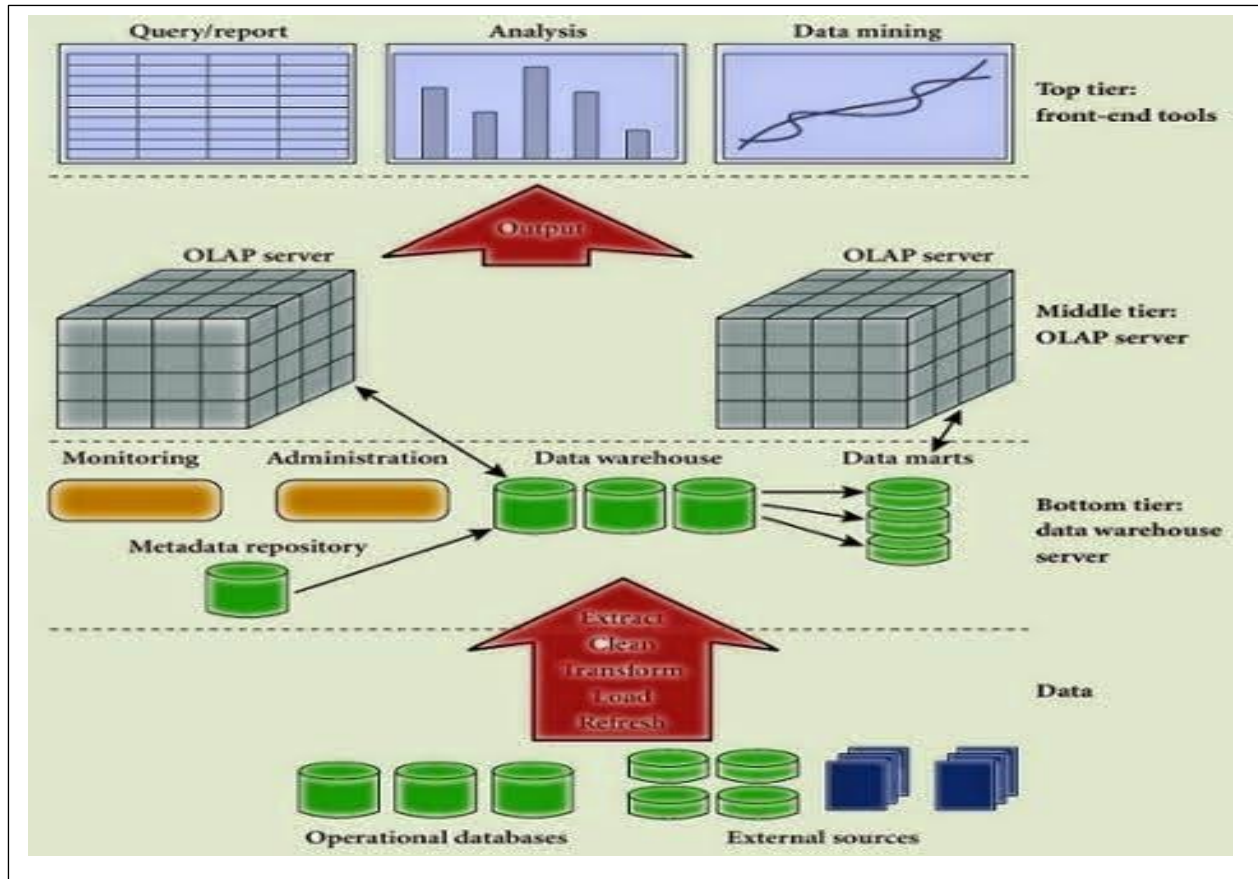
Snowflake schema



14. Give 3 tier architecture for the data warehouse.

Data Warehouses usually have a three-level (tier) architecture that includes:

1. Bottom Tier (Data Warehouse Server)
2. Middle Tier (OLAP Server)
3. Top Tier (Front end Tools)



- These tools and utilities perform data extraction, cleaning, and transformation as well as load and refresh functions to update the data warehouse
- The data are extracted using application program interfaces known as Gateways.
 - A gateway is supported by the underlying DBMS
 - It allows client programs to generate SQL code to be executed at a server
 - Examples: ODBC (Open Database Connection), OLEDB (Object Linking and Embedding Database), JDBC (Java Database Connection)
- This tier also contains a metadata repository, which stores information about the data warehouse and its contents
- It includes the following parameters and information for the middle and the top-tier applications:
 - A description of the DW structure, including the warehouse schema, dimension, hierarchies, data mart locations, and contents, etc.
 - Operational metadata, which usually describes the currency level of the stored data, i.e., active, archived or purged, and warehouse monitoring information, i.e., usage statistics, error reports, audit, etc.

- System performance data, which includes indices, used to improve data access and retrieval performance.
- Information about the mapping from operational databases, which provides source RDBMSs and their contents, cleaning and transformation rules, etc.

Middle tier: OLAP server

- Data Warehouse can have more than one OLAP server, and it can have more than one type of OLAP server model as well, which depends on the volume of the data to be processed and the type of data held in the bottom tier
- **OLAP server** is typically implemented using either:
 - **Relational OLAP(ROLAP) model**
 - An extended relational DBMS that maps operations on multidimensional data to standard relational operations
 - **Multidimensional OLAP (MOLAP) model**
 - A special-purpose server that directly implements multidimensional data and operations

Top tier: front-end client layer

- Includes Tools and APIs that connects and get data out from the data warehouse and present the same to the client
- Contains query and reporting tools, analysis tools, data mining tools (e.g., trend analysis, prediction etc.)
- The type of tool depends purely on the form of outcome expected

15)What is Data Lake? Differentiate between Data lake and Data warehouse.

A data lake is a centralized repository designed to store,process,and secure large amount of structured,semistructured,and unstructured data.

It is usually a single store of data including raw copies of source system data, sensor data, social data etc. And transformed data used for tasks such as reporting, visualization, advanced analytics and machine learning.

Difference between Data lake and Data warehouse:

Parameters	Data Lake	Data Warehouse
Data Structure	Data is raw and all types—structured, semi-structured, or unstructured—is captured in its original form.	Data is processed and only structured information is captured and organized in schemas.
Users	Ideal for users who carry out deep analysis such as data scientists and need advanced analytical tools.	Ideal for operational users such as business professionals and moguls since the data is structured and easy to use.
Storage Costs	Storing data is relatively inexpensive.	Storing data is time-consuming and costly.
Accessibility	Updates can be made quickly thus making it highly accessible	Costly to make changes, thereby quite complicated
Position of Schema	Schema is defined after data is stored, thus making it highly agile.	Schema is defined before data is stored, thus offering performance and security.
Data Processing	Uses ELT (Extract Load Transform) process.	Uses ETL (Extract Transform Load) process.

16. Define Metadata? Explain the contents of metadata repository

Meta data is simply defined as data about data. The data used to represent other data is known as metadata. (Ex: Index of a book serves as a metadata for the contents in the book.)

Metadata repository is an integral part of a data warehouse system. It contains,

- **Definition of data warehouse** – It includes the description of structure of data warehouse. The description is defined by schema, view, hierarchies, derived data definitions, and data mart locations and contents.

- **Business metadata** – It contains has the data ownership information, business definition, and changing policies.

- **Operational Metadata** – It includes currency of data and data lineage. Currency of data means whether the data is active, archived, or purged. Lineage of data means the history of data migrated and transformation applied on it.

- **Data for mapping from operational environment to data warehouse** – It includes the source databases and their contents, data extraction, data partition cleaning, transformation rules, data refresh and purging rules.

- **Algorithms for summarization** – It includes dimension algorithms, data on granularity, aggregation, summarizing, etc.

•**Data related to system performance** – which include indices and profiles that improve data access and retrieval performance, in addition to rules for the timing and scheduling of refresh, update, and replication cycles. Business metadata, which include business terms and definitions, data ownership.

Unit – 2

1) Explain how OLAP queries can be efficiently processed?

- The purpose of materializing cuboids and constructing OLAP index structures is to speed up query processing in data cubes
- Given materialized views, query processing should proceed as follows:
 - Determine which operations should be performed on the available cuboids**
- Involves transforming any selection, projection, roll-up (group-by), and drill-down operations specified in the query into corresponding SQL and/or OLAP operations
 - Determine to which materialized cuboid(s) the relevant operations should be applied**
- Involves identifying all of the materialized cuboids that may potentially be used to answer the query, pruning the set using knowledge of “dominance” relationships among the cuboids, estimating the costs of using the remaining materialized cuboids, and selecting the cuboid with the least cost

EXAMPLE:

- Consider a data cube for All Electronics of the form “sales cube [time, item, location]: sum(sales in dollars).”
 - dimension hierarchies:
 - day < month < quarter < year for time
 - item name < brand < type for item;
 - street < city < province or state < country for location
 - Query to be processed is on {brand, province_or_state}, with the selection constant “year = 2010.”
 - cuboid 1: {year, item name, city}
 - cuboid 2: {year, brand, country}
 - cuboid 3: {year, brand, province or state}
 - cuboid 4: {item name, province or state}, where year = 2010
 - “Which of these four cuboids should be selected to process the query?”
- Cuboid 2 cannot be used because country is a more general concept than province or state
- Cuboids 1, 3, and 4 can be used to process the query because
 - They have the same set or a superset of the dimensions in the query,
 - The selection clause in the query can imply the selection in the cuboid, and
 - The abstraction levels for the item and location dimensions in these cuboids are at a finer level than brand and province or state, respectively

2) What is OLAP? Explain OLAP server architecture.

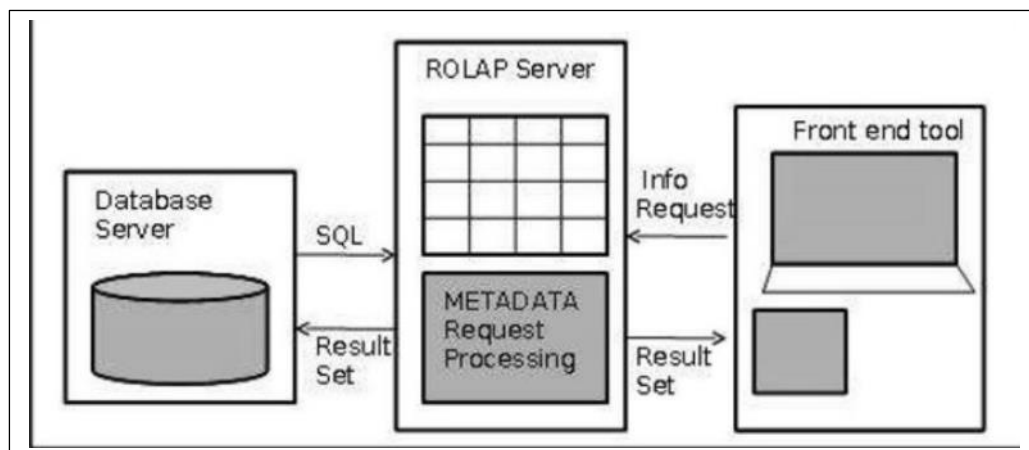
OLAP servers present business users with multidimensional data from data warehouses or data marts, without concerns regarding how or where the data are stored

Implementations of a warehouse server for OLAP processing include the following architectures:

- **Relational OLAP (ROLAP) servers**
- **Multidimensional OLAP (MOLAP) servers**
- **Hybrid OLAP (HOLAP) servers**

Relational OLAP (ROLAP) servers

- Relational OLAP servers are placed between relational back-end server and client front-end tools
 - To store and manage the warehouse data, the relational OLAP uses relational or extended-relational DBMS
 - ROLAP includes the following :
 - Implementation of aggregation navigation logic
 - Optimization for each DBMS back-end
 - Additional tools and services
 - A ROLAP database can be accessed through complex SQL queries to calculate information
- **Advantages:**
 - ROLAP servers can be easily used with existing RDBMS
 - ROLAP can handle large data volumes, but the larger the data, the slower the processing times
 - Because queries are made on-demand, ROLAP does not require the storage and pre-computation of information
- **Disadvantages:**
 - Potential performance constraints and scalability limitations that result from large and inefficient join operations between large tables



Multidimensional OLAP (MOLAP) servers

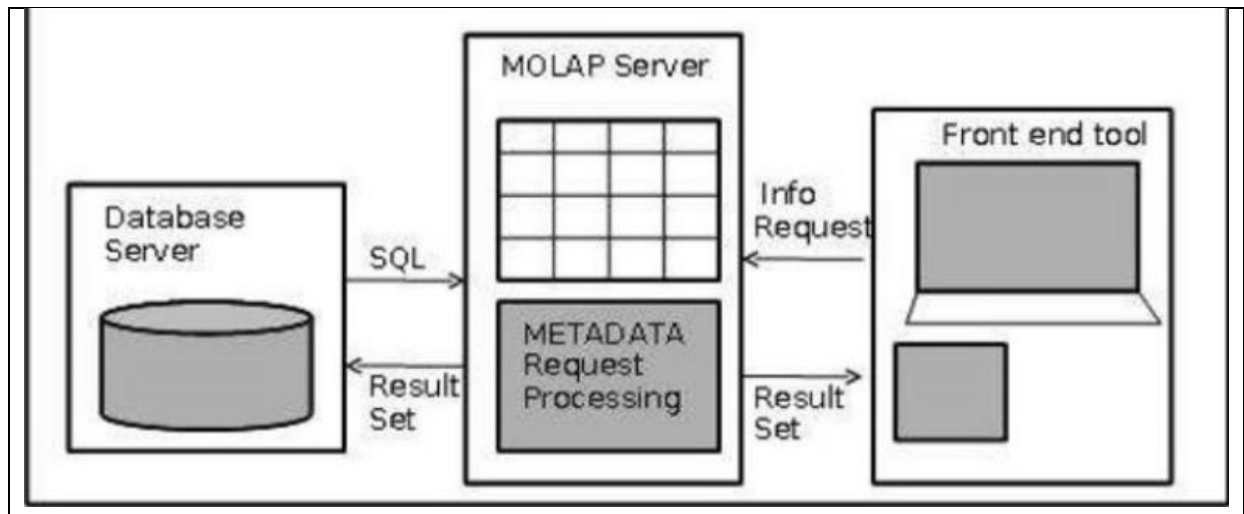
- MOLAP uses array-based multidimensional storage engines for multidimensional views of data
 - They map multidimensional views directly to data cube array structures
 - With multidimensional data stores, the storage utilization may be low if the dataset is sparse
 - Therefore, many MOLAP servers use two levels of data storage representation to handle dense and sparse datasets
 - Denser sub cubes are identified and stored as array structures, whereas sparse sub cubes employ compression technology for efficient storage utilization

Advantages:

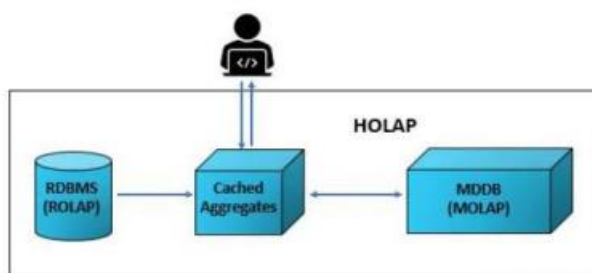
- Data is pre-computed, pre-summarized, and stored
- Its simple interface makes MOLAP easy to use, even for inexperienced users
- Its speedy data retrieval makes it the best for “slicing and dicing” operations

Disadvantages:

- Less scalable than ROLAP, as it can handle a limited amount of data
- Storage utilization may be low if the data set is sparse

**Hybrid OLAP (HOLAP) servers**

- Combines ROLAP and MOLAP technology
- Involves storing part of data in a ROLAP store and another part in a MOLAP store, thus developers get the benefits of both
- Gets benefited from the greater scalability of ROLAP and the faster computation of MOLAP
 - HOLAP server may allow large volumes of detailed data to be stored in a relational database, while aggregations are kept in a separate MOLAP store
- This setup allows much more flexibility for handling data

**3.Explain the process of data cube computation.**

When data is grouped or combined in multidimensional matrices called Data Cubes. The data cube method has a few alternative names or a few variants, such as "Multidimensional databases," "materialized views," and "OLAP (On-Line Analytical Processing)."

The compute cube Operator and the Curse of Dimensionality One approach to cube computation extends SQL so as to include a compute cube operator The compute cube operator computes aggregates over all subsets of the dimensions specified in the operation This can

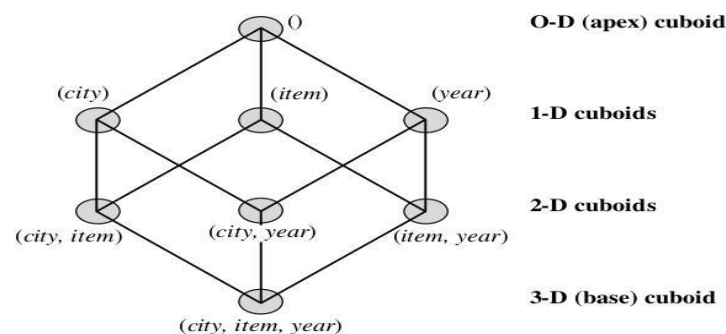
require excessive storage space, especially for large numbers of dimensions. A data cube is a lattice of cuboids.

Consider the example:

To create a data cube for All Electronic sales that contains the city, item, year, and sales in dollars:

Total number of cuboids, or group by that can be computed for this data cube is $2^3 = 8$

The possible group-by's are: $\{(city, item, year), (city, item), (city, year), (item, year), (city), (item), (year), ()\}$, where $()$ means that the group-by is empty (i.e., the dimensions are not grouped)



The base cuboid contains all three dimensions, city, item, and year. It can return the total sales for any combination of the three dimensions. The base cuboid is the least generalized (most specific) of the cuboids. The apex cuboid, or 0-D cuboid, refers to the case where the group-by is empty. It contains the total sum of all sales.

An SQL query containing no group-by is a zero dimensional operation e.g., “compute the sum of total sales”. An SQL query containing one group-by is a one-dimensional operation e.g., “compute the sum of sales, group-by city”. A cube operator on n dimensions is equivalent to a collection of group-by statements, one for each subset of the n dimensions. The cube operator is the n -dimensional generalization of the group-by operator. ■ Data cube definition: define cube sales cube [city, item, year]: sum(sales in dollars) ■ For a cube with n dimensions, there are a total of 2^n cuboids, including the base cuboid. ■ compute cube sales_cube ■ Statement explicitly instruct the system to compute the sales aggregate cuboids for all eight subsets of the set {city, item, year}, including the empty subset.

4) Explain different methods of indexing in OLAP data.

OLAP data can be indexed using

- Bitmap Index
- Join Index

Bitmap Index:

- Popular method because it allows quick searching in data cubes
- It is an alternative representation of the record ID (RID) list
- In the bitmap index for a given attribute, there is a distinct bit vector, B_v , for each value v in the attribute's domain
- If a given attribute's domain consists of n values, then n bits are

- needed for each entry in the bitmap index If the attribute has the value v for a given row in the data table, then the bit representing that value is set to 1 in the corresponding row of the bitmap index. All other bits for that row are set to

Example:

- Consider All Electronics data warehouse
- Let dimension item at the top level has four values- “home entertainment”, “computer”, “phone” and “security.”
- Each value (e.g., “computer”) is represented by a bit vector in the item bitmap index table
- If the cube is stored as a relation table with 100,000 rows, because the domain of item consists of four values, the bitmap index table requires four bit vectors (or lists), each with 100,000 bits
- The diagram shows a base (data) table containing the dimensions item and city, and its mapping to bitmap index tables for each of the dimensions.

Base Table:

RID	Item	City
R1	Computer	India
R2	Phone	India
R3	Laptop	India
R4	Mouse	India
R5	Computer	London
R6	Phone	London
R7	Laptop	London
R8	Mouse	London

Item Bitmap Index Table:

RID	Computer	Phone	Laptop	Mouse
R1	1	0	0	0
R2	0	1	0	0
R3	0	0	1	0
R4	0	0	0	1
R5	1	0	0	0
R6	0	1	0	0
R7	0	0	1	0
R8	0	0	0	1

City Bitmap Index Table:

RID	India	London
R1	1	0
R2	1	0
R3	1	0
R4	1	0
R5	0	1
R6	0	1
R7	0	1
R8	0	1

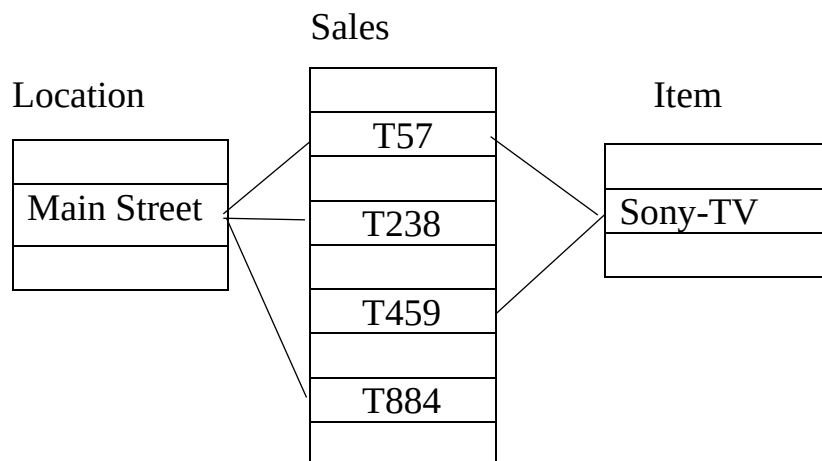
- It is especially useful for low-cardinality domains
- Comparison, join, and aggregation operations are reduced to bit arithmetic, which substantially reduces the processing time
- leads to significant reductions in space and input/output (I/O) since a string of characters can be represented by a single bit
- For higher-cardinality domains, the method can be adapted using compression technique

Join Index:

- Gained popularity from its use in relational database query processing
- Join indexing registers the joinable rows of two relations from a relational database
- If two relations R(RID, A) and S(B, SID) join on the attributes A and B, then the join index record contains the pair (RID, SID), where RID and SID are record identifiers from the R and S relations, respectively
- The join index records can identify joinable tuples without performing costly join operations
- It is especially useful for maintaining the relationship between a foreign key and its matching primary keys, from the joinable relation
- In data warehouses, join index relates the values of the dimensions of a star schema to rows in the fact table
- Join indices may span multiple dimensions to form composite join indices

Example:

- Consider a star schema for All Electronics of the form
“sales_star[time, item, branch, location]: dollars sold D sum (sales in dollars).”
- Example for join index relationship between the sales fact table and the location and item dimension tables
- Example: the “Main Street” value in the location dimension table joins with tuples T57, T238, and T884 of the sales fact table
- Similarly, the “Sony-TV” value in the item dimension table joins with tuples T57 and T459 of the sales fact table
- To further speed up query processing, the join indexing and the bitmap indexing methods can be integrated to form bitmapped join



Linkage between a sales fact table and location and item dimension tables.

Join index table location/sales

location	Sales_key
.....	
Main Street	T57
Main Street	T238
Main Street	T884
.....	

Join index table for item/sales

Item	Sales_key
.....	
Sony-TV	T57
Sony-TV	T459
.....	

Join index table linking location and item to sales

location	Item	Sales_key
.....		
Main Street	Sony-TV	T57
.....		

Join index table based on the linkage between the sales fact table and the location and item

5) What is an attribute ? Describe following Attribute types with appropriate examples.

a. Qualitative (Nominal and Ordinal) Attribute.

b. Quantitative (Interval and Ratio) Attribute.

c. Continuous attribute

d. Asymmetric attribute

e. Discrete attribute.

- An attribute is a property or characteristic of an object that may vary either from one object to another or from one time to another
- Examples: eye color of a person, temperature, etc.
- Attribute is also known as variable, field, characteristic, dimension, or feature
- A measurement scale is a rule (function) that associates a numerical or symbolic value with an attribute of an object
- Process of measurement is the application of a measurement scale to associate a value with a particular attribute of a specific object

a)Qualitative(Nominal and Ordinal) Attribute:

- Attribute where the values correspond to discrete categoriesDeals with characteristics and descriptors that can't be easily measured, but can be observed subjectively
- Lack most of the properties of numbers
- Even if they are represented by numbers, i.e., integers, they should be treated more like symbols

b) Quantitative(Interval and Ratio) Attribute:

- Represented by numbers
- Have most of the properties of numbers
- Can be integer-valued or continuous

c) Continuous attribute:

- Has real numbers as attribute values
- Examples: temperature, height, or weight.
- Practically, real values can only be measured and represented using a finite number of digits
- Continuous attributes are typically represented as floating-point variables

d)Asymmetric attribute

- Only presence (a non-zero attribute value) is regarded as important
- Binary attributes where only non-zero values are important are called asymmetric binary attributes
- A binary variable is said to be symmetric if both of its states are of the same value and carry the same weight, and there is no preference
- Example: attribute of gender
- An asymmetric binary attribute is one in which outcomes are not of equal importance

e)Discrete attribute.

- Has only a finite or countably infinite set of values
- Examples: zip codes, counts, or the set of words in a collection of documents
- Often represented as integer variables

Binary attributes

- Special case of discrete attributes
- Assume only two values, e.g., true/false, yes/no, male/female, or 0/1
- Often represented as Boolean variables, or as integer variables that only take the values 0 or 1

6)Show how one can measure Similarity and Dissimilarity for simple qualitative attributes.

(How similarity and dissimilarity for simple attribute can be measured?)

Ans: - Similarity and Dissimilarity are important because they are used by a number of data mining algorithms-clustering, nearest neighbour classification, clustering.

Similarity

- Numerical measure of the degree to which the two objects are alike
- Higher for pairs of objects that are more alike
- Usually, non-negative and are often between 0 (no similarity) and 1 (complete similarity)

Dissimilarity

- Numerical measure of the degree to which the two objects are different
- Lower for more similar pairs of objects
- Sometimes fall in the interval [0,1], but it is also common for them to range from 0 to ∞

Transformations

- Applied to convert a similarity to dissimilarity or vice versa
- Example: We may have similarities ranging from 1 to 10, but an algorithm may work only with dissimilarities.
- To transform a proximity measure to fall within a particular range
- We may have similarities ranging from 1 to 10, but an algorithm may work only with similarities in the interval [0,1].

Example:

1.
 - If the similarity falls in the interval [0,1], then the dissimilarity can be defined as $d = 1 - s$
2.
 - If similarities between objects range from 1(not at all similar) to 10 (completely similar), we can make them to fall within the range [0,1] by using transformation $s' = (s - 1) / 9$
 - In general, the transformation of similarities to the interval [0,1] is given by $s' = (s - \min_s) / (\max_s - \min_s)$
 - Similarly, the transformation of dissimilarities to the interval [0,1] is given by $d' = (d - \min_d) / (\max_d - \min_d)$

Similarity and Dissimilarity between Simple Attributes

Table 2.7. Similarity and dissimilarity for simple attributes

Attribute Type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$	$s = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$
Ordinal	$d = x - y / (n - 1)$ (values mapped to integers 0 to $n-1$, where n is the number of values)	$s = 1 - d$
Interval or Ratio	$d = x - y $	$s = -d, s = \frac{1}{1+d}, s = e^{-d},$ $s = 1 - \frac{d - \min_d}{\max_d - \min_d}$

7) What is Data preprocessing? Explain the steps of data preprocessing.

(Brief about Aggregation operation and motivations behind aggregation.)

(Define data preprocessing. Explain dimensionality reduction with respect to data preprocessing)

(Explain the following preprocessing techniques with suitable examples.

a) Sampling

b) Discretization)

(Describe various sampling approaches)

Ans:- Data pre-processing is the process of transforming raw data into a useful, understandable format. Real-world or raw data usually has inconsistent formatting, human errors, and can also be incomplete. Data pre-processing resolves such issues and makes datasets more complete and efficient to perform data analysis.

Categories:

- Selecting data objects and attributes for the analysis
- Creating/changing the attributes
- Aggregation
- Sampling

- Dimensionality Reduction
- Feature subset selection
- Feature creation
- Discretization and Binarization
- Attribute Transformation

Aggregation

Combining two or more attributes (or objects) into a single attribute (or object)

Purpose

- **Data reduction**

Smaller data sets resulting from data reduction require less memory and processing time, and hence, aggregation may permit the use of more expensive data mining algorithms

- **Change of scale**

Cities aggregated into regions, states, countries, etc. Days aggregated into weeks, months, or years.

- **More “stable” Data**

Aggregated data tends to have less variability.

Disadvantage:

- Potential loss of interesting details.
- Example: aggregating over months loses information about which day of the week has the highest sales

TID	Item	Store Location	Date	Price	...
..	..	..	..	..	..
101	Watch	Delhi	01/10/20	5000	
102	Bathey	Mumbai	01/10/20	2000	
103	Shoes	Delhi	02/10/20	2500	

(Define data preprocessing. Explain dimensionality reduction with respect to data preprocessing)

Dimensionality Reduction

- Data sets can have a large number of features
- Example: Word count in a document

Advantages:

- Data mining algorithms work better if the dimensionality is lower – We can eliminate irrelevant features and reduce noise.
- Can lead to a more understandable model because the model may involve fewer Attributes.
- May allow the data to be more easily visualized.
- Amount of time and memory required by the data mining algorithm is reduced with a reduction in dimensionality.
- Techniques that reduce the dimensionality of a data set by creating new attributes that are a combination of the old attributes - Dimensionality reduction

Curse of dimensionality Problems:

Classification – There are not enough data objects to allow the creation of a model that reliably assigns a class to all possible objects

Clustering – Definitions of density and the distance between points, which are critical for clustering, become less meaningful.

Linear Algebra Techniques for Dimensionality Reduction

1. Principal Components Analysis (PCA)

For continuous attributes that finds new attributes (principal components) that

- are linear combinations of the original attributes
- are orthogonal (perpendicular) to each other
- capture the maximum amount of variation in the data

2. Singular Value Decomposition (SVD)

#(Explain the following preprocessing techniques with suitable examples.

a) Sampling

b) Discretization

(Describe various sampling approaches)

Ans:

a. Sampling

- Approach for selecting a subset of the data objects to be analyzed
- Statisticians use sampling because obtaining the entire set of data of interest is too expensive or time consuming
- Data miners sample because it is too expensive or time consuming to process all the data

- **“Choose a sampling scheme that guarantees a high probability of getting a representative sample”**

Sampling Approaches

1. Simple Random Sampling

- There is an equal probability of selecting any particular item.

Sampling without replacement

- As each item is selected, it is removed from the population.

Sampling with replacement

- Objects are not removed from the population as they are selected for the sample
- In sampling with replacement, the same object can be picked up more than once
- **E x a m p l e** : when building classification models for rare classes, it is critical that the rare classes be adequately represented in the sample.

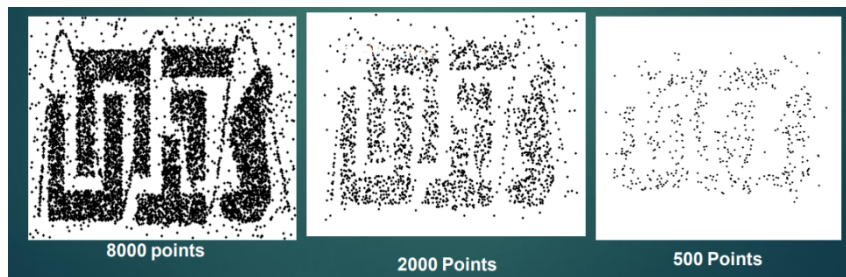
2. Stratified Sampling

- Split the data into several partitions; then draw random samples from each partition

Variations:

- Equal numbers of objects are drawn from each group even though the groups are of different size
- Number of objects drawn from each group is proportional to the size of that group
- Once a sampling technique has been selected, it is still necessary to choose the sample size
- Larger sample sizes increase the probability that a sample will be representative, but they also eliminate much of the advantage of sampling
- With smaller sample sizes, patterns may be missed or erroneous

patterns can be detected



1. Progressive Sampling

- Proper sample size can be difficult to determine Solution:
- Start with a small sample, and then increase the sample size until a sample of sufficient size has been obtained
- It requires that there be a way to evaluate the sample to judge if it is large enough
- Although the accuracy of predictive models increases as the sample size increases, at some point the increase in accuracy levels off
- We want to stop increasing the sample size at this leveling-off point
- by keeping track of the change in accuracy of the model as we take progressively larger samples
- by taking other samples close to the size of the current one

b) Discretization

- Transformation of a continuous attribute to a categorical attribute
- Involves two subtasks:
- Deciding how many categories to have
 - Determining how to map the values of the continuous attribute to these categories

Unsupervised Discretization

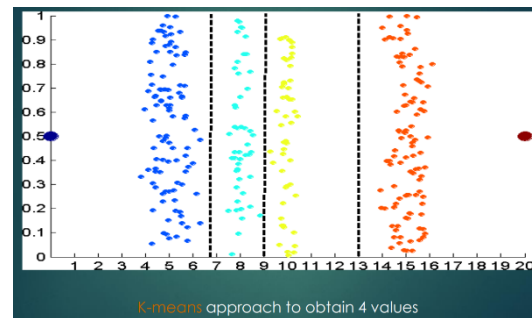
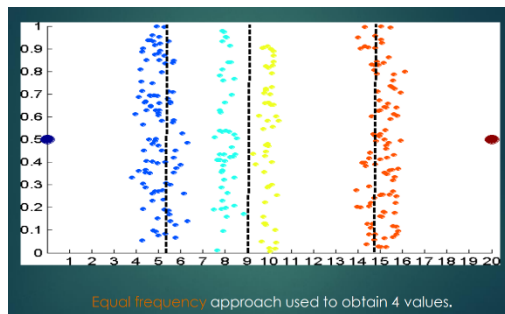
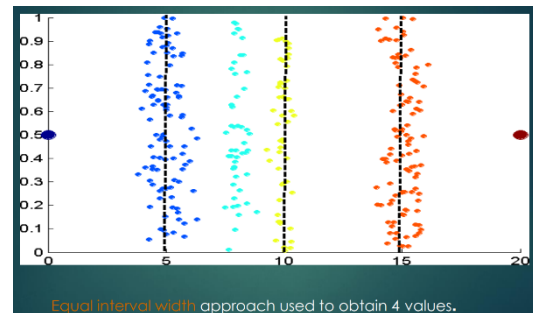
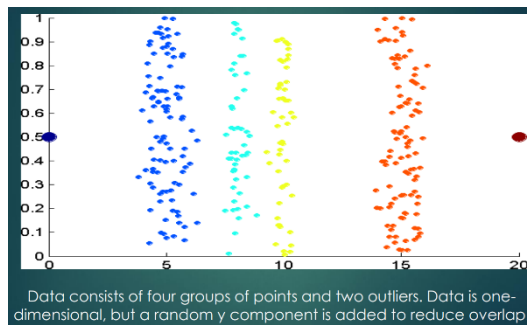
- Class information is not used

Equal width approach:

- divides the range of the attribute into a user-specified number of intervals each having the same width
- Can be badly affected by outliers

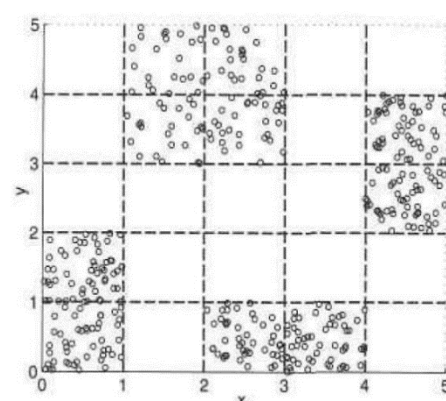
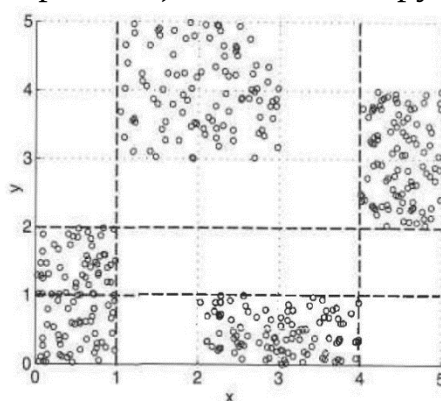
Equal frequency (equal depth) approach

- Tries to put the same number of objects into each interval



Supervised Discretization

- Class information is used
- An interval constructed with no knowledge of class labels often contains a mixture of class labels
- A conceptually simple approach is to place the splits in a way that maximizes the purity of the intervals
- Entropy based approaches are one of the most promising approaches to discretization
- Entropy of an interval is a measure of the purity of an interval
- If an interval contains only values of one class (is perfectly pure), then the entropy is 0
- If the classes of values in an interval occur equally often (the interval is as impure as possible), then the entropy is a maximum



8) Explain the various types of data sets with example

OR

Explain types of following data with suitable examples:

a) Record data

b) Ordered data

OR

What do you mean by record data? Explain its subtypes

OR

Classify different types of data on which mining can be performed

Ans:

Types of data sets

- Record
 - Data Matrix
 - Document Data
 - Transaction Data
- Graph
 - World Wide Web
 - Molecular Structures
- Ordered
 - Spatial Data
 - Temporal Data
 - Sequential Data
 - Genetic Sequence Data

Record Data

Record data is usually stored either in flat files or in relational databases

<i>Tid</i>	<i>Refund</i>	<i>Marital Status</i>	<i>Taxable Income</i>	<i>Cheat</i>
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No

Transaction Data(Market Basket Data)

- A special type of record data, where each record (transaction) involves a set of items
- Example: Grocery store (Market basket) data
- The set of products purchased by a customer during one shopping trip constitute a transaction, while the individual products that were purchased are the items:

TID	ITEMS
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Data Matrix

- Data objects have the same fixed set of numeric attributes
- Such a data set can be represented by an m by n matrix, where there are m rows, one for each object, and n columns, one for each attribute
- Because it consists of numeric attributes, standard matrix operation can be applied to transform and manipulate the data

Projection of x load	Projection of y load	Distance	Load	Thickness
10.23	5.27	15.22	27	1.2
12.65	6.25	16.22	22	1.1
13.54	7.23	17.34	23	1.2
14.27	8.43	18.45	25	0.9

Sparse Data Matrix

- Special case of a data matrix in which the attributes are of the same type and are asymmetric; i.e., only non-zero values are important
- Example: Transaction data

TID	Bread	Milk	Diapers	Beer	Eggs	Cola
1	1	1	0	0	0	0
2	1	0	1	1	1	0
3	0	1	1	1	0	1
4	1	1	1	1	0	0
5	1	1	1	0	0	1

- Each document becomes a ‘term’ vector
- Each term is a component (attribute) of the vector
- The value of each component is the number of times the corresponding term occurs in the document

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

Graph based data

- A graph can sometimes be a convenient and powerful representation for data
- Types:
 - Graph capturing relationships among data objects
 - Data objects themselves are represented as graph

Data with Relationships among Objects

- Relationships among objects frequently convey important information
- Data objects are mapped to nodes of the graph
- The relationships among objects are captured by the links between objects and link properties (such as direction and weight)
- Example: Web pages on the World Wide Web

Data with Objects That Are Graphs

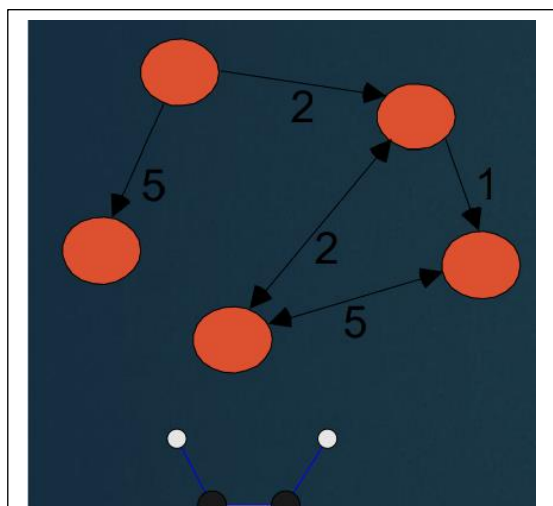
- If objects contain sub-objects that have relationships, then such objects are frequently represented as graphs
- structure of chemical compounds can be represented by a graph, where the nodes are atoms and the links between nodes are chemical bonds

- Graph representation makes it possible to
 - determine which substructures occur frequently in a set of compounds
 - ascertain whether the presence of any of these substructures is associated with the presence or absence of certain chemical properties, such as melting point or heat of formation

Graph Data

- Examples: Generic graph, a molecule, and webpages

Useful Links: • Bibliography • Other Useful Web sites ◦ ACM SIGKDD ◦ KDnuggets ◦ The Data Mine	Knowledge Discovery and Data Mining Bibliography (Gets updated frequently, so visit often!) • Books • General Data Mining
Book References in Data Mining and Knowledge Discovery Usama Fayyad, Gregory Piatetsky-Shapiro, Padhraic Smyth, and Ramasamy uthurasamy, "Advances in Knowledge Discovery and Data Mining", AAAI Press/the MIT Press, 1996. J. Ross Quinlan, "C4.5: Programs for Machine Learning", Morgan Kaufmann Publishers, 1993. Michael Berry and Gordon Linoff, "Data Mining Techniques (For Marketing, Sales, and Customer Support)", John Wiley & Sons, 1997.	General Data Mining Usama Fayyad, "Mining Databases: Towards Algorithms for Knowledge Discovery", Bulletin of the IEEE Computer Society Technical Committee on data Engineering, vol. 21, no. 1, March 1998. Christopher Matheus, Philip Chan, and Gregory Piatetsky-Shapiro, "Systems for knowledge Discovery in databases", IEEE Transactions on Knowledge and Data Engineering, 5(6):903-913, December 1993.



Ordered Data

- Attributes may have relationships that involve order in time or space

Sequential Data/Temporal data

- Extension of record data, where each record has a time associated with it
- Example: retail transaction data set that also stores the time at which the transaction took place

Time	Customer	Items Purchased
t1	C1	A, B
t2	C3	A, C
t2	C1	C, D
t3	C2	A, D
t4	C2	E
t5	C1	A, E

Customer	Time and Items Purchased
C1	(t1: A,B) (t2:C,D) (t5:A,E)
C2	(t3: A, D) (t4: E)
C3	(t2: A, C)

Sequence Data

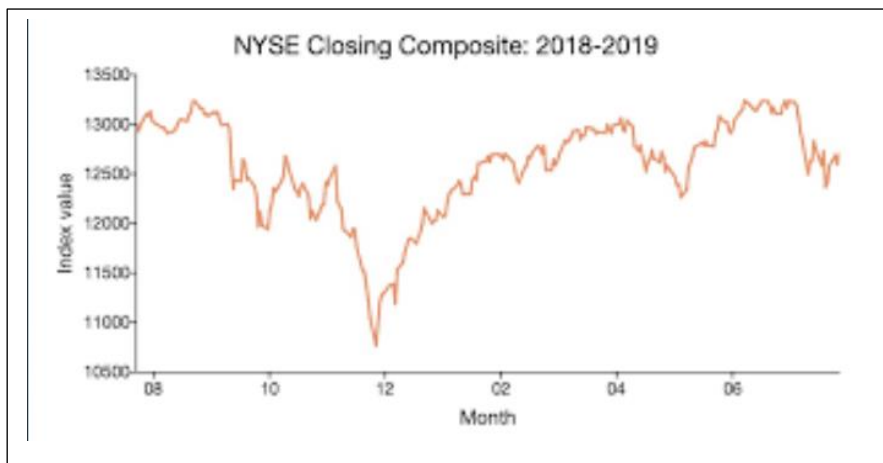
- Consists of a data set that is a sequence of individual entities, such as a sequence of words or letters
- Quite similar to sequential data, except that there are no time stamps; instead, there are positions in an ordered sequence

Example: Genomic sequence data

```
GGTTCCGCCTTCAGCCCCGCGCC
CGCAGGGCCCGCCCCGCGCCGTC
GAGAAGGGCCCGCCTGGCGGGCG
GGGGGAGGCGGGGCCCGCCCGAGC
CCAACCGAGTCCGACCAGGTGCC
CCCTCTGCTCGGCCTAGACCTGA
GCTCATTAGGCGGCAGCGGACAG
```


Time Series Data

- Time-series data contain values that are typically generated by continuous measurement over time
- If two measurements are close in time, then the values of those measurements are often very similar
- Financial data set might contain objects that are time series of the daily prices of various stocks



Spatial Data

- Some objects have spatial attributes, such as positions or areas
- Examples: Weather data (precipitation, temperature, pressure) that is collected for a variety of geographical locations
- Objects that are physically close tend to be similar in other ways also
 - Example: Data sets that are the result of measurements or model output taken at regularly or irregularly distributed points on a two or three-dimensional grid or mesh

9) For the following vectors x & y , calculate indicated similarity or distance measures.

a) $x=(1,1,1,1)$, $y=(2,2,2,2)$: Cosine, Euclidean

b) $x=(0,1,0,1)$, $y=(1,0,1,0)$: Cosine, Euclidean, Jacard

Ans:

a)

Cosine

$$\text{Cos}(x,y) = \frac{(x,y)}{\|x\|\|y\|}$$

$$x.y = (1*2)+(1*2)+(1*2)+(1*2)$$

$$= 2+2+2+2$$

$$= 8$$

$$\|x\| = \sqrt{(1*1)+(1*1)+(1*1)+(1*1)}$$

$$= \sqrt{1+1+1+1}$$

$$= \sqrt{4}$$

$$= 2$$

$$\|y\| = \sqrt{(2*2)+(2*2)+(2*2)+(2*2)}$$

$$= \sqrt{4+4+4+4}$$

$$= \sqrt{16}$$

$$= 4$$

$$\text{Cos}(x,y) = 8$$

$$(2*4)$$

$$=1$$

Euclidean distance

	X	y
P1	1	2
P2	1	2
P3	1	2
P4	1	2

	P1	P2	P3	P4
P1	0.0	0.0	0.0	0.0
P2	0.0	0.0	0.0	0.0
P3	0.0	0.0	0.0	0.0
P4	0.0	0.0	0.0	0.0

$$d(P2,P1) = \sqrt{(1-1)^2 + (2-2)^2}$$

$$= \sqrt{0+0}$$

$$=0$$

$$d(P3,P1) = \sqrt{(1-1)^2 + (2-2)^2}$$

$$= \sqrt{0+0}$$

$$=0$$

$$d(P4,P1)=0$$

$$d(P3,P2)=0$$

$$d(P4,P2)=0$$

$$d(P4,P3)=0$$

$$d(P1,P2)=0$$

$$d(P1,P3)=0$$

$$d(P1,P4)=0$$

$$d(P2,P3)=0$$

$$d(P2,P4)=0$$

$$d(P3,P4)=0$$

$$b) x = (0,1,0,1)$$

$$y = (1,0,1,0)$$

Cosine

$$\text{Cos}(x,y) = \frac{(x,y)}{\|x\|\|y\|}$$

$$x,y = (0*1)+(1*0)+(0*1)+(1*0)$$

$$= 0+0+0+0$$

$$= 0$$

$$\|x\| = \sqrt{(0*0)+(1*1)+(0*0)+(1*1)}$$

$$= \sqrt{0+1+0+1}$$

$$= \sqrt{2}$$

$$= 1.414$$

$$\|y\| = \sqrt{(1*1)+(0*0)+(1*1)+(0*0)}$$

$$= \sqrt{1+0+1+0}$$

$$\sqrt{2}$$

$$= \sqrt{2}$$

$$= 1.414$$

$$\text{Cos}(x,y) = 0$$

$$\frac{(1.414+1.414)}{2}$$

$$= 0$$

Jacard

$$x = (0,1,0,1)$$

$$y = (1,0,1,0)$$

$$f_{01}=2$$

$$f_{00}=0$$

$$f_{11}=0$$

$$f_{10}=2$$

$$J = \frac{f_{11}}{f_{01}+f_{10}+f_{11}}$$

$$= \frac{0}{2+2+0}$$

$$= 0$$

Euclidean distance

	X	y
P1	0	1
P2	1	0
P3	0	1
P4	1	0

	P1	P2	P3	P4
P1	0.0	1.414	0.0	1.414
P2	1.414	0.0	1.414	0.0

P3	0.0	1.414	0.0	1.414
P4	1.414	0.0	1.414	0.0

$$\begin{aligned}
 d(P2, P1) &= \sqrt{(1-0)^2 + (0-1)^2} \\
 &= \sqrt{1^2 + 1^2} \\
 &= 1.414
 \end{aligned}$$

$$\begin{aligned}
 d(P3, P1) &= \sqrt{(0-0)^2 + (1-1)^2} \\
 &= \sqrt{0+0} \\
 &= 0
 \end{aligned}$$

$$d(P4, P1) = 1.414$$

$$d(P3, P2) = 1.414$$

$$d(P4, P2) = 0$$

$$d(P4, P3) = 1.414$$

$$d(P1, P2) = 1.414$$

$$d(P1, P3) = 0$$

$$d(P1, P4) = 1.414$$

$$d(P2, P3) = 1.414$$

$$d(P2, P4) = 0$$

$$d(P3, P4) = 1.414$$

10) What are Data sets? Explain the general characteristics of data sets. Collection of data objects.

Important Characteristics of Data sets

Dimensionality

- Number of attributes that the objects in the data set possess
- High dimensional data brings a number of challenges
- Difficulties associated with analysing high-dimensional data

Curse of dimensionality

Sparsity

- For some data sets, most attributes of an object have values of 0
- Advantage because usually only the non-zero values need to be stored and manipulated
- Results in significant savings with respect to computation time and storage
- Some data mining algorithms work well only for sparse data

Resolution

- It is possible to obtain data at different levels of resolution
- Properties of the data are different at different resolutions
- Patterns in the data depend on the level of resolution
- If the resolution is too fine, a pattern may not be visible or may be buried in noise; if the resolution is too coarse, the pattern may disappear
- Example: Variations in atmospheric pressure on a scale of hours reflect the movement of storms and other weather systems. On a scale of months, such phenomena are not detectable

11) List and explain the strategies for dealing with missing data.

- i. Eliminate Data Objects or Attributes
 - Simple and effective strategy is to eliminate objects with missing values
 - Issues: If many objects have missing values, reliable analysis can be difficult if data objects are eliminated; also partially specified data will be lost
 - If a dataset has only few objects, that have missing values, removing objects is not a good idea
 - As an alternative, attributes that have missing values can be eliminated
 - Should be done with caution, as attributes may be ones that are critical to the analysis
- ii. Estimate Missing Values
 - Sometimes, missing data can be reliably be estimated by using the remaining values
 - If we consider a data set that ha many similar data points, attribute values of the points closest to the point with missing value are often used to estimate the missing value
 - For continuous attribute, average attribute value of the nearest neighbor is used; for categorical attribute, most commonly occurring attribute value can be taken
 - Example: for areas not containing a ground station, the precipitation can be estimated using values observed at nearby ground stations
- iii. Ignore the Missing value during Analysis
 - Many data mining approaches can be modified to ignore missing values
 - Example: if objects are clustered, and the similarity between pairs of data objects to be calculated-
 - If one or both objects of a pair have missing values, for some attributes, similarity can be calculated using only the attributes that do not have missing values

12) With example discuss feature subset selection and feature extraction. (Explain Feature construction.)

Feature Subset Selection

- Reducing dimensionality by using only a subset of features
- Involves removing:
 - Redundant features (duplication of much or all of the information contained in one or more other attributes)
 - Irrelevant features (contain almost no useful information for the data mining task at hand) are removed
- Ideal approach: Try all possible subsets of features as input to the data mining algorithm of interest, and then take the subset that produces the best results
- Number of subsets involving n attributes is 2^n , such an approach is impractical in most situations

Standard approaches to Feature Selection:

Embedded Approach

- Feature selection occurs naturally as part of the data mining algorithm
- During the operation of the data mining algorithm, the algorithm itself decides which attributes to use and which to ignore
- Example: Algorithms for building decision tree classifier

Filter approaches

- Features are selected before the data mining algorithm is run, using some approach that is independent of the data mining task
- Example: selecting sets of attributes whose pairwise correlation is as low as possible

Wrapper approaches

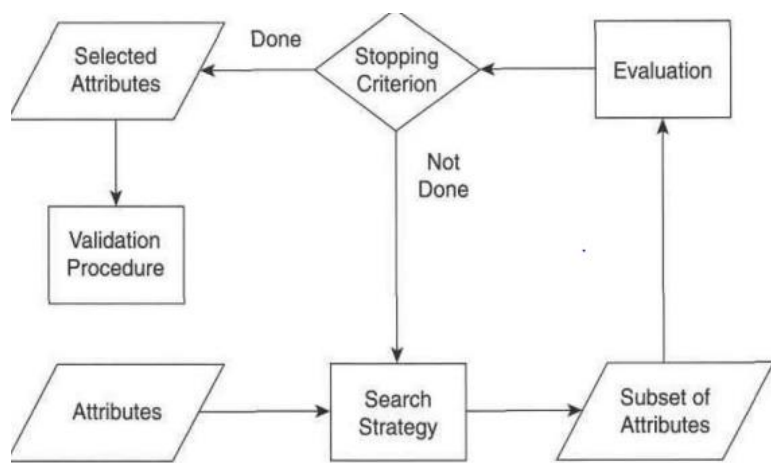
- Use the target data mining algorithm as a black box to find the best subset of attribute

An Architecture for Feature Subset Selection

It is possible to encompass both the filter and wrapper approaches within a common architecture

The Feature Selection process consists of four parts:

- Measure for evaluating a subset
- Search strategy that controls the generation of a new subset of features



- Stopping criterion
- Validation procedure

Feature Extraction

- Creation of a new set of features from the original raw data
- Photographs are classified according to whether or not it contains a human face
- Highly domain-specific
- Whenever data mining is applied to a relatively new area, a key task is the development of new features and feature extraction method

Feature Construction

- Sometimes the features in the original data sets have the necessary information, but it is not in a form suitable for the data mining algorithm
- One or more new features constructed out of the original features can be more useful than the original features
- Example: density feature constructed from the mass and volume features
density = mass/volume

13. Define data mining. Explain the challenges that motivated the development of data mining.

Ans: Data Mining is a non-trivial extraction of implicit, previously unknown, and potentially useful information from data.

Exploration and analysis, by automatic or semi-automatic means, of large quantities of data in order to discover meaningful patterns.

Motivating Challenges

1. Scalability
2. High Dimensionality
3. Heterogeneous and Complex Data
4. Data Ownership and Distribution
5. Non-traditional Analysis

Scalability

- Data sets are sizes of gigabytes, terabytes, or even petabytes are becoming common.
- Data mining algorithms are scalable.
- DM algorithms employ special search strategies to handle exponential search problems
- Implements novel data structures to access individual records efficiently.
- Can be improved by using sampling or developing parallel and distributed algorithms.

High Dimensionality

- Data sets with a large number of attributes are common.
- E.g: In bioinformatics, progress in microarray technology has produced gene expression data involving thousands of features.
- Data sets with temporal or spatial components also tend to have high dimensionality.

Heterogeneous and Complex Data

- Traditional data analysis methods often deal with data sets containing attributes of the same type, either continuous or categorical.
- But need not techniques that can handle heterogeneous attributes has grown.
- Emergence of more complex data objects.
- Examples of non-traditional types of data include
 - Collections of Web pages containing semi-structured text and hyperlinks.
 - DNA data with sequential and three-dimensional structure.
 - Climate data.

Data Ownership and Distribution

- Data is geographically distributed among resources belonging to multiple entities.
- Distributed data mining algorithms challenges:
 - How to reduce the amount of communication needed to perform the distributed computation.
 - How to effectively consolidate the data mining results obtained from multiple sources.
 - How to address data security issues.

Non-traditional Analysis

- Hypothesis is proposed, an experiment is designed to gather the data, and then the data is analyzed with respect to the hypothesis-labor intensive.
- Current data analysis tasks often require the generation and evaluation of thousands of hypotheses.
- Development of some data mining techniques has been motivated by the desire to automate the process of hypothesis generation and evaluation.
- Data sets analyzed in data mining are typically not the result of a carefully designed experiment.
- Often represent opportunistic samples of the data.
- Data sets frequently involved non-traditional types of data and data distribution.

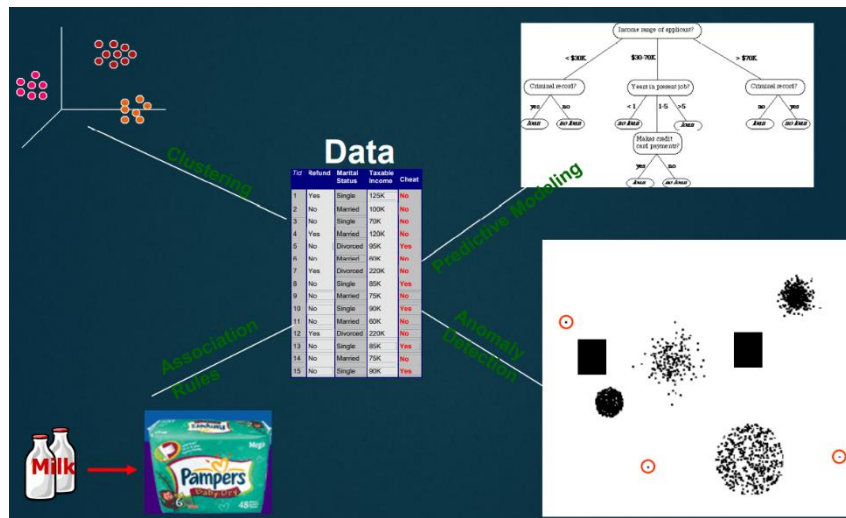
14. Explain different data mining tasks in detail.

➤ Predictive Methods

- The objective of these tasks is to predict the value of a particular attribute based on the values of other attributes.
- Use some attributes to predict unknown or future values of other variables.
- The attribute to be predicted is commonly known as the target or dependent variable.
- The attributes used for making the prediction are known as the explanatory or independent variables.

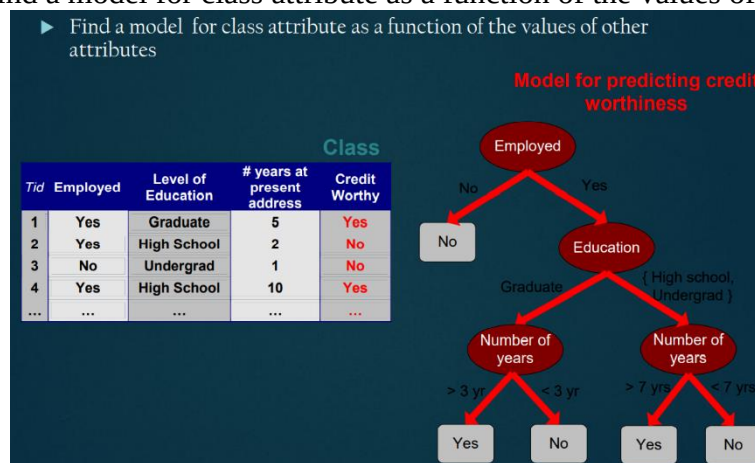
➤ Descriptive Methods

- Objective is to derive patterns (correlation, trends, clusters, trajectories, and anomalies) that summarize the underlying relationships in data.
- Often exploratory in nature and frequently require postprocessing techniques to validate and explain the results.
- Find human-interpretable patterns that describe the data.



➤ Predictive Modeling

- task of building a model for the target variable as a function of the explanatory variables.
- Classification**
 - ✓ Used for discrete target variables
 - ✓ Predicting whether a Web user will make a purchase at an online bookstore is a classification task because the target variable is binary-valued.
- Regression**
 - ✓ Used for continuous target variables
 - ✓ Forecasting the future price of a stock is a regression task because the price is a continuous-valued attribute.
- Find a model for class attribute as a function of the values of other attributes.

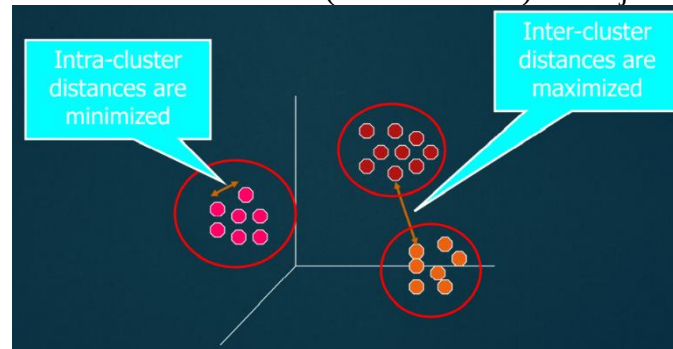


Regression

- Predict a value of a given continuous valued variable based on the values of other variables, assuming a linear or nonlinear model of dependency.
- Extensively studied in statistics, neural network fields.
- Examples:
 - ✓ Predicting sales amounts of new product based on advertising expenditure.
 - ✓ Predicting wind velocities as a function of temperature, humidity, air pressure, etc.
 - ✓ Time series prediction of stock market indices.

➤ **Clustering**

- Finding groups of objects such that the objects in a group will be similar (or related) to one another and different from (or unrelated to) the objects in other groups



Applications of Cluster Analysis

- Understanding
 - ✓ Custom profiling for targeted marketing
 - ✓ Group related documents for browsing
 - ✓ Group genes and proteins that have similar functionality
 - ✓ Group stocks with similar price fluctuations
- Summarization
 - ✓ Reduce the size of large data.

➤ **Association analysis**

- Used to discover patterns that describe strongly associated features in the data.
- The discovered patterns are typically represented in the form of implication rules or feature subsets
- Because of the exponential size of its search space, the goal of association analysis is to extract the most interesting patterns in an efficient manner.

Association Rule Discovery: Definition

- Given a set of records each of which contain some number of items from a given collection
- Produce dependency rules which will predict occurrence of an item based on occurrences of other items.

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Rules Discovered:

$\{Milk\} \rightarrow \{Coke\}$

$\{Diaper, Milk\} \rightarrow \{Beer\}$

Association Analysis: Applications

- Market-basket analysis
 - ✓ Rules are used for sales promotion, shelf management, and inventory management
- Telecommunication alarm diagnosis
 - ✓ Rules are used to find the combination of alarms that occur together frequently in the same time period
- Medical Informatics

- ✓ Rules are used to find the combination of patient symptoms and test results associated with certain diseases.

➤ **Deviation/Anomaly/Change Detection**

- Detect significant deviations from normal behaviour
- Applications:
 - ✓ Credit Card Fraud Detection
 - ✓ Network Intrusion Detection
 - ✓ Identify anomalous behaviour from sensor networks for monitoring and surveillance.
 - ✓ Detecting changes in the global forest cover.

15. How do you measure similarity for binary data? Explain.

Compare the measures.

Similarity Coefficients

- Similarity measures between objects that contain only binary attributes
- typically have values between 0 and 1
- 1 indicates that the two objects are completely similar; 0 indicates that the objects are not at all similar
- Let x and y be two objects that consist of n binary attributes. The comparison of two such objects, i.e., two binary vectors, leads to the following four quantities(frequencies)

f_{00} = the number of attributes where x is 0 and y is 0

f_{01} = the number of attributes where x is 0 and y is 1

f_{10} = the number of attributes where x is 1 and y is 0

f_{11} = the number of attributes where x is 1 and y is 1

Simple Matching Coefficient

$$SMC = \frac{\text{number of matching attribute values}}{\text{number of attributes}} = \frac{f_{11} + f_{00}}{f_{01} + f_{10} + f_{11} + f_{00}}.$$

- Counts both presences and absences equally.

Jaccard Coefficient

Used to handle objects consisting of asymmetric binary attributes.

$$J = \frac{\text{number of matching presences}}{\text{number of attributes not involved in 00 matches}} = \frac{f_{11}}{f_{01} + f_{10} + f_{11}}.$$

$$\mathbf{x} = (1, 0, 0, 0, 0, 0, 0, 0, 0)$$

$$\mathbf{y} = (0, 0, 0, 0, 0, 0, 1, 0, 0, 1)$$

$f_{01} = 2$ the number of attributes where $\mathbf{x}$ was 0 and $\mathbf{y}$ was 1

$f_{10} = 1$ the number of attributes where $\mathbf{x}$ was 1 and $\mathbf{y}$ was 0

$f_{00} = 7$ the number of attributes where $\mathbf{x}$ was 0 and $\mathbf{y}$ was 0

$f_{11} = 0$ the number of attributes where $\mathbf{x}$ was 1 and $\mathbf{y}$ was 1

$$SMC = \frac{f_{11} + f_{00}}{f_{01} + f_{10} + f_{11} + f_{00}} = \frac{0 + 7}{2 + 1 + 0 + 7} = 0.7$$

$$J = \frac{f_{11}}{f_{01} + f_{10} + f_{11}} = \frac{0}{2 + 1 + 0} = 0$$

Cosine Similarity

- One of the most common measures of document similarity
- Can handle non-binary vectors
- If $\mathbf{x}$ and $\mathbf{y}$ are two document vectors, then

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|},$$

- Where $\cdot$ Indicates the vector dot product
- $\|\mathbf{x}\|$ is the length of vector $\mathbf{x}$,

$$\mathbf{x} \cdot \mathbf{y} = \sum_{k=1}^n x_k y_k$$

$$\|\mathbf{x}\| = \sqrt{\sum_{k=1}^n x_k^2} = \sqrt{\mathbf{x} \cdot \mathbf{x}}$$

$$\mathbf{x} = (3, 2, 0, 5, 0, 0, 0, 2, 0, 0)$$

$$\mathbf{y} = (1, 0, 0, 0, 0, 0, 0, 1, 0, 2)$$

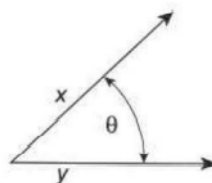
$$\mathbf{x} \cdot \mathbf{y} = 3 * 1 + 2 * 0 + 0 * 0 + 5 * 0 + 0 * 0 + 0 * 0 + 0 * 0 + 2 * 1 + 0 * 0 + 0 * 2 = 5$$

$$\|\mathbf{x}\| = \sqrt{3 * 3 + 2 * 2 + 0 * 0 + 5 * 5 + 0 * 0 + 0 * 0 + 0 * 0 + 2 * 2 + 0 * 0 + 0 * 0} = 6.48$$

$$\|\mathbf{y}\| = \sqrt{1 * 1 + 0 * 0 + 0 * 0 + 0 * 0 + 0 * 0 + 0 * 0 + 0 * 0 + 1 * 1 + 0 * 0 + 2 * 2} = 2.24$$

$$\cos(\mathbf{x}, \mathbf{y}) = 0.31$$

- Cosine similarity really is a measure of the (cosine of the) angle between $\mathbf{x}$ and $\mathbf{y}$



- if the cosine similarity is 1, the angle between x and y is 0 degrees, and x and y are the same except for magnitude (length)
- If the cosine similarity is 0, then the angle between x and y is 90 degrees, and they do not share any terms (words)
- Dividing x and y by their lengths normalizes them to have a length of 1

$$\cos(\mathbf{x}, \mathbf{y}) = \frac{\mathbf{x}}{\|\mathbf{x}\|} \cdot \frac{\mathbf{y}}{\|\mathbf{y}\|} = \mathbf{x}' \cdot \mathbf{y}'$$

- For vectors with a length of 1, the cosine measure can be calculated by taking a simple dot product
- when many cosine similarities between objects are being computed, normalizing the objects to have unit length can reduce the time required.

16. Illustrate the simple matching coefficient and jaccard coefficient to measure similarity for binary data?

Ans:

Similarity Measures for Binary Data Similarity coefficients

- Similarity measures between objects that contain only binary attributes
- typically have values between 0 and 1 ▪

1 indicates that the two objects are completely similar; 0 indicates that the objects are not at all similar ▪

Let x and y be two objects that consist of n binary attributes. The comparison of two such objects, i.e., two binary vectors, leadsto the following four quantities(frequencies)

f_{00} = the number of attributes where **x** is 0 and **y** is 0

f_{01} = the number of attributes where **x** is 0 and **y** is 1

f_{10} = the number of attributes where **x** is 1 and **y** is 0

f_{11} = the number of attributes where **x** is 1 and **y** is 1

Simple matching coefficient

$$SMC = \frac{\text{number of matching attribute values}}{\text{number of attributes}} = \frac{f_{11} + f_{00}}{f_{01} + f_{10} + f_{11} + f_{00}} \quad (2.5)$$

Jaccard Coefficient

- Used to handle objects consisting of asymmetric binary attributes

$$\mathbf{x} = (1, 0, 0, 0, 0, 0, 0, 0, 0, 0)$$

$$\mathbf{y} = (0, 0, 0, 0, 0, 0, 1, 0, 0, 1)$$

$f_{01} = 2$ the number of attributes where $\mathbf{x}$ was 0 and $\mathbf{y}$ was 1

$f_{10} = 1$ the number of attributes where $\mathbf{x}$ was 1 and $\mathbf{y}$ was 0

$f_{00} = 7$ the number of attributes where $\mathbf{x}$ was 0 and $\mathbf{y}$ was 0

$f_{11} = 0$ the number of attributes where $\mathbf{x}$ was 1 and $\mathbf{y}$ was 1

$$SMC = \frac{f_{11} + f_{00}}{f_{01} + f_{10} + f_{11} + f_{00}} = \frac{0 + 7}{2 + 1 + 0 + 7} = 0.7$$

$$J = \frac{f_{11}}{f_{01} + f_{10} + f_{11}} = \frac{0}{2 + 1 + 0} = 0$$

17. Calculate the Minkowski distance for the below four two dimensional points P1(0,2), P2((2,0), P3(3,1) and P4(5,1).

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

L1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

L2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

L_{∞}	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

Distance Matrix

1. Write the Apriori principle and explain the pseudo code for the frequent itemset generation part of the Apriori algorithm.
State and illustrate Apriori Principle. Mention the characteristics of same.
Illustrate frequent itemset generation using apriori algorithm.

Apriori principle:

- Used to Reduce the number of candidate item sets
- If an itemset is frequent, then all of its subsets must also be frequent

Pseudo code:

```

1:  $k = 1$ .
2:  $F_k = \{ i \mid i \in I \wedge \sigma(\{i\}) \geq N \times \text{minsup} \}$ .    {Find all frequent 1-itemsets}
3: repeat
4:    $k = k + 1$ .
5:    $C_k = \text{apriori-gen}(F_{k-1})$ .    {Generate candidate itemsets}
6:   for each transaction  $t \in T$  do
7:      $C_t = \text{subset}(C_k, t)$ .    {Identify all candidates that belong to  $t$ }
8:     for each candidate itemset  $c \in C_t$  do
9:        $\sigma(c) = \sigma(c) + 1$ .    {Increment support count}
10:    end for
11:  end for
12:   $F_k = \{ c \mid c \in C_k \wedge \sigma(c) \geq N \times \text{minsup} \}$ .    {Extract the frequent  $k$ -itemsets}
13: until  $F_k = \emptyset$ 
14: Result =  $\bigcup F_k$ .
```

Characteristics:

- Level-wise algorithm:
 It traverses the itemset lattice one level at a time, from frequent 1 item sets to the maximum size of frequent item set.
- Employs a generate-and-test strategy for finding frequent itemset:
 At each iteration, new candidate item sets are generated from the frequent item sets found in the previous iteration. The support for each candidate is then counted and tested

against the min sup threshold.

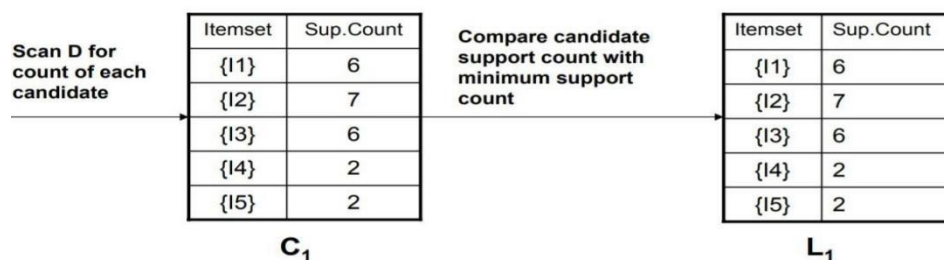
- ❖ Illustrate frequent item set generation using apriori algorithm.

The Apriori Algorithm: Example

TID	List of Items
T100	I1, I2, I5
T100	I2, I4
T100	I2, I3
T100	I1, I2, I4
T100	I1, I3
T100	I2, I3
T100	I1, I3
T100	I1, I2, I3, I5
T100	I1, I2, I3

- Consider a database, D , consisting of 9 transactions.
- Suppose min. support count required is 2 (i.e. $\text{min_sup} = 2/9 = 22\%$)
- Let minimum confidence required is 70%.
- We have to first find out the frequent itemset using Apriori algorithm.
- Then, Association rules will be generated using min. support & min. confidence.

Step 1: Generating 1-itemset Frequent Pattern

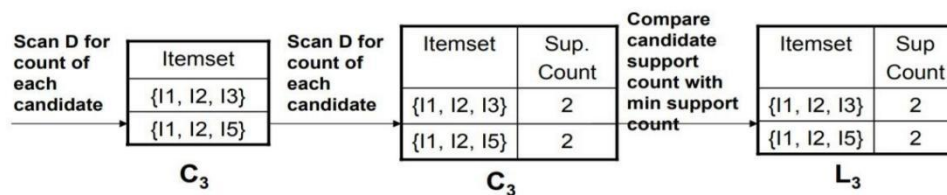


- The set of frequent 1-itemsets, L_1 , consists of the candidate 1-itemsets satisfying minimum support.
- In the first iteration of the algorithm, each item is a member of the set of candidate.

Step 2: Generating 2-itemset Frequent Pattern

- To discover the set of frequent 2-itemsets, L_2 , the algorithm uses $L_1 \text{ Join } L_1$ to generate a candidate set of 2-itemsets, C_2 .
- Next, the transactions in D are scanned and the support count for each candidate itemset in C_2 is accumulated (as shown in the middle table).
- The set of frequent 2-itemsets, L_2 , is then determined, consisting of those candidate 2-itemsets in C_2 having minimum support.
- **Note:** We haven't used Apriori Property yet.

Step 3: Generating 3-itemset Frequent Pattern



- The generation of the set of candidate 3-itemsets, C_3 , involves use of the Apriori Property.
- In order to find C_3 , we compute $L_2 \text{ Join } L_2$.
- $C_3 = L_2 \text{ Join } L_2 = \{\{I1, I2, I3\}, \{I1, I2, I5\}, \{I1, I3, I5\}, \{I2, I3, I4\}, \{I2, I3, I5\}, \{I2, I4, I5\}\}$.
- Now, Join step is complete and Prune step will be used to reduce the size of C_3 . Prune step helps to avoid heavy computation due to large C_k .

Step 4: Generating 4-itemset Frequent Pattern

- The algorithm uses $L_3 \text{ Join } L_3$ to generate a candidate set of 4-itemsets, C_4 . Although the join results in $\{\{I1, I2, I3, I5\}\}$, this itemset is pruned since its subset $\{\{I2, I3, I5\}\}$ is not frequent.
- Thus, $C_4 = \emptyset$, and algorithm terminates, having found all of the frequent items. This completes our Apriori Algorithm.
- What's Next ?
These frequent itemsets will be used to generate strong association rules (where strong association rules satisfy both minimum support & minimum confidence).

2. Define Association rule. Explain different algorithm for association rule mining. Write the use of support and confidence.

Association rule:

An implication expression of the form $X \Rightarrow Y$, where X and Y are disjoint item sets.

- Common strategy adopted by many association rule mining algorithms is to decompose the problem into two major subtasks:
- Frequent Itemset Generation



- whose objective is to find all the item sets that satisfy the min sup threshold. These item sets are called frequent item sets
- Rule Generation
- whose objective is to extract all the high-confidence rules from the frequent item sets found in the previous step. These rules are called strong rules

The computational requirements for frequent itemset generation are generally more expensive than those of rule generation

Use of support and confidence:

- A rule that has very low support may occur simply by chance
- A low support rule is also likely to be uninteresting from a business perspective because it may not be profitable to promote items that customers seldom buy together
- For these reasons, support is often used to eliminate uninteresting rules
- Support also has a desirable property that can be exploited for the efficient discovery of association rule
- Confidence measures the reliability of the inference made by a rule
- For a given rule $X \rightarrow Y$, the higher the confidence, the more likely it is for Y to be present in transactions that contain X
- Confidence also provides an estimate of the conditional probability of Y given X

3. Illustrate the brute force method with reference to the association analysis

A brute-force approach for mining association rules is to compute the support and confidence for every possible rule

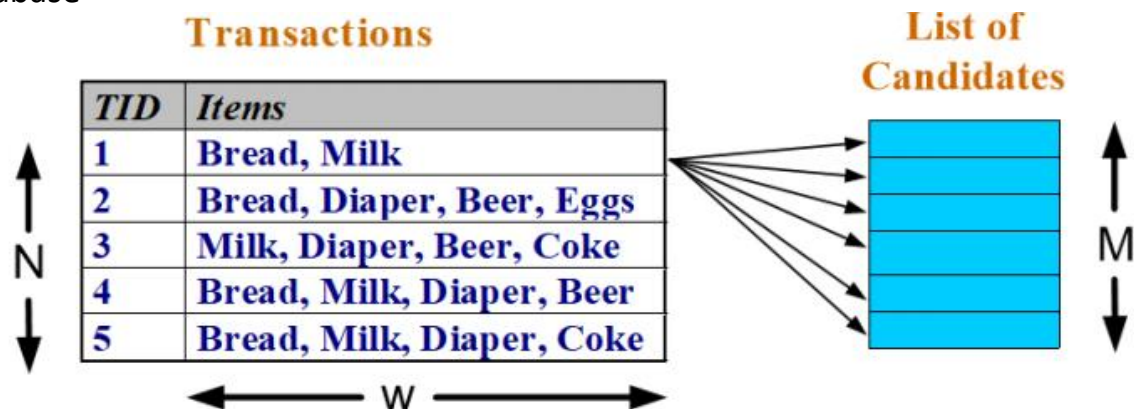
- This approach is prohibitively expensive because there are exponentially many rules that can be extracted from a data set
- Total number of possible rules extracted from a data set that contains

d items is $R = 3^d - 2^{d+1} + 1$

Brute-force approach:

- Each itemset in the lattice is a candidate frequent itemset

- Count the support count of each candidate by scanning the database



- Match each transaction against every candidate
- Complexity $\sim O(NMw) \Rightarrow$ Expensive since $M = 2^d$!!!

Candidate Generation: Brute-force method

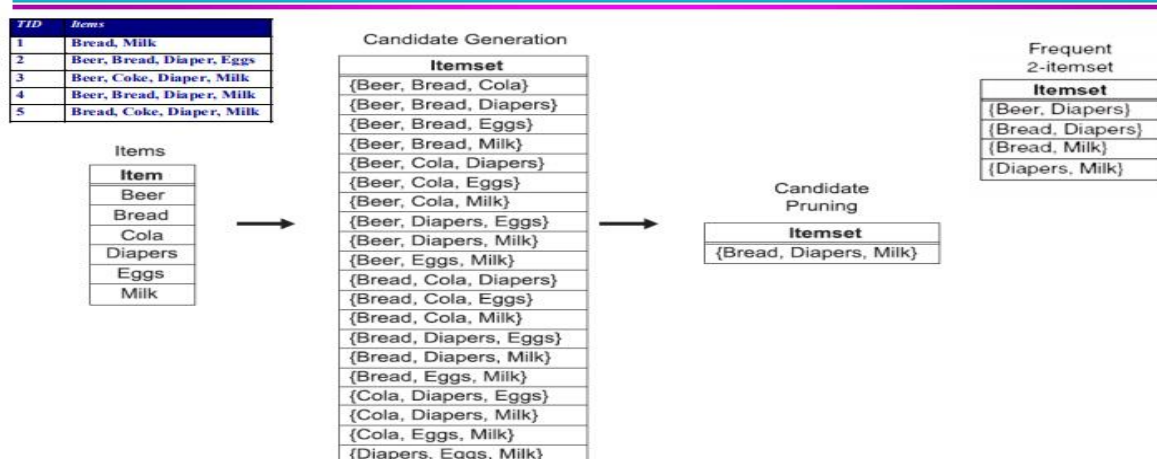
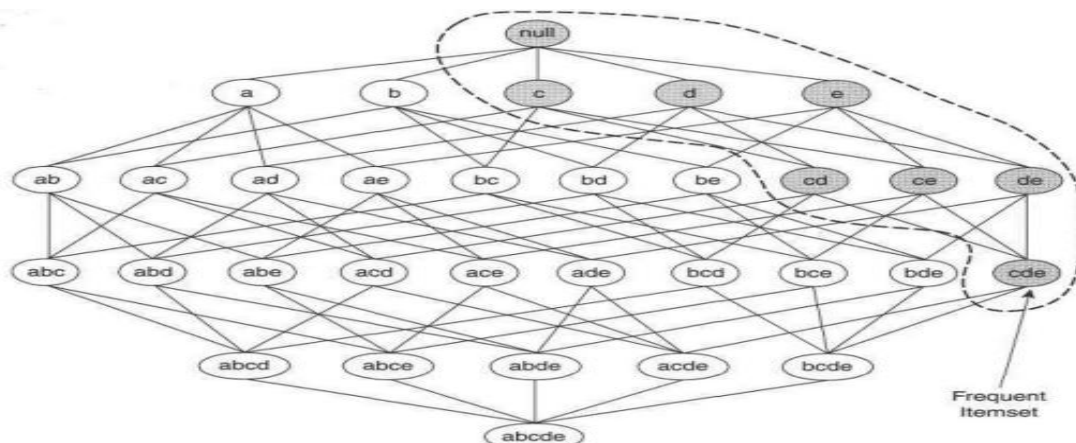


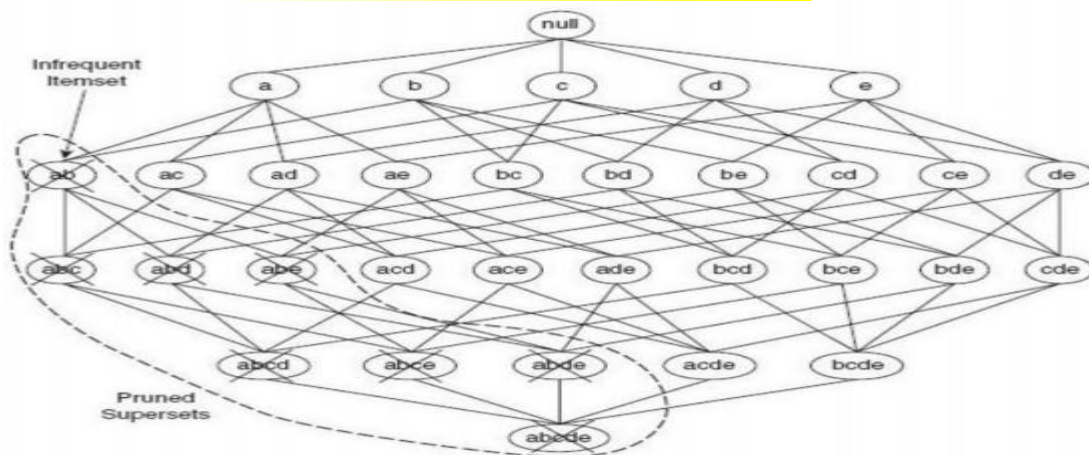
Figure 5.6. A brute-force method for generating candidate 3-itemsets.

4. Explain support based pruning strategy with suitable example. Also brief on Monotonicity property.

- Suppose $\{c, d, e\}$ is a frequent itemset
- Any transaction that contains $\{c, d, e\}$ must also contain its subsets, $\{c, d\}$, $\{c, e\}$, $\{d, e\}$, $\{c\}$, $\{d\}$, and $\{e\}$
- As a result, if $\{c, d, e\}$ is frequent, then all subsets of $\{c, d, e\}$ must also be frequent



- Conversely, if an itemset such as {a, b} is infrequent, then all of its supersets must be infrequent too
- the entire subgraph containing the supersets of {a, b} can be pruned immediately once {a, b} is found to be infrequent
- This strategy of trimming the exponential search space based on the support measure is known as **Support-based pruning**



Such a pruning strategy is made possible by a key property of the support measure

- **Anti-monotone property of the support measure:** The support for an itemset never exceeds the support for its subsets Monotonicity and Antimonotonicity Property
- Let I be a set of items, and $J = 2^I$ be the power set of I
 - A measure f is monotone (or upward closed) if i.e. if X is a subset of Y , then $f(X)$ must not exceed $f(Y)$
 - f is anti-monotone (or downward closed) if i.e.

$$\forall X, Y \in J : (X \subseteq Y) \longrightarrow f(Y) \leq f(X)$$

if X is a subset of Y , then $f(Y)$ must not exceed $f(X)$

Monotonicity:

Let I be a set of items, and $J = 2^I$ be the power set of I

A measure f is monotone (or upward closed) if

$$\forall X, Y \in J : (X \subseteq Y) \longrightarrow f(X) \leq f(Y)$$

i.e. if X is a subset of Y , then $f(X)$ must not exceed $f(Y)$

support based pruning strategy suitable example:

TID	Items
1	Bread, Milk
2	Beer, Bread, Diaper, Eggs
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Bread, Coke, Diaper, Milk



Items (1-itemsets)

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Minimum Support = 3

If every subset is considered,
 ${}^6C_1 + {}^6C_2 + {}^6C_3$
 $6 + 15 + 20 = 41$
 With support-based pruning,
 $6 + 6 + 4 = 16$

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Items (1-itemsets)



Itemset
{Bread, Milk}
{Bread, Beer }
{Bread, Diaper}
{Beer, Milk}
{Diaper, Milk}
{Beer, Diaper}

Pairs (2-itemsets)

(No need to generate candidates involving Coke or Eggs)

Minimum Support = 3

If every subset is considered,
 ${}^6C_1 + {}^6C_2 + {}^6C_3$
 $6 + 15 + 20 = 41$
 With support-based pruning,
 $6 + 6 + 4 = 16$

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Items (1-itemsets)



Itemset	Count
{Bread,Milk}	3
{Beer,Bread}	2
{Bread,Diaper}	3
{Beer,Milk}	2
{Diaper,Milk}	3
{Beer,Diaper}	3

Pairs (2-itemsets)

(No need to generate candidates involving Coke or Eggs)

Minimum Support = 3

If every subset is considered,

$${}^6C_1 + {}^6C_2 + {}^6C_3$$

$$6 + 15 + 20 = 41$$

With support-based pruning,

$$6 + 6 + 4 = 16$$

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Items (1-itemsets)



Itemset	Count
{Bread,Milk}	3
{Bread,Beer}	2
{Bread,Diaper}	3
{Milk,Beer}	2
{Milk,Diaper}	3
{Beer,Diaper}	3

Pairs (2-itemsets)

(No need to generate candidates involving Coke or Eggs)

Minimum Support = 3

If every subset is considered,

$${}^6C_1 + {}^6C_2 + {}^6C_3$$

$$6 + 15 + 20 = 41$$

With support-based pruning,

$$6 + 6 + 4 = 16$$



Itemset
{Beer,Diaper,Milk}
{Beer,Bread,Diaper}
{Bread,Diaper,Milk}
{Beer,Bread,Milk}

Triplets (3-itemsets)

Item	Count
Bread	4
Coke	2
Milk	4
Beer	3
Diaper	4
Eggs	1

Items (1-itemsets)



Itemset	Count
{Bread,Milk}	3
{Bread,Beer}	2
{Bread,Diaper}	3
{Milk,Beer}	2
{Milk,Diaper}	3
{Beer,Diaper}	3

Pairs (2-itemsets)

(No need to generate candidates involving Coke or Eggs)

Minimum Support = 3

If every subset is considered,

$${}^6C_1 + {}^6C_2 + {}^6C_3$$

$$6 + 15 + 20 = 41$$

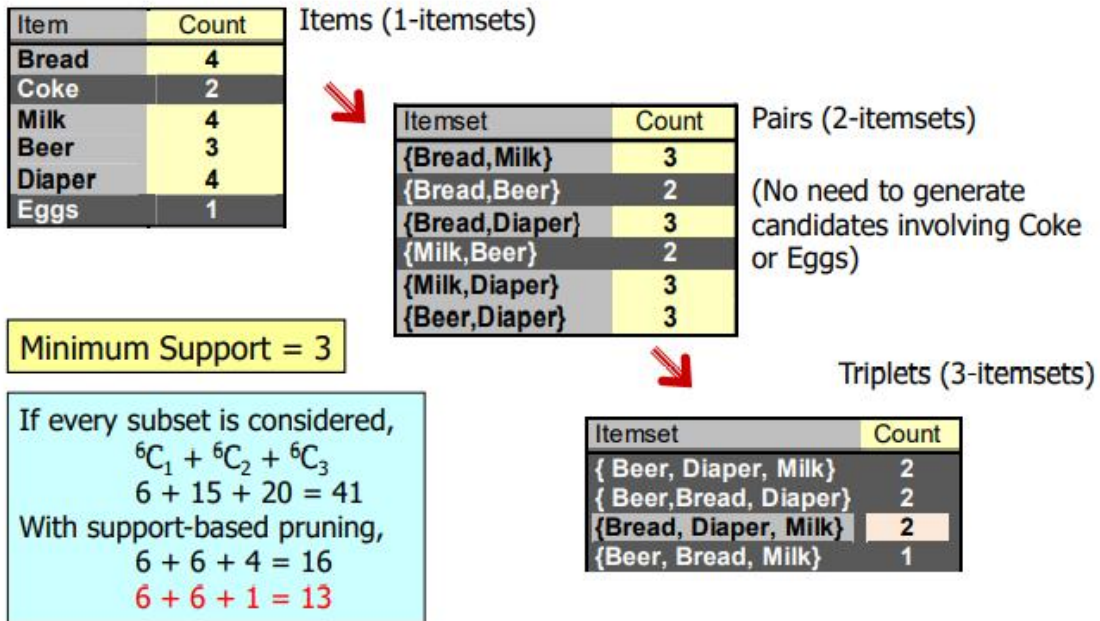
With support-based pruning,

$$6 + 6 + 4 = 16$$



Itemset	Count
{Beer,Diaper,Milk}	2
{Beer,Bread,Diaper}	2
{Bread,Diaper,Milk}	2
{Beer,Bread,Milk}	1

Triplets (3-itemsets)



5.What do you mean by Association analysis? Mention its applications.

What are the advantages.

Provide the meaning of terms: Item set and Support count?

- Association analysis is the methodology useful for discovering interesting relationships hidden in large data sets.
- Association analysis is the task of finding interesting relationships in large data set.
- These interesting relationships can take two forms: Frequent item set or Association rule
- Frequent item sets are a collection of items that frequently occur together.
- The second way to view interesting relationships is Association rule.
- Association rules suggest that a strong relationship exists between two items.
- Uncovered relationships can be represented in the form of association rule.

Application:



- **Market basket analysis**
A typical and widely-used example of association rule mining is market basket

analysis. For example, data are collected using bar-code scanners in supermarkets. Such 'Market basket' databases consist of a large number of transaction records. Each record lists all items bought by a customer on a single purchase transaction. Managers would be interested to know if certain groups of items are consistently purchased together. They could use this data for adjusting store layouts (placing items optimally with respect to each other), for cross-selling, for promotions, for catalog design and to identify customer segments based on buying patterns.

- **Medical diagnosis**

Applying association rules in medical diagnosis can be used for assisting physicians to cure patients. The general problem of the induction of reliable diagnostic rules is hard because theoretically no induction process by itself can guarantee the correctness of induced hypotheses

- **Protein sequences**

Proteins are sequences made up of 20 types of amino acids. Each protein has a unique 3-dimensional structure, which depends on amino-acid sequence; slight change in sequence may change the functioning of protein. The heavy dependence of protein functioning on its amino acid sequence has been a subject of great anxiety. Lot of research has gone into understanding the composition.

- **Census data**

Censuses make a huge variety of general statistical information on society available to both researchers and the general public [3]. The information related to population and economic census can be forecasted in planning public services (education, health,

transport, funds) as well as in public business (for setup new factories, shopping malls or banks and even marketing particular products)

- **CRM of credit card business**

Customer Relationship Management (CRM), through which, banks hope to identify the preference of different customer groups, products and services tailored to their liking to enhance the cohesion between credit card customers and the bank, has become a topic of great interest. Show mainly describes how to incorporate data mining into the framework of marketing knowledge management

- **Advantages:**

- **Medicine.**

Doctors can use association rules to help diagnose patients. There are many variables to consider when making a diagnosis, as many diseases share symptoms. By using association rules and machine learning-fuelled data analysis, doctors can determine the conditional probability of a given illness by comparing symptom relationships in the data from past cases. As new diagnoses get made, the machine learning model can adapt the rules to reflect the updated data.

- **Retail.**

Retailers can collect data about purchasing patterns, recording purchase data as item barcodes are scanned by point-of-sale systems. Machine learning models can look for co-occurrence in this data to determine which products are most likely to be purchased together. The retailer can then adjust marketing and sales strategy to take advantage of this information.

- **User experience (UX) design.**

Developers can collect data on how consumers use a website they create. They can then use associations in the data to optimize the website user interface -- by analysing where users tend to click and what maximizes the chance that they engage with a call to action, for example.

- **Entertainment.**

Services like Netflix and Spotify can use association rules to fuel their content recommendation engines. Machine learning models analyse past user behaviour data for frequent patterns, develop association rules and use those rules to recommend content that a user is likely to engage with, or organize content in a way that is likely to put the most interesting content for a given user first.

- **Item Set:**

An item set is **any grouping of one or more items**. Frequent item sets are individual item sets that meet or exceed a given minimum support threshold. Support between item sets is an association rule that gives the frequency at which both item sets occur in the same basket.

- **Support Count:**

The number of transactions that include items in the {X} and {Y} parts of the rule as a percentage of the total number of transactions. It is a measure of how frequently the collection of items occur together as a percentage of all transactions

6. Define Association Rule. How do you measure the strength of an Association rule?

Association rule mining is a procedure which is meant to find frequent patterns, correlations, associations, or causal structures from data sets found in various kinds of databases such as relational databases, transactional databases, and other forms of data repositories

An implications of the form $X \rightarrow Y$, where x and y are disjoint item sets. It is used for searching interesting relationships among items in a given data set.

Strength of an association rule can be measured in terms of its Support and Confidence.

Support:

- Determines how often a rule is applicable to given data set
- Function of transaction that contains both X and Y

$$\text{Support, } s(X \rightarrow Y) = \sigma(X \cup Y) / N$$

Confidence:

- Determines how frequently items in Y appear in transactions that contains X

Confidence, $c(X \rightarrow Y) = \sigma(X \cup Y) / \sigma(X)$

Example:

TID	ITEMS
1	Bread, Milk
2	Bread, Diaper, Beer, Eggs
3	Milk, Diaper, Beer, Coke
4	Bread, Milk, Diaper, Beer
5	Bread, Milk, Diaper, Coke

Consider the rule: {Milk, Diaper} => {Beer}

$$s = \sigma(\text{Milk, Diaper, Beer}) = \frac{2}{5} = 0.4$$
$$c = \frac{\sigma(\text{Milk, Diaper, Beer})}{\sigma(\text{Milk, Diaper})} = \frac{2}{3} = 0.67$$

8. Explain the process of FP tree construction with an example
Discuss the FP growth algorithm in detail.

1. The data set is scanned once to determine the support count of each item. Infrequent items are discarded, while the frequent items are sorted in decreasing support counts. For the data set shown in Figure 6.24, a is the most frequent item, followed by b, c, d, and e.

2. The algorithm makes a second pass over the data to construct the FPtree. After reading the first transaction, {a, b}, the nodes labeled as a and b are created. A path is then formed from null $\rightarrow$ a $\rightarrow$ b to encode the transaction. Every node along the path has a frequency count of 1.

3. After reading the second transaction, {b,c,d}, a new set of nodes is created for items b, c, and d. A path is then formed to represent the transaction by connecting the nodes null $\rightarrow$ b $\rightarrow$ c $\rightarrow$ d. Every node along this path also has a frequency count equal to one. Although the first two transactions have an item in common, which is b, their paths are disjoint because the transactions do not share a common prefix.

4. The third transaction, {a,c,d,e}, shares a common prefix item (which is a) with the first transaction. As a result, the path for the third transaction, null $\rightarrow$ a $\rightarrow$ c $\rightarrow$ d $\rightarrow$ e, overlaps with the path for the first transaction, null $\rightarrow$ a $\rightarrow$ b. Because of their overlapping path, the frequency count for node a is incremented to two, while the frequency counts for the newly created nodes, c, d, and e, are equal to one.

5. This process continues until every transaction has been mapped onto one of the paths given in the FP-tree. The resulting FP-tree after reading all the transactions is shown at the bottom of Figure

Example:

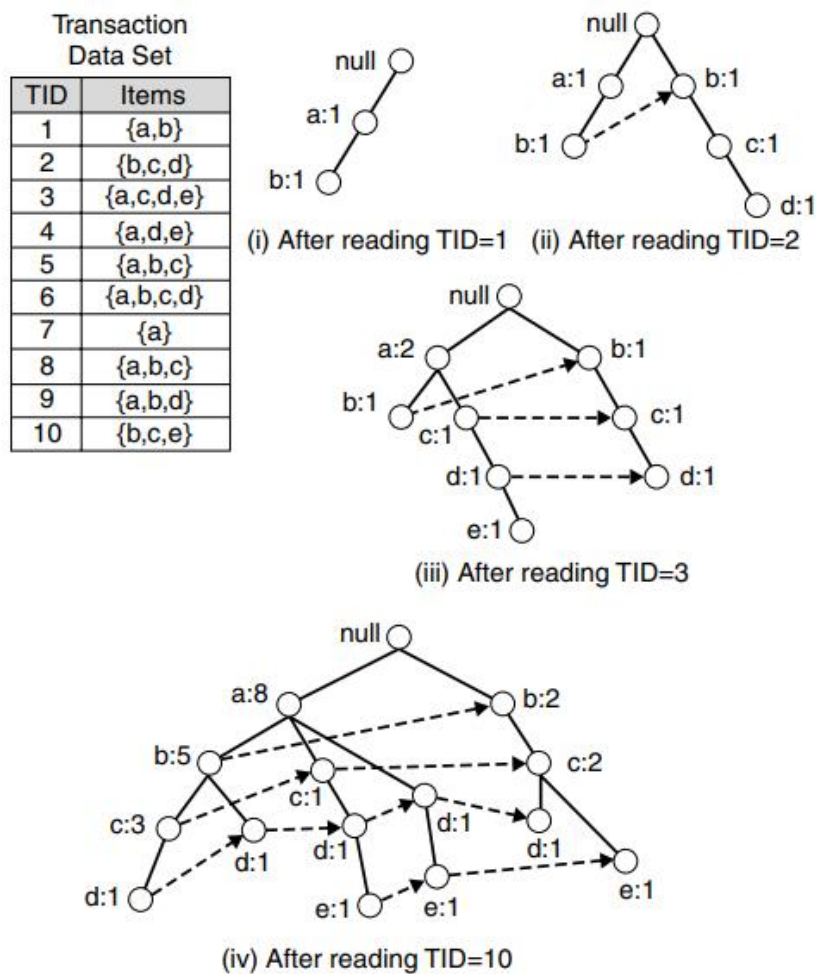


Figure 6.24. Construction of an FP-tree.

9. What is candidate generation and candidate pruning discuss ?

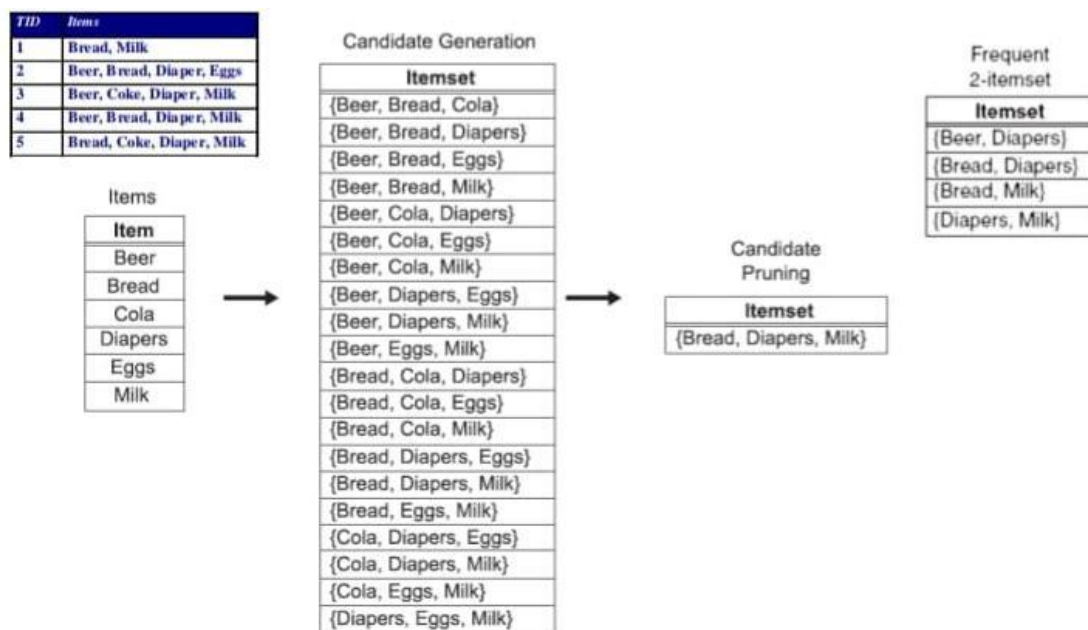
List of requirements for an effective candidate generation procedure:

- It should avoid generating too many unnecessary candidates (at least one of its subsets is infrequent)
 - It must ensure that the candidate set is complete (no frequent item sets are left out)
 - It should not generate same candidate item set more than once
- Candidate generation:

This operation generates new candidate k-itemsets based on the frequent (k - 1)-itemsets found in the previous iteration

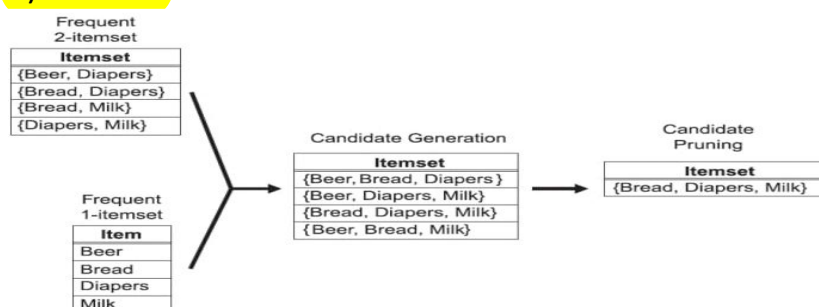
1.Brute-Force Method :

The brute-force method considers every k-itemset as a potential candidate and then applies the candidate pruning step to remove any unnecessary candidates whose subsets are infrequent. The number of candidate itemsets generated at level k is equal to d^k , where d is the total number of items. Although candidate generation is rather trivial, candidate pruning becomes extremely expensive because a large number of itemsets must be examined.



2. Merge $F_{k-1} \times F_1$ Method:

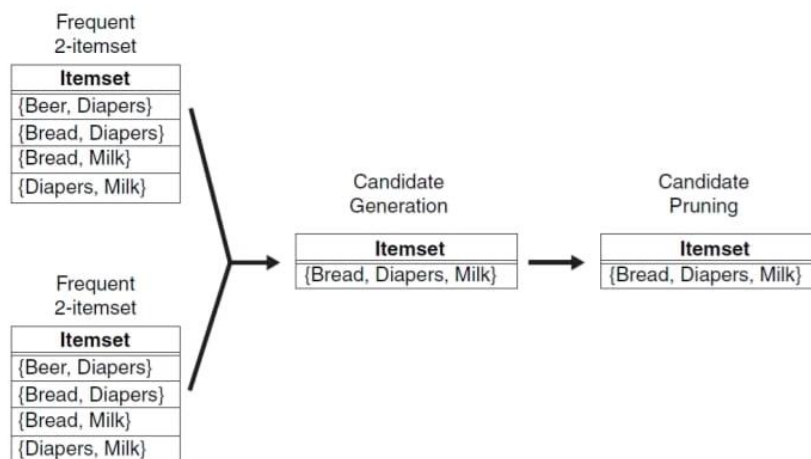
An alternative method for candidate generation is to extend each frequent $(k - 1)$ -itemset with frequent items that are not part of the $(k - 1)$ -itemset.



Generating and pruning candidate k-itemsets by merging a frequent(k-1) itemset with a frequent item. Some of the candidates are unnecessary because their subsets are infrequent.

3.Fk-1 x Fk-1 Method:

This candidate generation procedure, which is used in the candidate-gen function of the Apriori algorithm.



Generating and pruning candidate k-itemsets by merging pairs of frequent(k-1) itemsets.

10. Specify and discuss two way to reduce the computational complex of itemset generation.

Ways to reduce computational complexity

1) Reduce the number of candidate itemsets(M) .

The Apriori Principle is an effective way to eliminate some of the candidate itemsets

2) Reduce the number of comparisons.

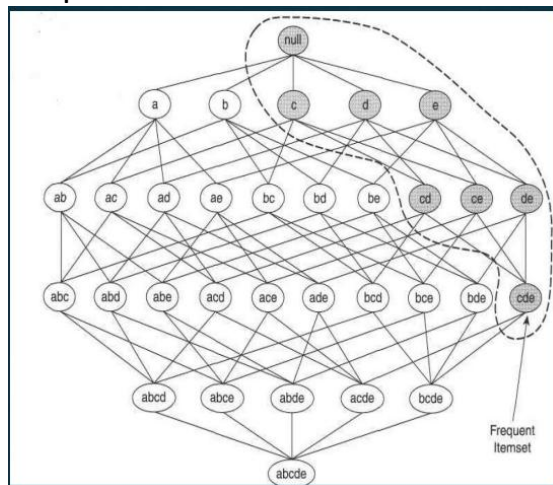
Use more advanced data structures either to store the candidate itemsets or to compress the data set

1) Reduce the number of candidate itemsets(M) .

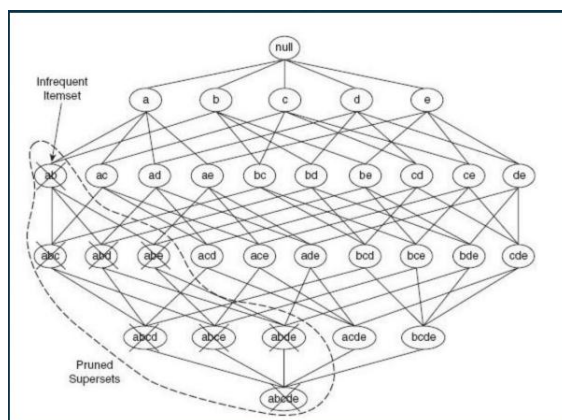
The Apriori Principle

- Used to Reduce the number of candidate itemsets
 - If an itemset is frequent, then all of its subsets must also be frequent •
- Suppose {c, d, e} is a frequent itemset
- Any transaction that contains {c,d,e} must also contain its subsets, {c,d}, {c,e}, {d,e}, {c}, {d}, and {e}

- As a result, if $\{c,d,e\}$ is frequent, then all subsets of $\{c, d,e\}$ must also be frequent



- Conversely, if an itemset such as $\{a, b\}$ is infrequent, then all of its supersets must be infrequent too
- the entire subgraph containing the supersets of $\{a, b\}$ can be pruned immediately once $\{a, b\}$ is found to be infrequent
 - This strategy of trimming the exponential search space based on the support measure is known as Support-based pruning



- Such a pruning strategy is made possible by a key property of the support measure
- Anti-monotone property of the support measure: The support for an itemset never exceeds the support for its subsets Monotonicity and Antimonotonicity Property
- Let I be a set of items, and $J = 2^I$ be the power set of I
- A measure f is monotone (or upward closed) if i.e.
 - if X is a subset of Y , then $f(X)$ must not exceed $f(Y)$
- f is anti-monotone (or downward closed) if i.e.
 - if X is a subset of Y , then $f(Y)$ must not exceed $f(X)$

2) Reduce the number of comparisons.

Candidate Generation List of requirements for an effective candidate generation procedure:

- It should avoid generating too many unnecessary candidates(atleast one of its subsets is infrequent)
- It must ensure that the candidate set is complete(no frequent itemsets are left out)
- It should not generate same candidate itemset more than once

12.Discuss the steps of FP tree construction by considering the following transaction of Market Basket data

TID	items
1	milk, bread, cookies, juice
2	milk, juice
3	milk, eggs
4	bread, cookies
5	milk, juice, bread
6	juice, eggs
7	eggs

Discuss the steps of FP-tree construction by considering the following transactions of market basket data. “FP-growth is an algorithm that generates frequent item sets from an FP-tree by exploring the tree in a bottom-up fashion”. Illustrate

- The frequency of each individual item is computed

Item	Frequency
Milk	4
Bread	3
Cookies	2
Juice	4
Egg	3

Let the minimum support be 1.

- A Frequent Pattern set is built which will contain all the elements whose frequency is greater than or equal to the minimum support.

▪ These elements are stored in descending order of their respective frequencies. After insertion of the relevant items, the set L looks like this:-

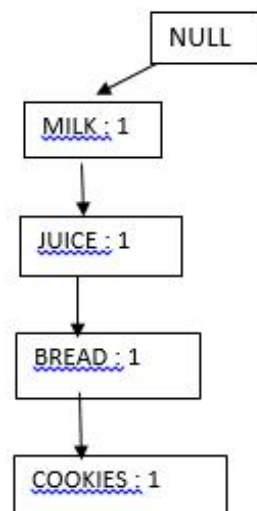
$L = \{\text{milk} : 4, \text{juice} : 4, \text{bread} : 3, \text{egg} : 3, \text{cookies} : 2\}$

- Now, for each transaction, the respective Ordered-Item set is built.

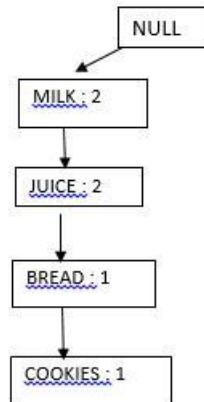
TID	items	Ordered-Item Set
1	{milk, bread, cookies, juice}	{Milk, juice, bread, cookies}
2	{milk, juice}	{milk, juice}
3	{milk, eggs}	{milk, eggs}
4	{bread, cookies}	{bread, cookies}
5	{milk, juice, bread}	{milk, juice, bread}
6	{juice, eggs}	{juice, eggs}
7	{Eggs}	{Eggs}

Construction of FP Tree :

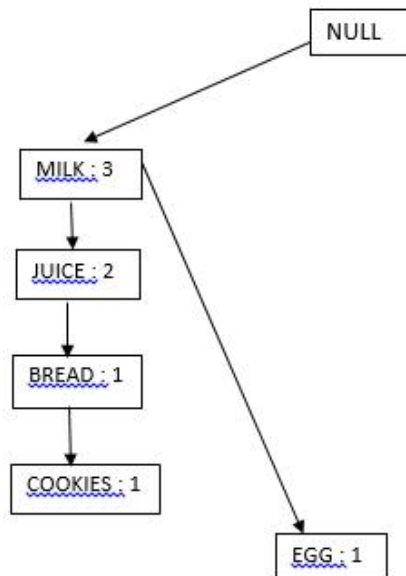
Step 1 : {Milk, juice, bread, cookies}



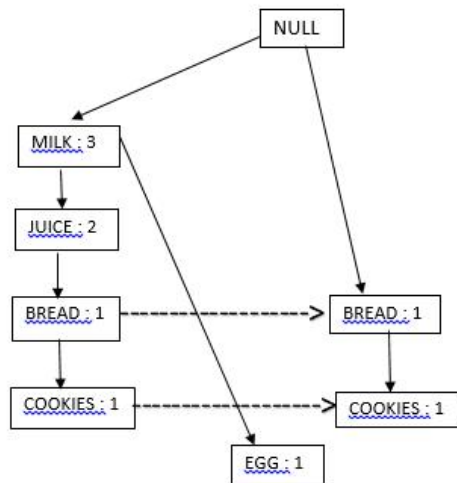
Step 2 : {milk, juice}



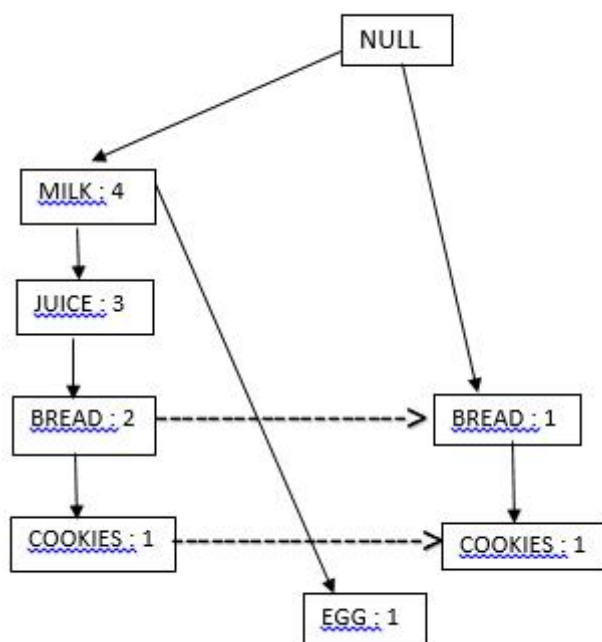
Step 3 : {milk,egg}



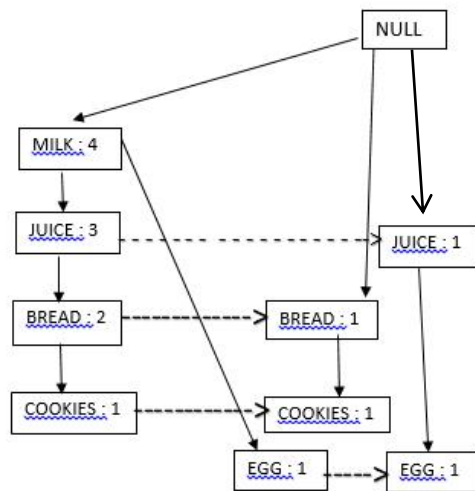
Step 4 : {bread, cookies}



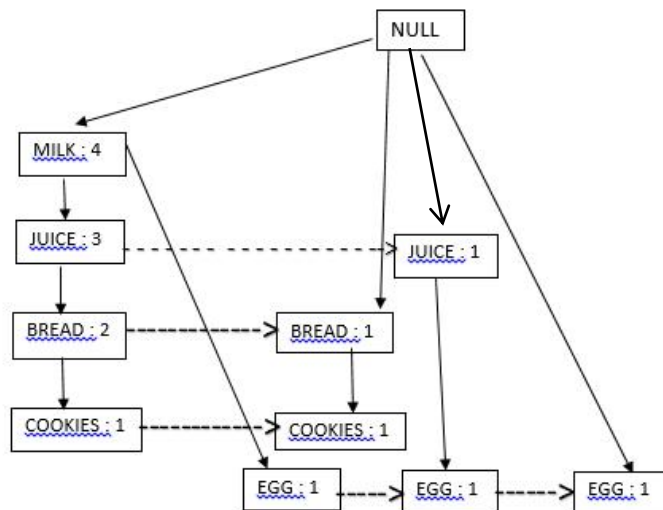
Step 5 : {milk, juice, bread}



Step 6 : {juice, egg}



Step 7 : {egg}



Note: Only write if they ask to find the rules.

Now, for each item, the Conditional Pattern Base is computed which is path labels of all the paths which lead to any node of the given item in the frequent-pattern tree

Item	Conditional Pattern Base
Egg	{juice : 1} {milk : 1}
Cookies	{milk, juice, bread : 1} {bread : 1}
Bread	{milk, juice : 2}
Juice	{milk : 3}
Milk	

Now for each item the Conditional Frequent Pattern Tree is built. It is done by taking the set of elements which is common in all the paths in the Conditional Pattern Base of that item and calculating it's support count by summing the support counts of all the paths in the Conditional Pattern Base

Item	Conditional Pattern Base	Conditional Pattern Tree
Egg	{juice : 1} {milk : 1}	Null
Cookies	{milk, juice, bread : 1} {bread : 1}	{ bread : 2}
Bread	{milk, juice : 2}	{milk, juice : 2}
Juice	{milk : 3}	{milk : 3}
Milk		

--	--	--

From the Conditional Frequent Pattern tree, the Frequent Pattern rules are generated by pairing the items of the Conditional Frequent Pattern Tree set to the corresponding to the item

Items	Frequent Pattern Generated
Egg	Null
Cookies	{<bread,cookies : 2>}
Bread	{<milk,bread : 2>,<juice,bread : 2>,<milk,juice,bread : 2>}
Juice	{<milk,juice :3>}
Milk	

13. Construct the FP tree for the following transaction data set and explain

tid	items
1	{a,b}
2	{b,c,d}
3	{a,c,d,e}
4	{a,d,e}
5	{a,b,c}
6	{a,b,c,d}
7	{a}
8	{a,b,c}
9	{a,b,d}
10	{b,c,e}

Calculating the frequency of each item

Item	Frequency
a	8
b	7
c	6
d	5
e	3

Let the minimum support be 1.

- A Frequent Pattern set is built which will contain all the elements whose frequency is greater than or equal to the minimum support.

▪ These elements are stored in descending order of their respective frequencies. After insertion of the relevant items, the set L looks like this:-

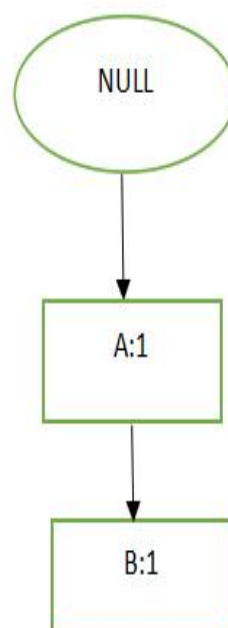
$L = \{a : 8, b : 7, c : 6, d : 5, e : 3\}$

tid	items	Ordered items
1	{a,b}	{a,b}
2	{b,c,d}	{b,c,d}
3	{a,c,d,e}	{a,c,d,e}
4	{a,d,e}	{a,d,e}
5	{a,b,c}	{a,b,c}
6	{a,b,c,d}	{a,b,c,d}
7	{a}	{a}
8	{a,b,c}	{a,b,c}
9	{a,b,d}	{a,b,d}
10	{b,c,e}	{b,c,e}

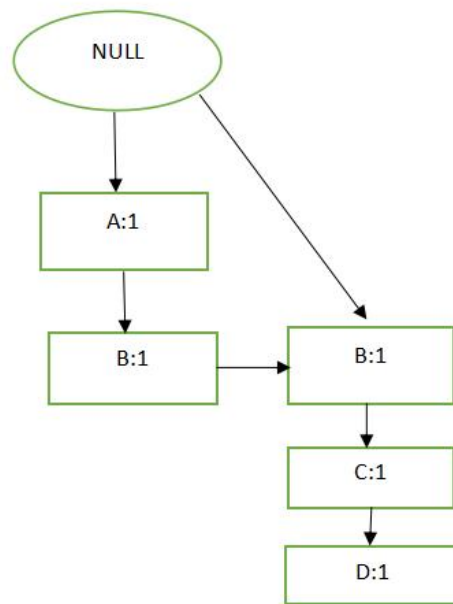
TD:1



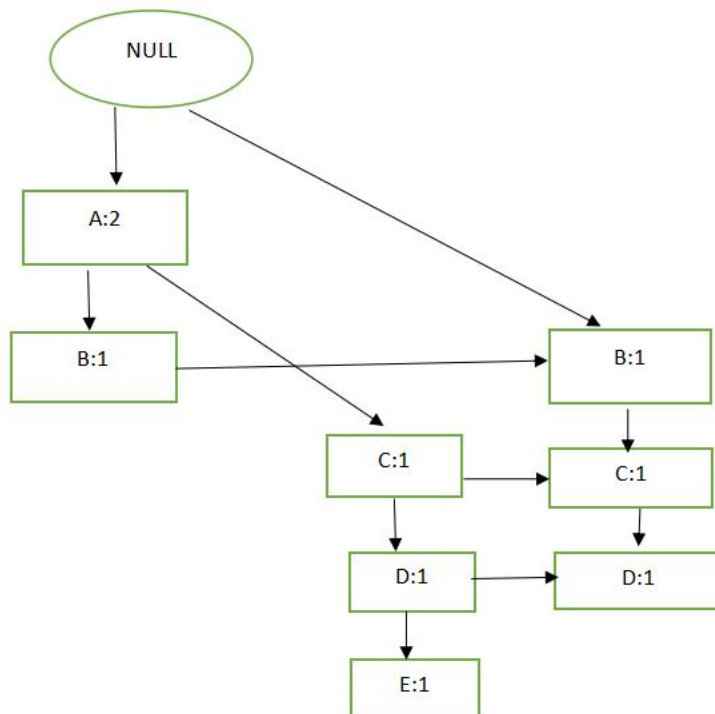
|



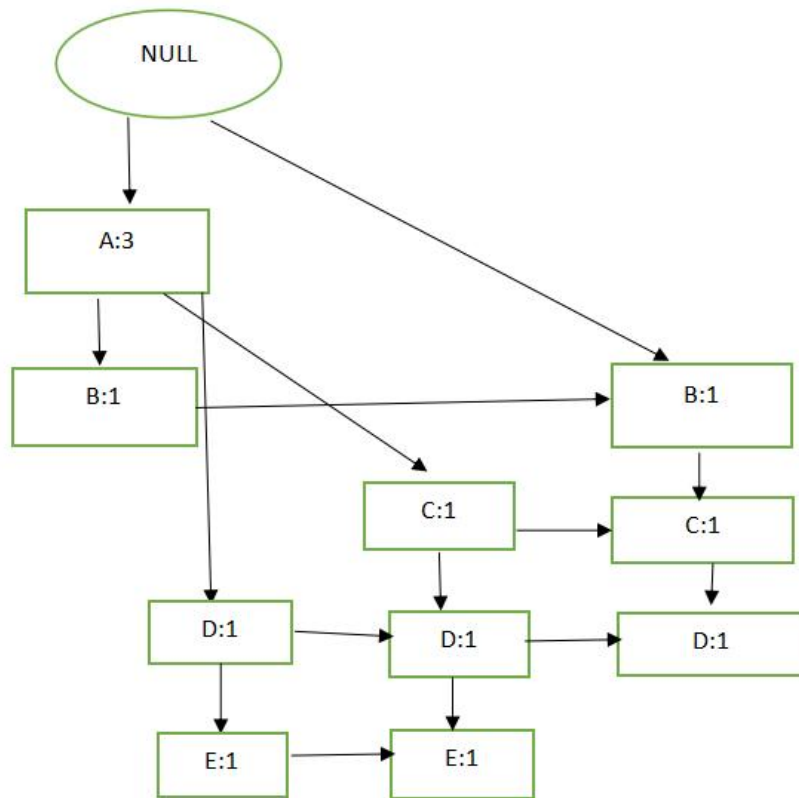
TD:2



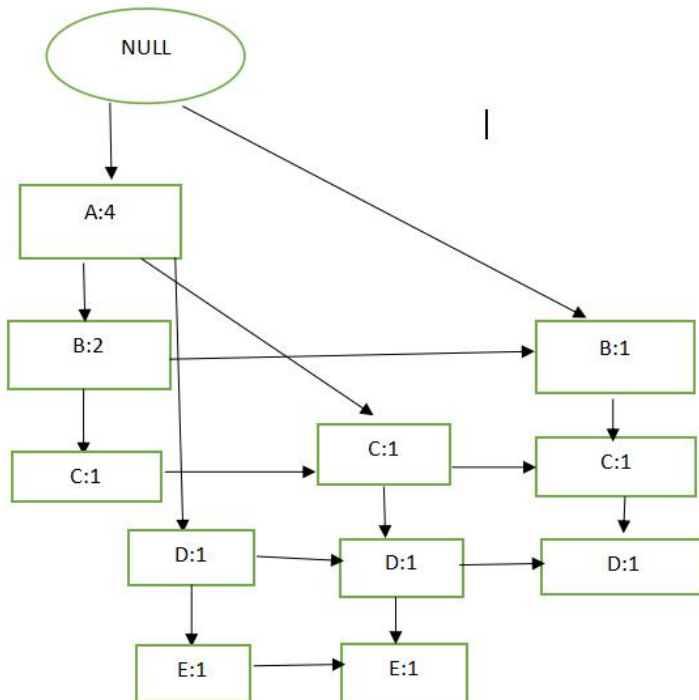
TD:3



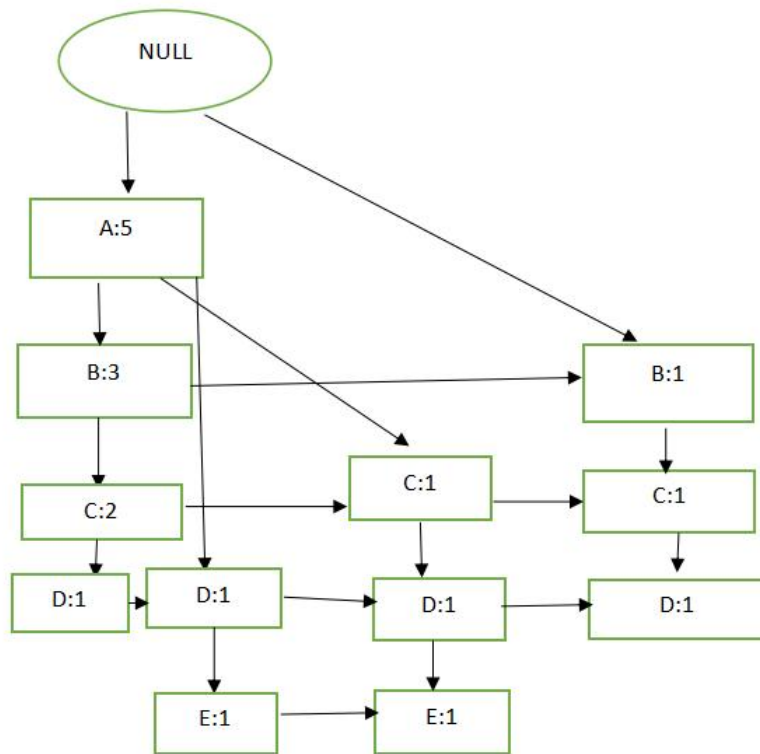
TD:4



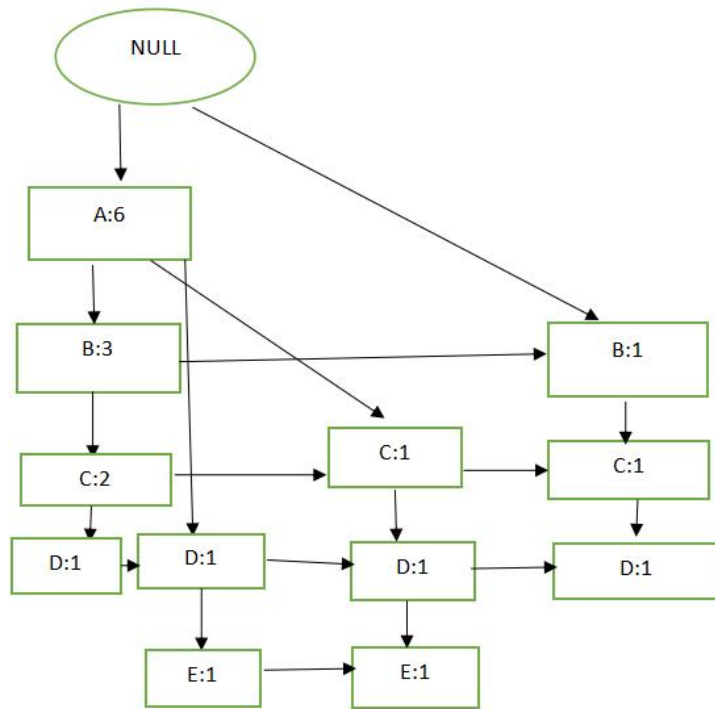
TD:5



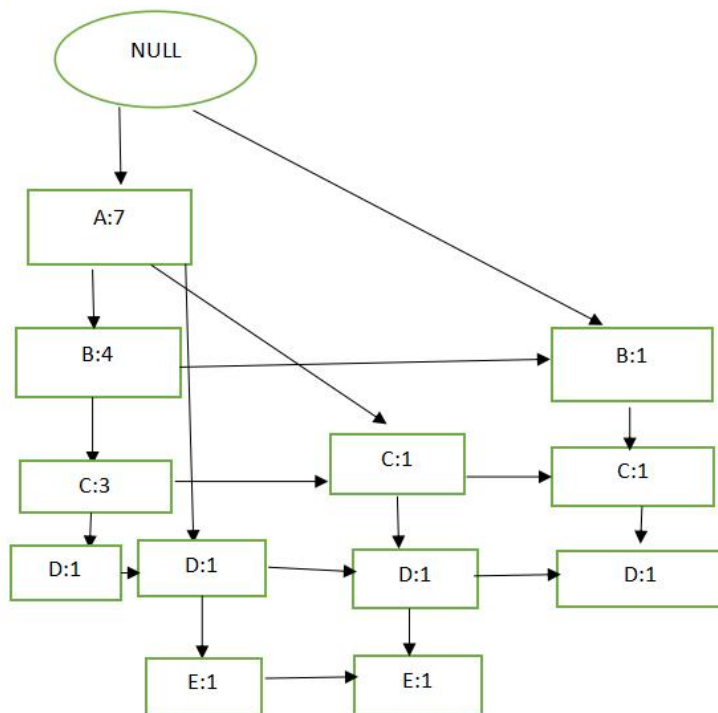
TD:6



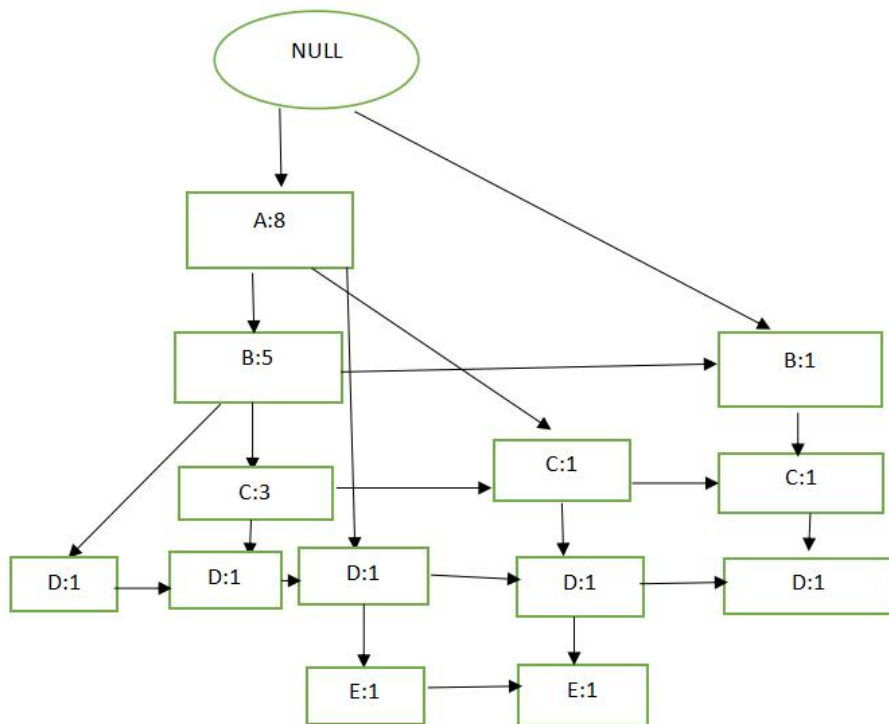
TD:7



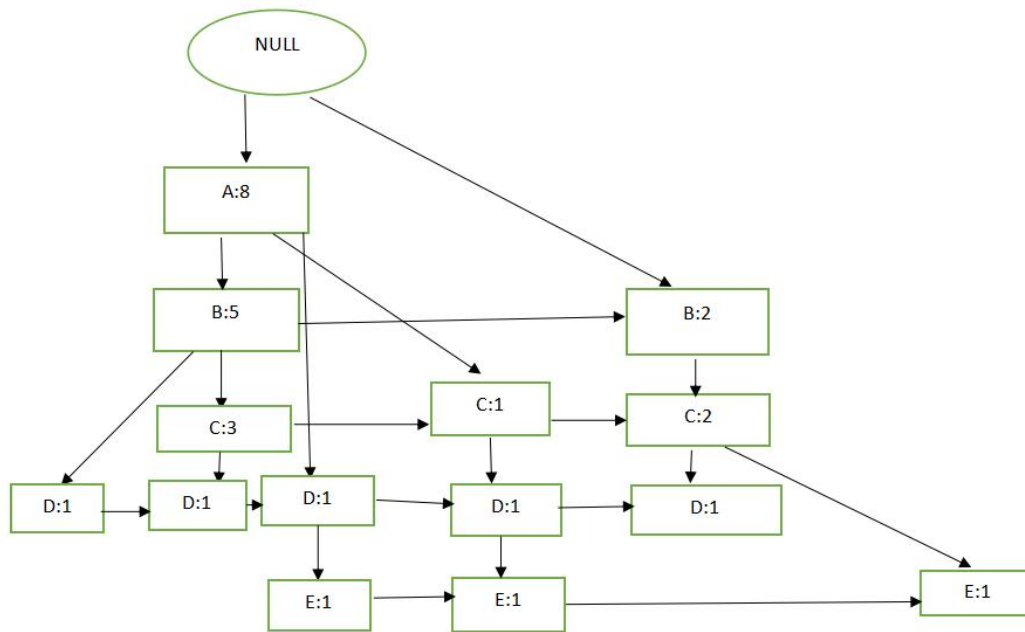
TD:8



TD:9



TD:10



14. Consider the following transaction data set.

tid	items
1	{f,c,a,m,p }
2	{f,c,a,b,m }
3	{f,b }
4	{c,p,b }
5	{f,c,a,m,p }

Construct the FP tree. Show the trees separately after reading each transaction.

(Minimum support =3)

Frequency of the each individual item

ITEM	FREQUENCY
f	4
c	4
a	3
m	3
p	3
b	3

Frequency pattern set is

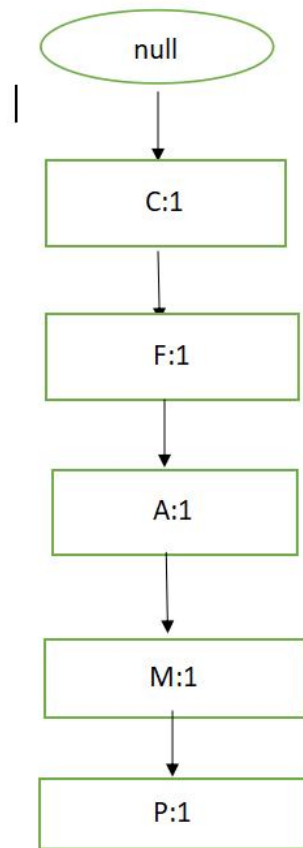
$L = \{c:4, f:4, a:3, b:3, m:3, p:3\}$

Now for each item set respective ordered item set is

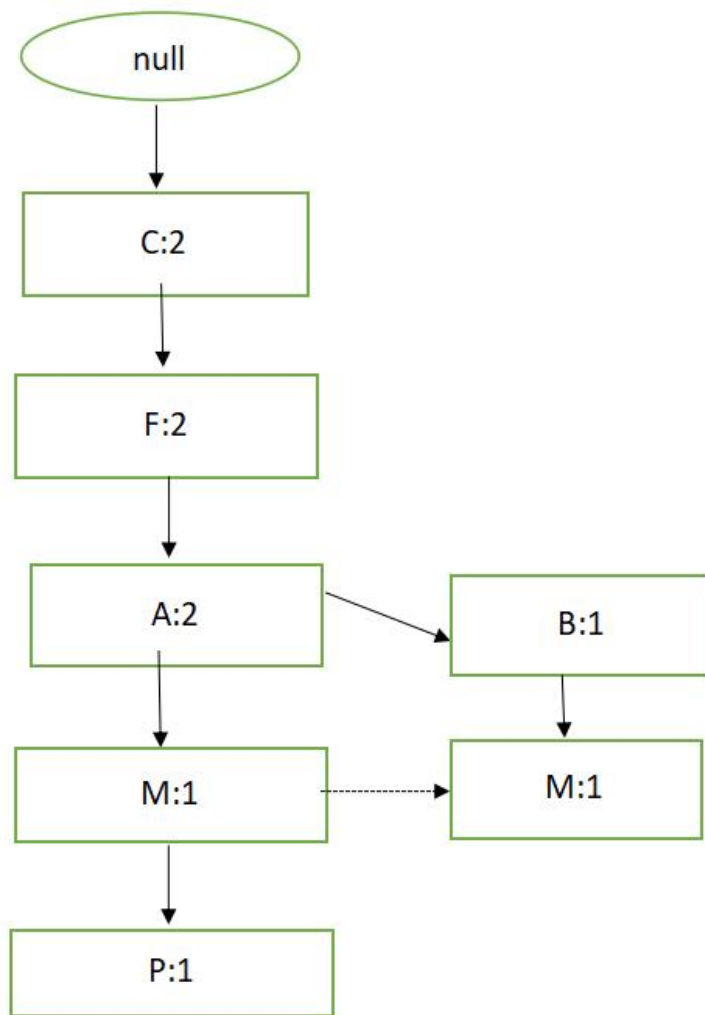
TID	ITEM SET	ORDERED ITEM SET
1	{f, c, a, m, p}	{c, f, a, m, p}
2	{f, c, a, b, m}	{c, f, a, b, m}
3	{f, b}	{f, b}
4	{c, p, b}	{c, b, p}
5	{f, c, a, m, p}	{c, f, a, m, p}

CONSTRUCTION OF FP TREE

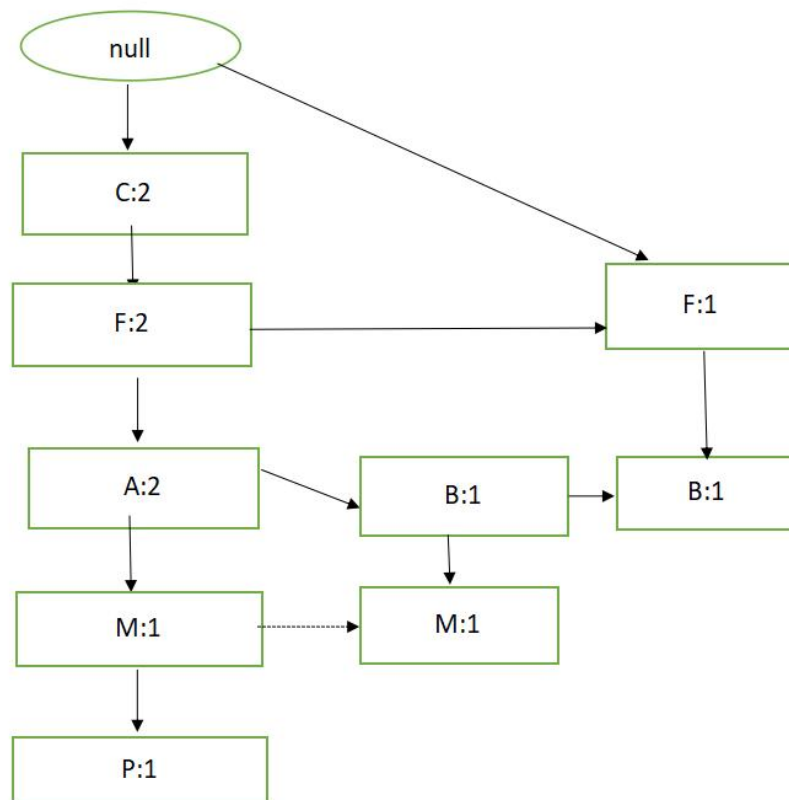
TID:1



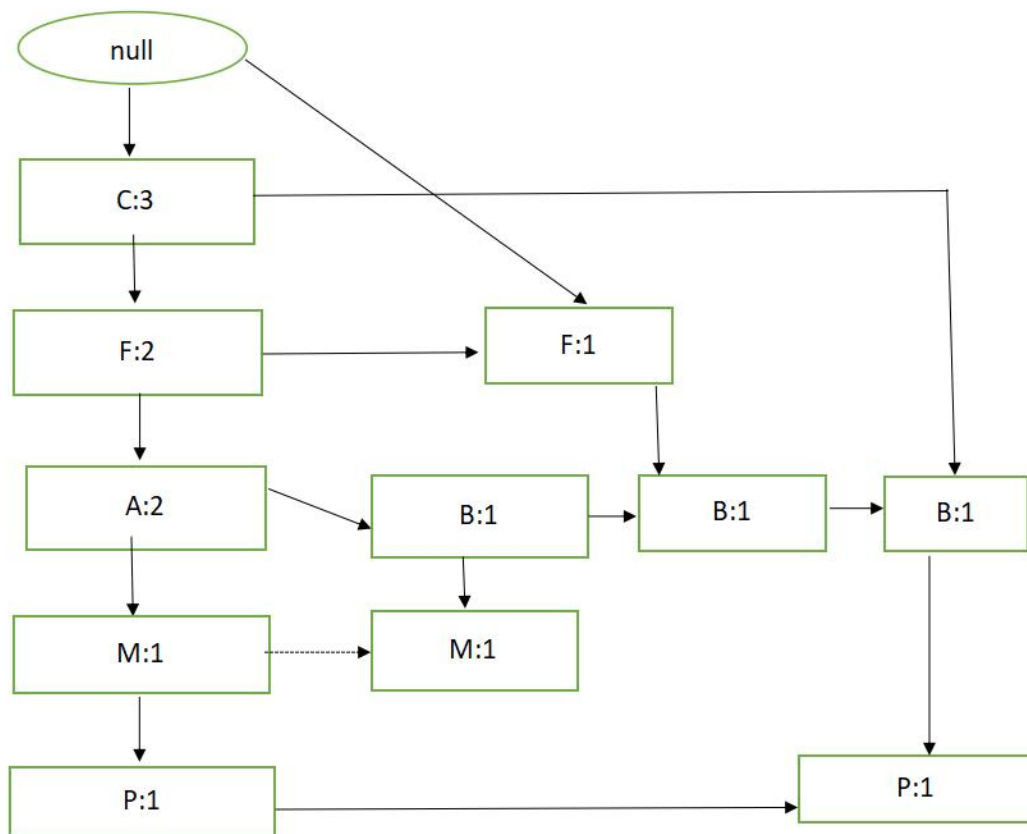
TID:2



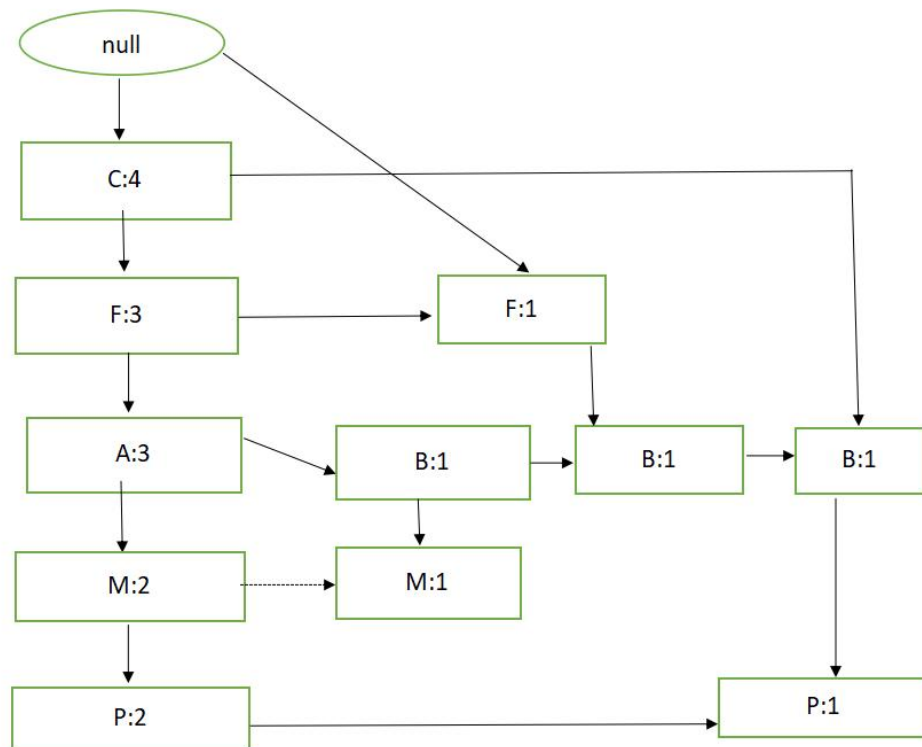
TID :3



TID:4



TID:5



15 . Consider the following data set

Customer ID	TRANSCATION ID	Item Bought
1	0001	{a,d,e }
1	0024	{a,b,c,e }
2	0012	{a,b,d,e }
2	0031	{a,c,d,e }
3	0015	{b,c,e }
3	0022	{b,d,e }
4	0029	{c,d }
4	0040	{a,b,e }
5	0033	{a,d,e }
5	0038	{a,b,e }

- I. Compute the support for item sets {e}, {b, d}, and {b, d, e} by treating each transaction ID as a market basket.

- II. Use the results in (i) to compute the confidence for the association rules $\{b, d\} \rightarrow \{e\}$ and $\{e\} \rightarrow \{b, d\}$ Is confidence a symmetric measure?
- III. Repeat (i) by treating each customer ID as a market basket. Each item should be treated as a binary variable (1 if an item appears in at least one transaction bought by the customer, and 0 otherwise).

Solution:

- support determines how often a rule is applicable to a given dataset.

$$\text{Support, } s(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{N}$$

- confidence determines how frequently items in appear in transactions that contain .

$$\text{Confidence, } c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)}$$

Where, is total number of transactions and () is the support count of (support count refers to the number of transactions that contain a particular itemset).

1.The support for itemsets:

$$S(\{e\})=9/10=0.9$$

$$S(\{b,d\})=2/10=0.2$$

$$S(\{b,d,e\})=2/10=0.2$$

1.The confidence for the association rules:

$$C(\{b,d\} \rightarrow \{e\}) = \frac{\sigma(\{b,d,e\})}{\sigma(\{b,d\})} = \frac{2}{2} = 1$$

$$C(\{e\} \rightarrow \{b,d\}) = \frac{\sigma(\{b,d,e\})}{\sigma(\{e\})} = \frac{2}{9} = 0.22$$

1. Repeat part 1 by treating each CustomerID as a market basket

Customer ID	Items Bought
1	{a, b,c,d,e}
2	{a,b,c,d,e}
3	{b,c,d,e}
4	{a,b,c,d,e}
5	{a,b,d,e}

The support for itemsets:

$$S(\{e\})=5/5=1$$

$$S(\{b,d\})=5/5=1$$

$$S(\{b,d,e\})=5/5=1$$

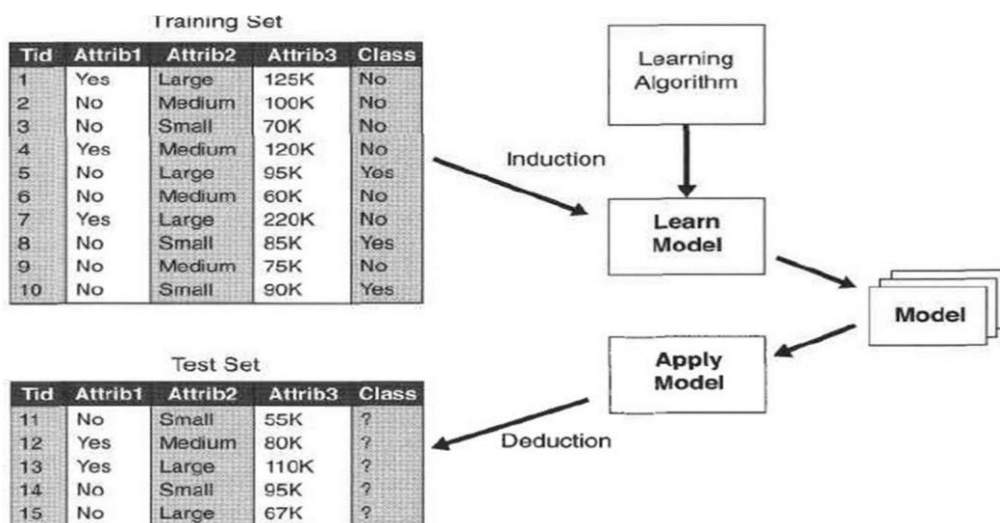
DM

UNIT – IV IMORTANT QUESTIONS & ANSWERS

1) Discuss the general approaches used for solving classification problems.

Ans:

- A classification technique (or classifier) is a systematic approach to building classification models from an input data set.
- Examples: Decision tree classifiers, Rule-based classifiers, Neural Networks, Support Vector Machines, Naive Bayes classifiers.
- Each technique employs a learning algorithm to identify a model that best fits the relationship between the attribute set and class label of the input data.
- The model generated by a learning algorithm should both fit the input data well and correctly predict the class labels of records it has never seen before.
- Key objective of the learning algorithm is to build models with good generalization capability (models that accurately predict the class labels of previously unknown records).
- First, a training set consisting of records whose class labels are known must be provided.
- The training set is used to build a classification model which is subsequently applied to the test set, which consists of records with unknown class labels.



- Evaluation of the performance of a classification model is based on the counts of test records correctly and incorrectly predicted by the model.
- Counts are tabulated in a table known as a Confusion matrix.

		Predicted Class	
		$Class = 1$	$Class = 0$
Actual Class	$Class = 1$	f_{11}	f_{10}
	$Class = 0$	f_{01}	f_{00}

- Each entry f_{ij} in this table denotes the number of records from class i predicted to be of class j .
- f_{01} is the number of records from class 0 incorrectly predicted as class 1.
- Total number of correct predictions made by the model is $(f_{11} + f_{00})$.
- Total number of incorrect predictions is $(f_{01} + f_{10})$.

2) Define and explain the concepts of root node, internal node and leaf node in decision tree algorithm with suitable example. (How to build a decision tree that is used in classification?)

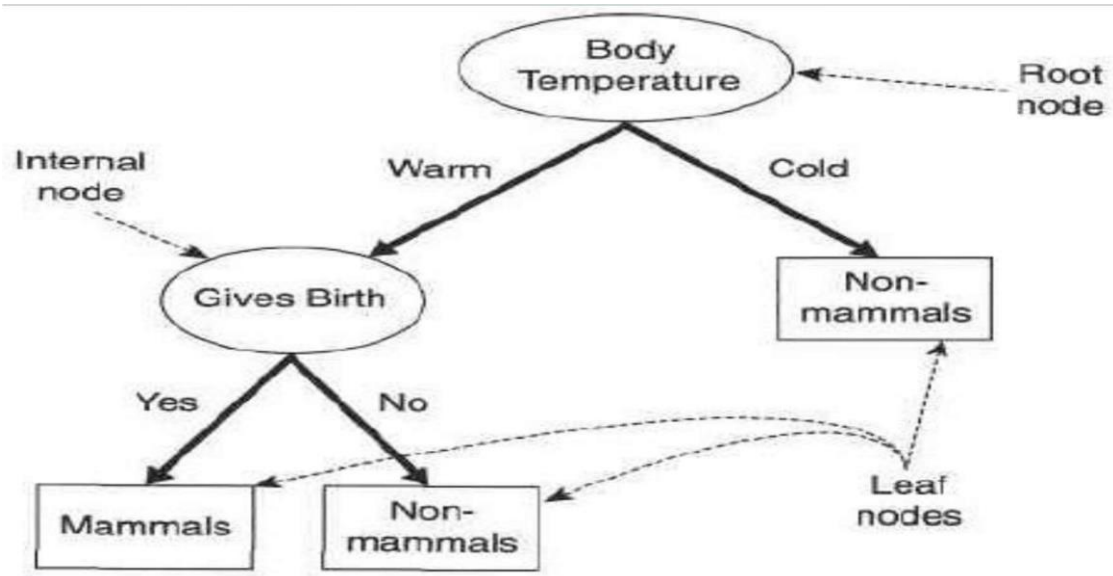
Ans:

- Solves a classification problem by asking a series of carefully crafted questions about the attributes of the test record.
- Each time we receive an answer, a follow-up question is asked until we reach a conclusion about the class label of the record.
- Series of questions and their possible answers can be organized in the form of a decision tree.
- Hierarchical structure consisting of nodes and directed edge.

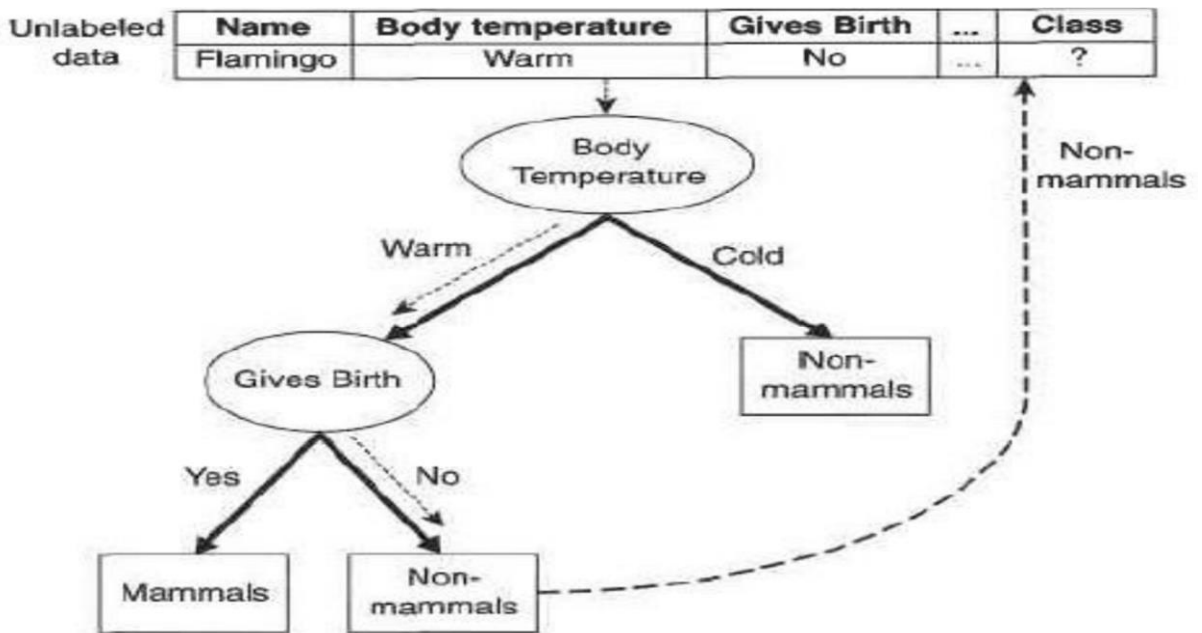
Three types of nodes:

- Root node that has no incoming edges and zero or more outgoing edge.

- Internal nodes each of which has exactly one incoming edge and two or more outgoing edge.
- Leaf or terminal nodes, each of which has exactly one incoming edge and no outgoing edge.
- Each leaf node is assigned a class label.
- Non-terminal nodes, which include the root and other internal nodes, contain attribute test conditions to separate records that have different characteristics.



- Classifying a test record is straightforward once a decision tree has been constructed.
- Starting from the root node, apply the test condition to the record and follow the appropriate branch based on the outcome of the test.
- This will lead either to another internal node, for which a new test condition is applied, or to a leaf node. The class label associated with the leaf node is then assigned to the record.



How to Build a Decision Tree:

- Finding the optimal tree is computationally infeasible because of the exponential size of the search space.
- Efficient algorithms have been developed to induce a reasonably accurate decision tree in a reasonable amount of time.

3) Define classification. Write and explain the Hunt's algorithm for classification by taking the example of loan repayment prediction. (Explain Hunt's algorithm for inducing decision trees.)

Ans:

A classification technique (or classifier) is a systematic approach to building classification models from an input data set

Examples: Decision tree classifiers, Rule-based classifiers, Neural Networks, Support Vector Machines, Naive Bayes classifiers.

HUNTS ALGORITHM:

A decision tree is grown in a recursive fashion by partitioning the training records into successively purer subsets

Let D_t be the set of training records that are associated with node t

- $y = \{y_1, y_2, \dots, y_c\}$ be the class labels

RECURSIVE DEFINATION OF HUNTS ALGORITHM:

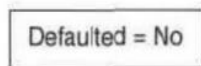
Step 1: If all the records in D_t belong to the same class y_t , then t is a leaf node labelled as y_t

- Step 2: If D_t contains records that belong to more than one class, an attribute test condition is selected to partition the records into smaller subsets
- A child node is created for each outcome of the test condition and the records in D_t are distributed to the children based on the outcomes

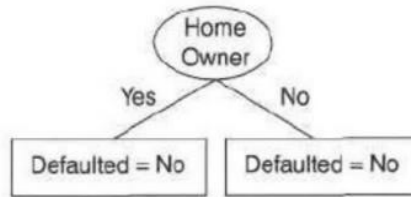
The algorithm is then recursively applied to each child node

EXAMPLE:

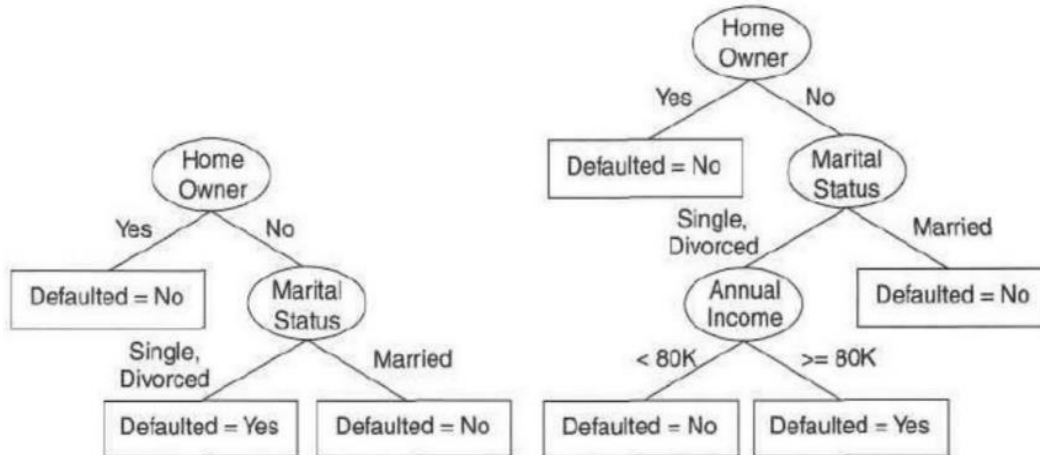
	binary	categorical	continuous	class
Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



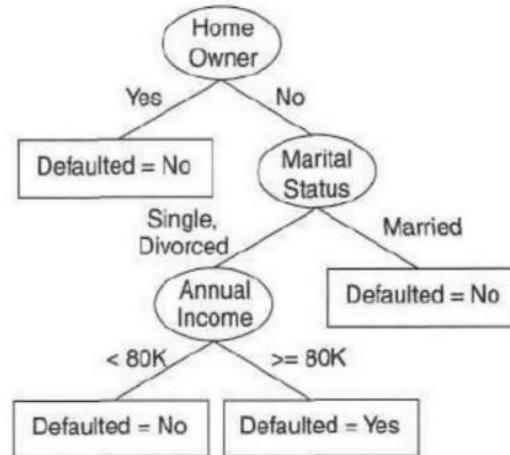
(a)



(b)



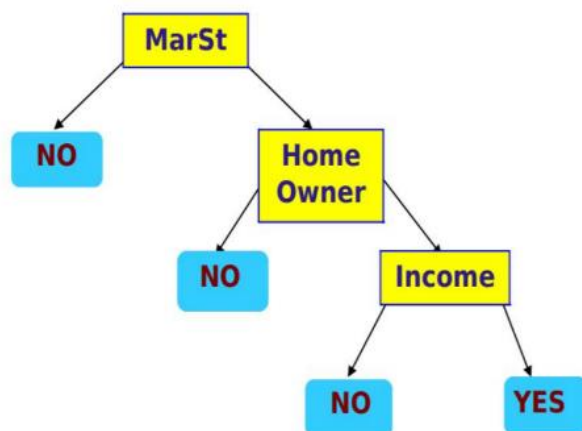
(c)



(d)

categorical
categorical
continuous
class

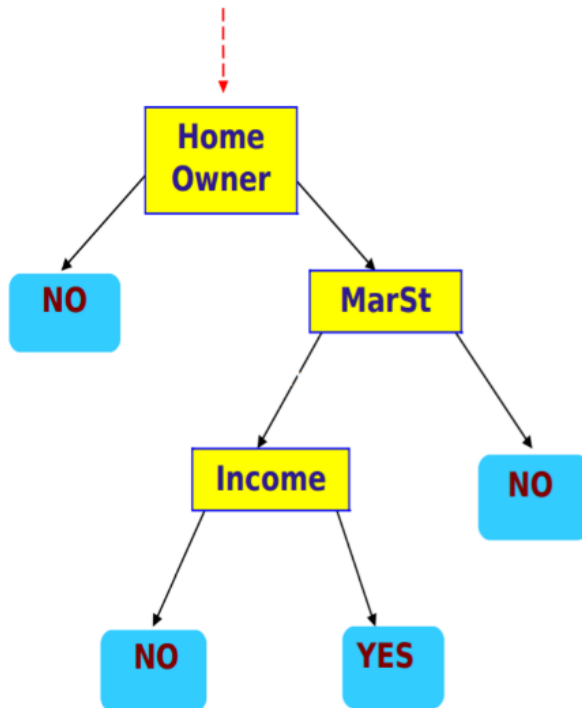
ID	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes



There could be more than one tree that fits the same data!

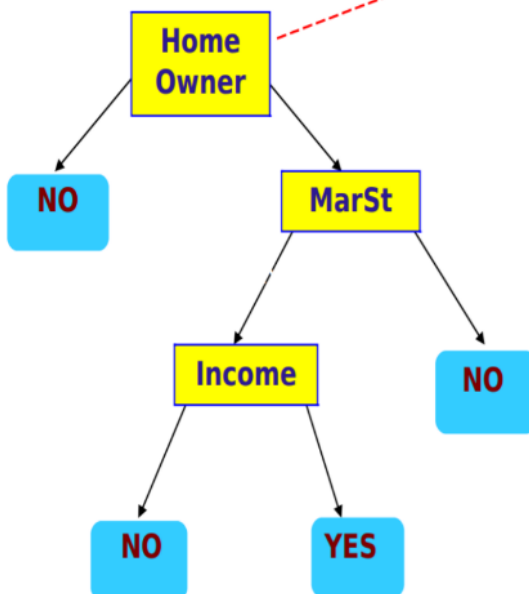
Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



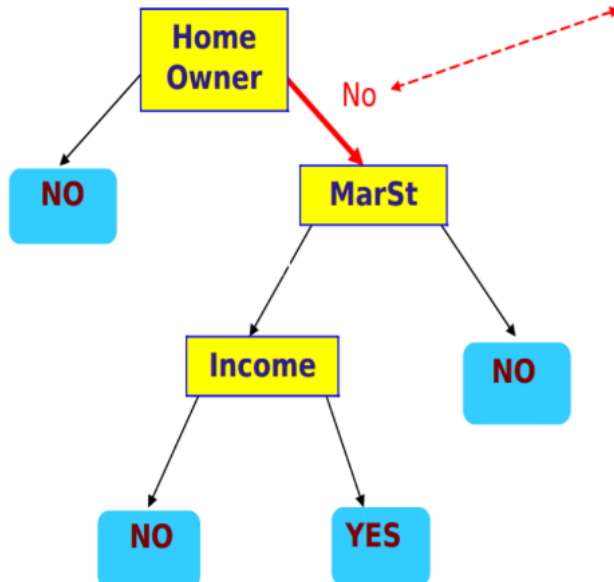
Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



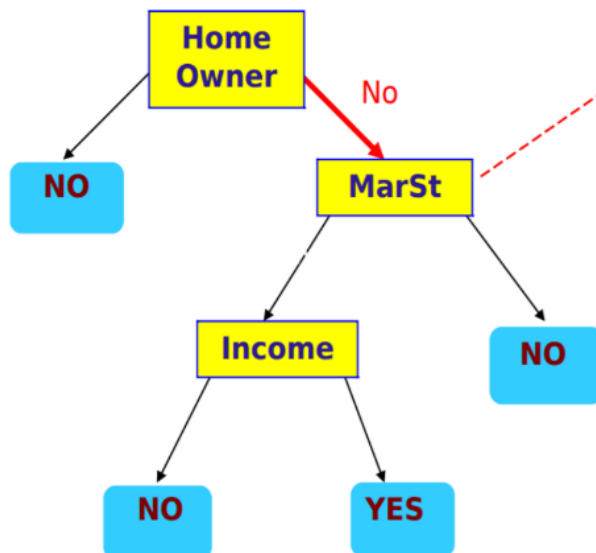
Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



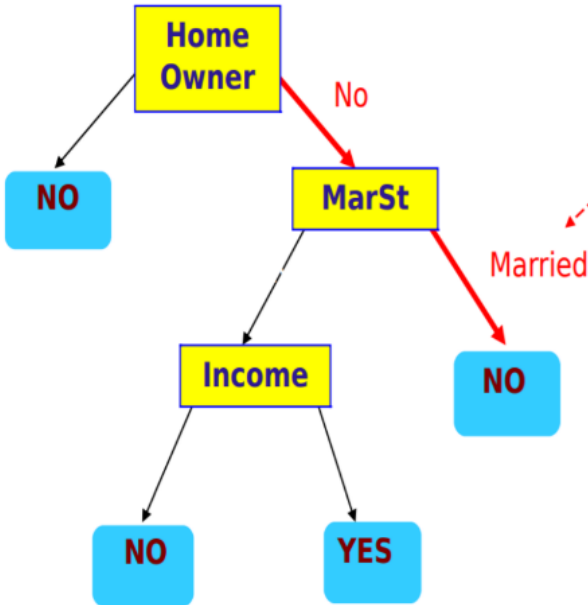
Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



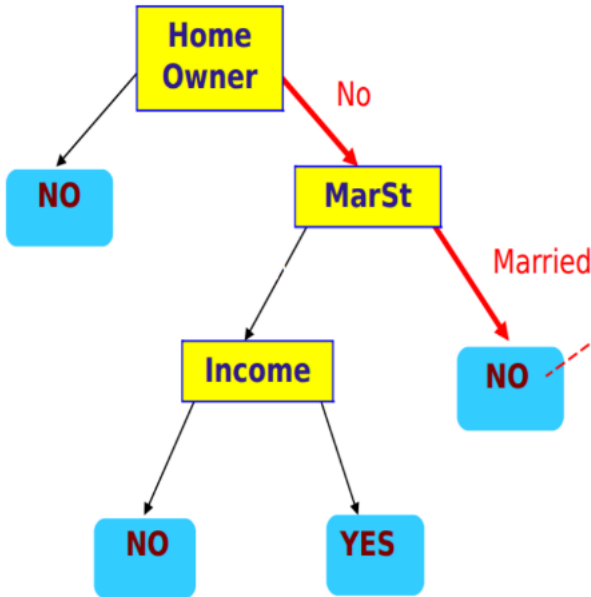
Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



Test Data

Home Owner	Marital Status	Annual Income	Defaulted Borrower
No	Married	80K	?



- 4) Explain nearest neighbor classifiers. Explain the K-Nearest Neighbor classification algorithm with an example. List the characteristics of nearest neighbor classifiers.

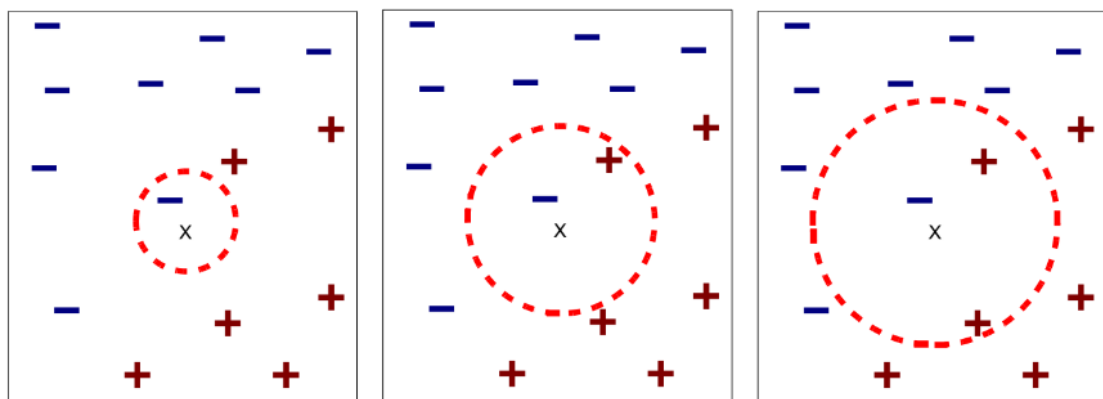
Ans:

A nearest neighbour classifier represents each example as a data point in a d-dimensional space, where d is the number of attributes.

Given a test example, we compute its proximity to the rest of the data points in the training set, using one of the proximity measures.

K-nearest neighbours of a given example z refer to the k points that are closest to z .

Data point is classified based on the class labels of its neighbors. In the case where the neighbors have more than one label, the data point is assigned to the majority class of its nearest neighbors.



(a) 1-nearest neighbor

(b) 2-nearest neighbor

(c) 3-nearest neighbor

K-nearest neighbors of a record x are data points that have the k smallest distances to x

If the number of nearest neighbors is one, data point is a negative example.

Therefore, the data point is assigned to the negative class

If the number of nearest neighbors is three, then the neighbourhood contains two positive examples and one negative example

Using the majority voting scheme, the data point is assigned to the positive class

In the case where there is a tie between the classes, we may randomly choose one of them to classify the data point. Choose the right value for k .

ALGORITHM:

Algorithm computes the distance (or similarity) between each test example and all the training examples (x, y) belongs D to determine its nearest-neighbour list, D_z

Such computation can be costly if the number of training examples is large

```
1: Let  $k$  be the number of nearest neighbors and  $D$  be the set of training examples.
2: for each test example  $z = (x', y')$  do
3:   Compute  $d(x', x)$ , the distance between  $z$  and every example,  $(x, y) \in D$ .
4:   Select  $D_z \subseteq D$ , the set of  $k$  closest training examples to  $z$ .
5:    $y' = \operatorname{argmax}_v \sum_{(x_i, y_i) \in D_z} I(v = y_i)$ 
6: end for
```

Once the nearest-neighbor list is obtained, the test example is classified based on the majority class of its nearest neighbors

$$\text{Majority Voting: } y' = \operatorname{argmax}_v \sum_{(x_i, y_i) \in D_z} I(v = y_i)$$

Where:

v is a class label

y_i is the class label for one of the nearest neighbors

I is an indicator function that returns the value 1 if its argument is true and 0 otherwise

CHARACTERISTICS OF NEAR NEIGHBOR CLASSIFIER:

- NN classification is part of instance-based learning, which uses specific training instances to make predictions without having to maintain model derived from data
- Do not require model building
- Predictions are based on local information
- Can produce wrong predictions unless the appropriate proximity measure and data pre-processing steps are taken
- Can produce arbitrary shaped decision boundaries

- 5) Write a note on
- a) Training set
 - b) Test set
 - c) Confusion matrix
 - d) Performance matrix.

Ans:

a) Training Set: -

A training set is a portion of a data set used to fit (train) a model for prediction or classification of values that are known in the training set, but unknown in other (future) data. The training set is used in conjunction with validation and/or test sets that are used to evaluate different models. Training sets are used in supervised learning procedures in data mining (i.e., classification of records, or prediction of target values that are continuous.)

b) Test Set:

A test set is a portion of a data set used in data mining to assess the likely future performance of a single prediction or classification model that has been selected from among competing models, based on its performance with the validation set. While the validation set provides a useful guide to selecting the best-performing model, that best-performing model may still do better with the validation data than it will with unseen future data. This is particularly the case when many different models are assessed - in such cases there is a greater chance that the "winning" model emerges on top because it just happens (by chance) to fit some particularities of the validation set quite well.

c) Confusion Matrix:

The confusion matrix is a matrix used to determine the performance of the classification models for a given set of test data. It can only be determined if the true values for test data are known. The matrix itself can be easily understood, but the related terminologies may be confusing. Since it shows the errors in the model performance in the form of a matrix, hence also known as an **error matrix**.

d) Performance Matrix:

Performance metric measures how well your data mining algorithm is performing on a given dataset. Performance matrix determines accuracy and error rate of a prediction.

- 1) Accuracy: Accuracy is the ratio between the number of correct predictions to the total number of predictions.

Accuracy = No. of correct predictions / Total No. of predictions

$$= (f_{11}+f_{00}) / (f_{11}+f_{10}+f_{01}+f_{00})$$

- 2) Error rate: Error rate is the ratio between the number of wrong predictions to the total number of predictions.

Error rate = No. of wrong predictions / Total No. of predictions

$$= (f_{01}+f_{10})/(f_{11}+f_{10}+f_{01}+f_{00})$$

6) Write a note on a) Accuracy b) Error Rate

(What are the different techniques used in evaluating the performance of classifiers.)

Ans:

Performance metric measures how well your data mining algorithm is performing on a given dataset. Performance matrix determines accuracy and error rate of a prediction.

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Techniques used in evaluating performance of classifiers:

- 1) Confusion matrix

The confusion matrix is a matrix used to determine the performance of the classification models for a given set of test data. It can only be

determined if the true values for test data are known. The matrix itself can be easily understood, but the related terminologies may be confusing. Since it shows the errors in the model performance in the form of a matrix, hence also known as an **error matrix**.

- 2) Accuracy: Accuracy is the ratio between the number of correct predictions to the total number of predictions.

$$\text{Accuracy} = \text{No. of correct predictions} / \text{Total No. of predictions} \\ = (f_{11} + f_{00}) / (f_{11} + f_{10} + f_{01} + f_{00})$$

- 3) Error rate: Error rate is the ratio between the number of wrong predictions to the total number of predictions.

$$\text{Error rate} = \text{No. of wrong predictions} / \text{Total No. of predictions} \\ = (f_{01} + f_{10}) / (f_{11} + f_{10} + f_{01} + f_{00})$$

- 7) Define rule-based classifier with vertebrate classification example
(Explain how rule-based classifier works with an example)

Ans:

Rule based classifier classifies a test record based on the rule triggered by the record

R1: (Give Birth = no) $\wedge$ (Can Fly = yes) $\rightarrow$ Birds

R2: (Give Birth = no) $\wedge$ (Live in Water = yes) $\rightarrow$ Fishes

R3: (Give Birth = yes) $\wedge$ (Blood Type = warm) $\rightarrow$ Mammals

R4: (Give Birth = no) $\wedge$ (Can Fly = no) $\rightarrow$ Reptiles

R5: (Live in Water = sometimes) $\rightarrow$ Amphibians

Name	Blood Type	Give Birth	Can Fly	Live in Water	Class
lemur	warm	yes	no	no	?
turtle	cold	no	no	sometimes	?
dogfish shark	cold	yes	no	yes	?

A lemur triggers rule R3, so it is classified as a mammal A turtle triggers both R4 and R5 A dogfish shark triggers none of the rules

- 8) Compare

- i) Mutual exclusive rule with exhaustive rule
- ii) Ordered rule with Unordered rule

Ans:

i) Mutually Exclusive Rules

- Rules in a rule set R are mutually exclusive if no two rules in R are triggered by the same record
- Property ensures that every record is covered by at most one rule in R

Exhaustive Rules

- A rule set -R has exhaustive coverage if there is a rule for each combination of attribute values
- This property ensures that every record is covered by at least one rule in R

ii) Ordered Rules

- Rules in a rule set are ordered in decreasing order of their priority (based on accuracy, coverage etc).
- An ordered rule set is also known as a decision list
- When a test record is presented, it is classified by the highest ranked rule that covers the record.
- This avoids the problem of having conflicting classes predicted by multiple classification rules.

Unordered Rules

- Approach allows a test record to trigger multiple classification rules
- It then considers the consequent of each rule as a vote for a particular class
- Votes are then tallied to determine the class label of the test record
- The record is usually assigned to the class that receives the highest number of votes

9) Explain various method used to evaluate the performance of classifier.

Ans:

Evaluating the performance of Classifiers

Holdout Methods:

- The original data with labelled examples is partitioned into two disjoint sets- training and test sets
- Classification model is induced from training set
- Its performance is evaluated on the test set
- The accuracy of the classifier can be estimated based on the accuracy of the induced model on the test set

Limitations:

- Fewer labelled examples are available for training
- Model may be highly dependent on the composition of training and test sets
- Training and test set are not independent of each other.

Random Subsampling:

- Holdout method can be repeated several times to improve the estimation of classifier's performance
- This approach is known as random subsampling

Limitations:

- Suffers from the problems associated with holdout method
- It also has no control over the number of times each record is used for testing and training

Cross validation:

- Each record is used the same number of times for training and exactly once for testing
- K fold cross validation segments the data into k equal size partitions
- During each run, one of the partitions is chosen for testing, rest of them are used for training
- Procedure is repeated k times so that each partition is used for testing exactly once
- Total error is found by summing up the errors for all k runs
- Leave one out approach: special case where $k=N$, the size of the data
- Each test set contains only one record

- **Advantages:**

- Utilizes as much as data as possible for training
- Test sets are mutually exclusive and effectively cover the entire data set

- **Drawback:** computationally expensive to repeat the procedure N times

Since each test set contains only one record, the variance of the estimated performance metric tends to be high.

10) a) Write and explain the algorithm for decision tree induction. Illustrate with an example.

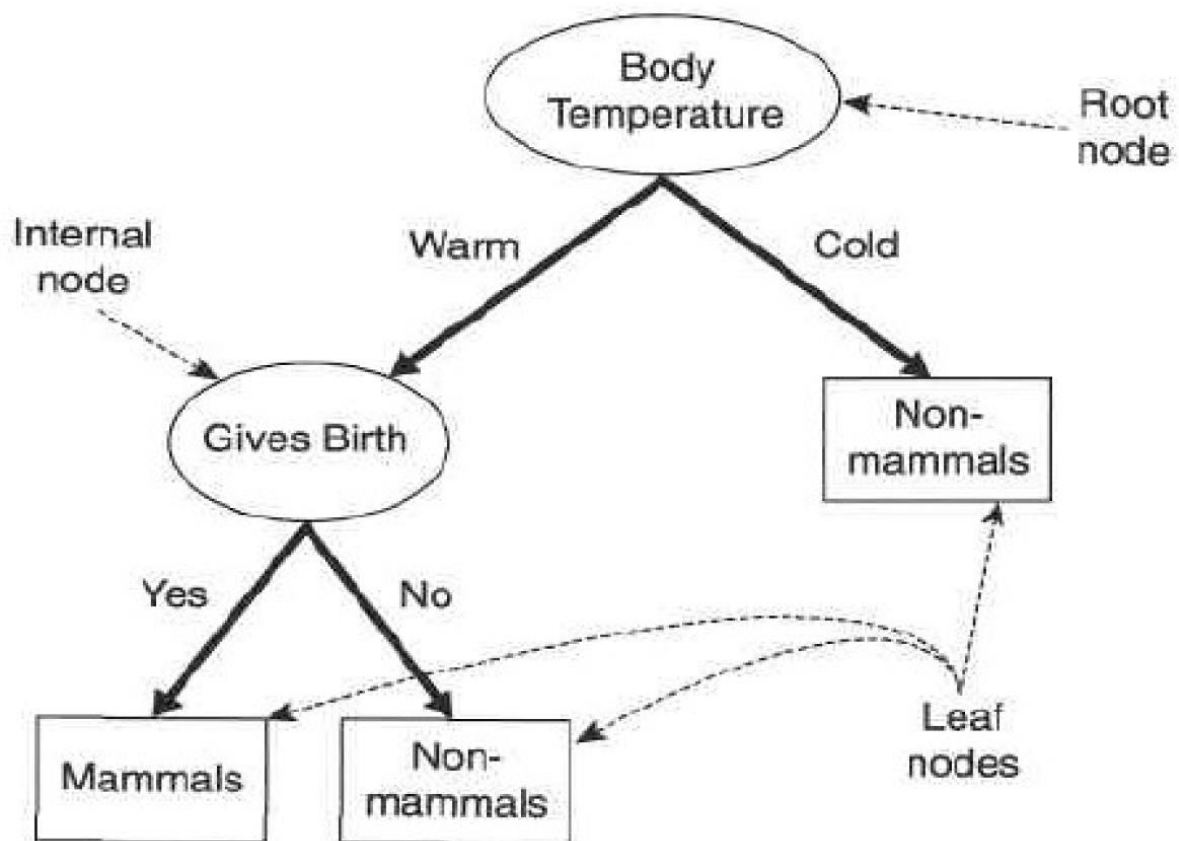
Ans:

How a Decision Tree Works:

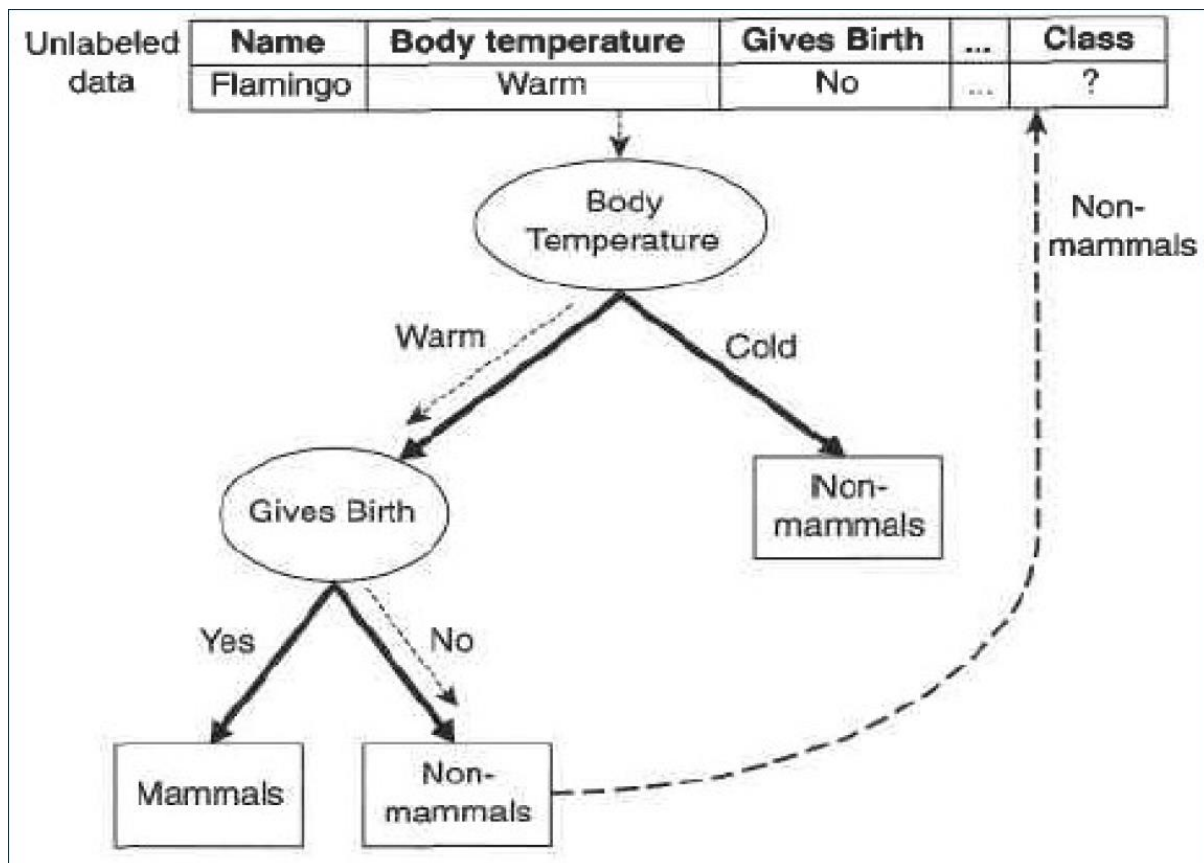
- Solves a classification problem by asking a series of carefully crafted questions about the attributes of the test record
- Each time we receive an answer, a follow-up question is asked until we reach a conclusion about the class label of the record
- Series of questions and their possible answers can be organized in the form of a decision tree
- Hierarchical structure consisting of nodes and directed edge

Three types of nodes:

- Root node that has no incoming edges and zero or more outgoing edge
- Internal nodes each of which has exactly one incoming edge and two or more outgoing edge
- Leaf or terminal nodes, each of which has exactly one incoming edge and no outgoing edge
- Each leaf node is assigned a *class label*
- Non-terminal nodes, which include the root and other internal nodes, contain *attribute test conditions* to separate records that have different characteristics



- Classifying a test record is straightforward once a decision tree has been constructed
- Starting from the root node, apply the test condition to the record and follow the appropriate branch based on the outcome of the test
- This will lead either to another internal node, for which a new test condition is applied, or to a leaf node
- The class label associated with the leaf node is then assigned to the record



How to Build a Decision Tree:

- Finding the optimal tree is computationally infeasible because of the exponential size of the search space
- Efficient algorithms have been developed to induce a reasonably accurate decision tree in a reasonable amount of time

11) Define classification. Explain how classification is done by decision tree induction. What are the purpose of classification model?

Ans:

Classification is a Task of assigning objects to one of several predefined categories. Input data for a classification task is a collection of records.

Eg:

Categorizing email messages (labeling it as spam or non-spam)

Cataloging galaxies (labeling it as Elliptical, spiral, or irregular-shaped galaxies)

Classification model is useful for the following purposes:

- **Descriptive Modeling**

It can serve as an explanatory tool to distinguish between objects of different classes

- **Predictive Modeling**

It can be used to predict the class label of unknown records

It can be treated as a black box that automatically assigns a class label when presented with the attribute set of an unknown record.

How a Decision Tree Works:

- Solves a classification problem by asking a series of carefully crafted questions about the attributes of the test record.
- Each time we receive an answer, a follow-up question is asked until we reach a conclusion about the class label of the record.
- Series of questions and their possible answers can be organized in the form of a decision tree.
- Hierarchical structure consisting of nodes and directed edge

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1. Root node that has no incoming edges and zero or more outgoing edge
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- Non-terminal nodes, which include the root and other internal nodes, contain attribute test conditions to separate records that have different characteristics.
- Classifying a test record is straightforward once a decision tree has been constructed.
- Starting from the root node, apply the test condition to the record, and follow the appropriate branch based on the outcome of the test
- This will lead either to another internal node, for which a new test condition is applied, or to a leaf node.
- The class label associated with the leaf node is then assigned to the record.

12) What is a Rule-Based classifier? Write and explain the sequential covering algorithm with example.
 (Explain the algorithm used to extract rules directly from data.)
 (How does Rule-based classifier work?)

Ans:

Rule-based classifier makes use of a set of IF-THEN rules for classification.

We can express a rule in the following form –

“IF condition THEN conclusion”

Eg:

r1: (Gives Birth = no) A (Aerial Creature = yes) → Birds

r2: (Gives Birth = no) A (Aquatic Creature = yes) → Fishes

r3: (Gives Birth = yes) A (Body Temperature = warm-blooded)
 → Mammals

r4: (Gives Birth = no) A (Aerial Creature = no) → Reptiles

r5: (Aquatic Creature = semi) → Amphibians

- The rules for the model are represented in a disjunctive normal form.
- R is known as the rule set.
- r_i 's are the classification rules or disjuncts.

Sequential covering algorithm is often used to extract rules directly from data:

- Algorithm extracts the rules one class at a time for data sets that contain more than two classes
- Criterion for deciding which class should be generated first depends on a number of factors, such as the class prevalence or the cost of misclassifying records from a given class.

Sequential covering algorithm

- Algorithm starts with an empty decision list R
- Learn-One-Rule function is then used to extract the best rule for class y that covers the current set of training records
- During rule extraction, all training records for class y are considered to be positive examples, while those that belong to other classes are considered to be negative examples
- A rule is desirable if it covers most of the positive examples and none (or very few) of the negative examples
- Once such a rule is found, the training records covered by the rule are eliminated
- The new rule is added to the bottom of the decision list R
- This procedure is repeated until the stopping criterion is met
- The algorithm then proceeds to generate rules for the next class
- Start from an empty rule
- Grow a rule using the Learn-One-Rule function
- Remove training records covered by the rule
- Repeat Step (2) and (3) until stopping criterion is met

Algorithm Sequential covering algorithm.

- 1: Let E be the training records and A be the set of attribute-value pairs, $\{(A, v)\}$
- 2: Let Y_0 be an ordered set of classes $\{y_1, y_2, \dots, y_k\}$.
- 3: Let $R = \{\}$ be the initial rule list.
- 4: for each class y belongs to $Y_0 - \{y_k\}$ do
- 5: while stopping condition is not met do
- 6: $r \leftarrow \text{Learn-one-rule}(E, A, y)$.
- 7: Remove training records from E that are covered by r.
- 8: Add r to the bottom of the rule list: $R \rightarrow R \vee r$

9: end while

10: end for

11: Insert the default rule, $\{\} \rightarrow y_k$, to the bottom of the rule list R.

- 13) a) Explain bayes theorem. How exactly Bayesian classifier works on below training set for predicting the loan default problem, explain.(Illustrate how it can be used to solve prediction problem.)

TID	Home owner	Marital status	Annual income	Defaulted borrower
1	Yes	Single	125k	No
2	No	Married	100k	No
3	No	Single	70k	No
4	Yes	Married	120k	No
5	No	Divorced	95k	Yes
6	No	Married	60k	No
7	Yes	Divorced	220k	No
8	No	Single	85k	Yes
9	No	Married	75k	No
10	No	Single	90k	Yes

Ans:

an approach for modelling probabilistic relationships between the attribute set and the class variable

Let X and Y be a pair of random variables

Joint probability, $P(X=x, Y=y)$

Probability that variable X will take on the value x and variable Y will take on the value y

Conditional probability, $P(Y=y | X=x)$

Probability that a random variable will take on a particular value given that the outcome for another random variable is known

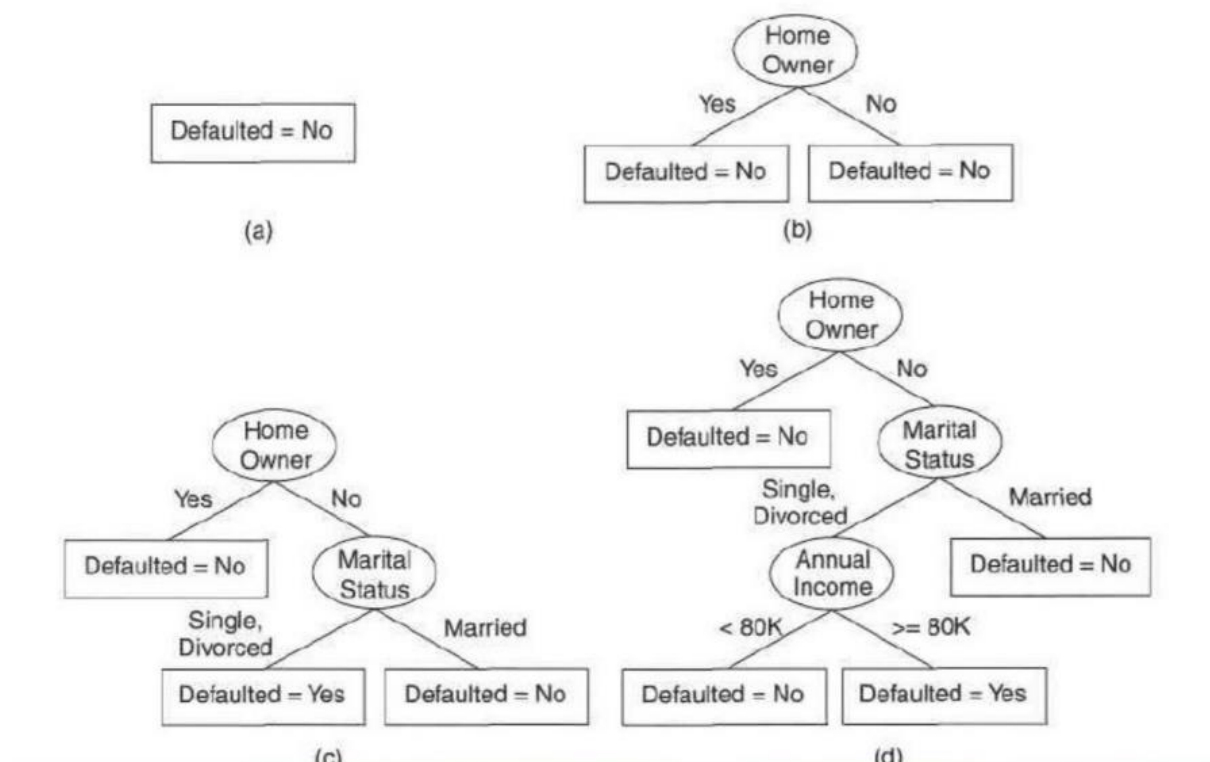
Probability that the variable Y will take on the value y, given that the variable X is observed to have the value x

The joint and conditional probabilities for X and Y are related in the following way:

$$P(X,Y) = P(Y \mid X) \times P(X) = P(X \mid Y) \times P(Y)$$

Rearranging the last two expression, we get a formula, known as the Bayes theorem:

$$P(Y \mid X) = \frac{P(X \mid Y) \times P(Y)}{P(X)}$$



14) Consider the training examples shown below:

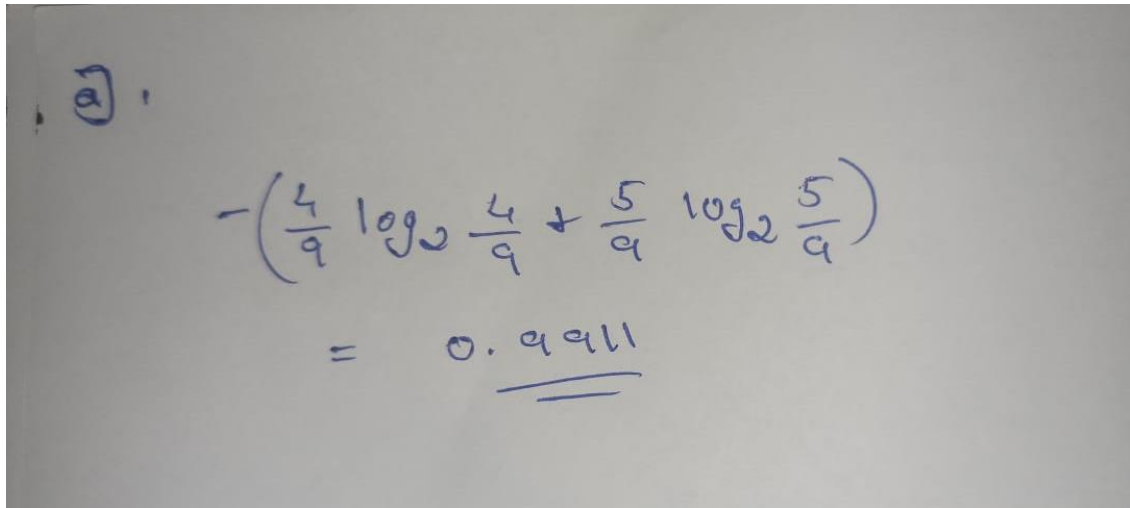
Instance	a_1	a_2	a_3	Target Class
1	T	T	1.0	+
2	T	T	6.0	+
3	T	F	5.0	—
4	F	F	4.0	+
5	F	T	7.0	—
6	F	T	3.0	—
7	F	F	8.0	—
8	T	F	7.0	+
9	F	T	5.0	—

a) Calculate the entropy of this collection of training examples.

- b) Calculate the information gains of a_1 and a_2
 c) For a_3 , which is a continuous attribute, compute the information gain for every possible split

Ans:

a)

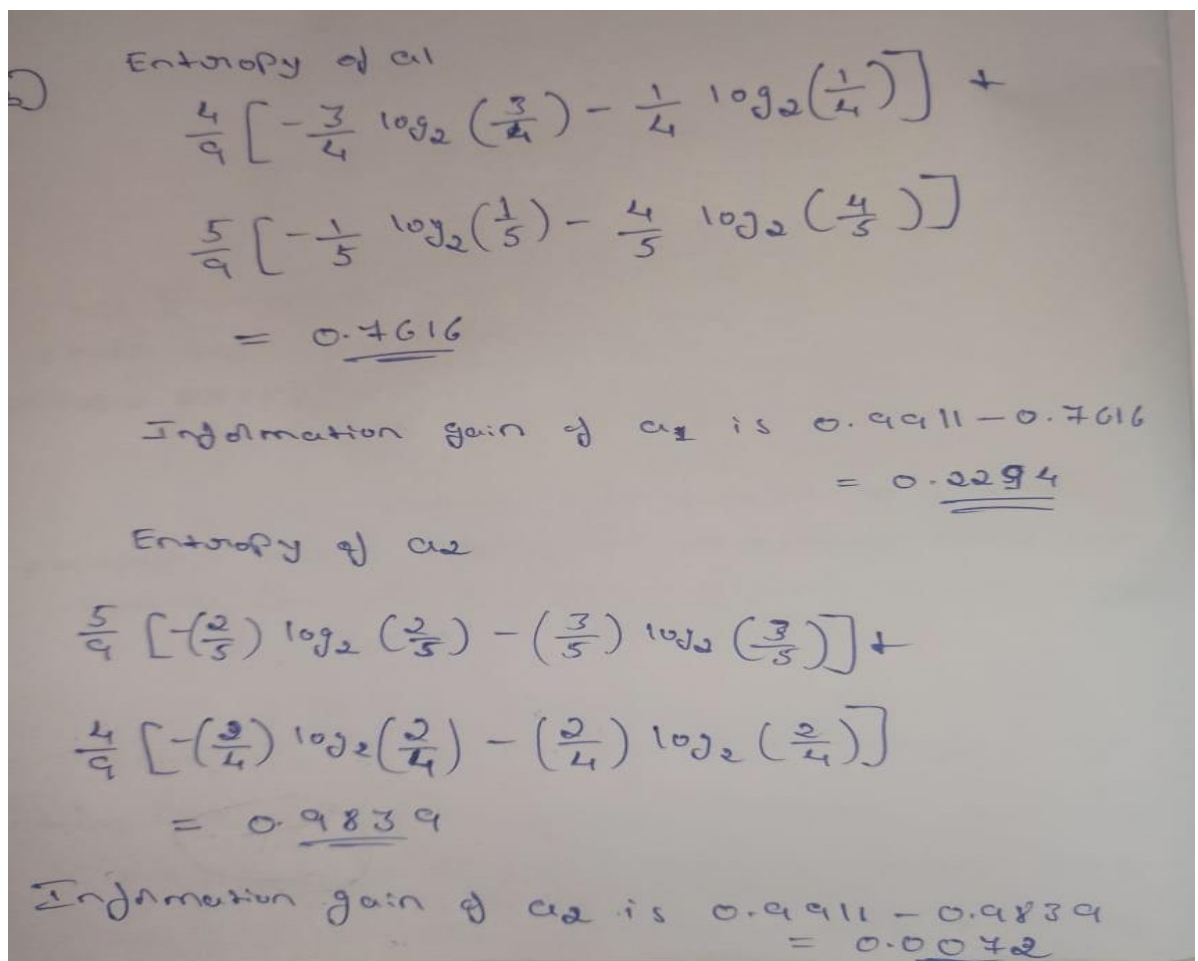


Handwritten calculation for entropy of a_1 :

$$-\left(\frac{4}{9} \log_2 \frac{4}{9} + \frac{5}{9} \log_2 \frac{5}{9}\right)$$

$$= \underline{\underline{0.9911}}$$

b)



Handwritten calculations for entropy and information gain of a_1 and a_2 :

Entropy of a_1

$$\frac{4}{9} \left[-\frac{3}{4} \log_2 \left(\frac{3}{4} \right) - \frac{1}{4} \log_2 \left(\frac{1}{4} \right) \right] +$$

$$\frac{5}{9} \left[-\frac{1}{5} \log_2 \left(\frac{1}{5} \right) - \frac{4}{5} \log_2 \left(\frac{4}{5} \right) \right]$$

$$= \underline{\underline{0.7616}}$$

Information gain of a_1 is $0.9911 - 0.7616$

$$= \underline{\underline{0.2294}}$$

Entropy of a_2

$$\frac{5}{9} \left[-\left(\frac{2}{5} \right) \log_2 \left(\frac{2}{5} \right) - \left(\frac{3}{5} \right) \log_2 \left(\frac{3}{5} \right) \right] +$$

$$\frac{4}{9} \left[-\left(\frac{3}{4} \right) \log_2 \left(\frac{3}{4} \right) - \left(\frac{2}{4} \right) \log_2 \left(\frac{2}{4} \right) \right]$$

$$= \underline{\underline{0.9839}}$$

Information gain of a_2 is $0.9911 - 0.9839$

$$= \underline{\underline{0.0072}}$$

c) what is the best split according to classification error?

Classification error of a_1 is $2/9$ and a_2 is $5/9$

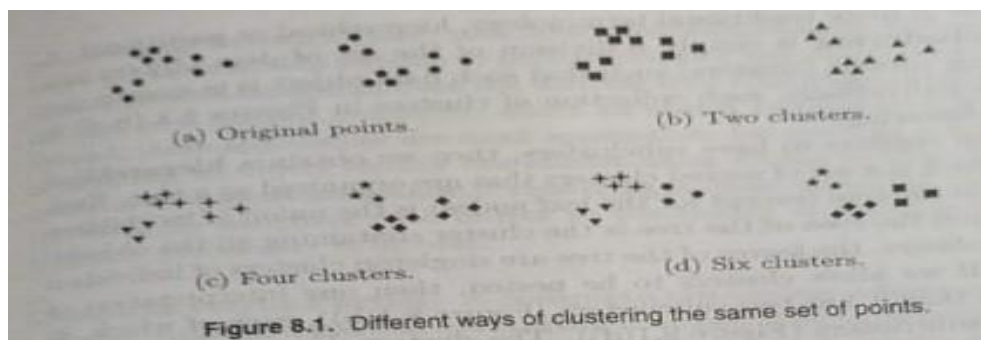
So a_1 is considered as a best split

UNIT 5

1. Define cluster analysis. Explain how it is related to other data mining techniques?

Ans:

- Cluster analysis groups data objects based only on information found in the data that describes the objects and their relationships.
- The goal is that the objects within a group be similar to one another and different from objects in other groups. The greater the similarity within a group and the greater the difference between groups, the better or more distinct the clustering.
- In many applications, the notation of clusters is not well defined. To better understand the difficulty of deciding what constitutes a cluster, consider the following figure which shows 20 points and three different ways of dividing them.



- Clustering analysis is related to other techniques that are used to divide data groups. For instances, clustering can be regarded as a form of classification in that it creates a labeling of objects with class labels. However, it derives these labels only from the data.
- In contrast, classification in the sense is supervised classification i.e., new unlabeled objects are assigned a class label using a model developed from objects with known class labels. For this

reason, cluster analysis is sometimes referred to as unsupervised classification.

2. Mention the applications of cluster analysis.

Ans:

- Clustering analysis is broadly used in many applications such as market research, pattern recognition, data analysis, and image processing.
- Clustering can also help marketers discover distinct groups in their customer base. And they can characterize their customer groups based on the purchasing patterns.
- In the field of biology, it can be used to derive plant and animal taxonomies, categorize genes with similar functionalities and gain insight into structures inherent to populations.
- Clustering also helps in identification of areas of similar land use in an earth observation database. It also helps in the identification of groups of houses in a city according to house type, value and geographic location.
- Clustering also helps in classifying documents on the web for information discovery.
- Clustering is also used in outlier detection applications such as detection of credit card fraud.
- As a data mining function, cluster analysis serves as a tool to gain insight into the distribution of data to observe characteristics of each cluster.



3. What is Clustering? Explain Different Types of Clustering Distinguish between (A) Hierarchical and Partitioning (B) Complete and Partial (C) Exclusive and Overlapping.

Ans:

- Clustering or Cluster analysis is the method of grouping the entities based on similarities. Defined as an unsupervised learning problem that aims to make training data with a given set of inputs but without any target values.
- It is the process of finding similar structures in a set of unlabeled data to make it more understandable and manipulative.
- Clustering helps to splits data into several subsets. Each of these subsets contains data similar to each other, and these subsets are called clusters.
- Now that the data from our customer base is divided into clusters, we can make an informed decision about who we think is best suited for this product.



The Various methods of clustering are:

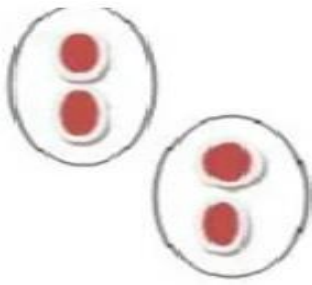
- **Partitioning based Method**

The partition algorithm divides data into many subsets. Let's assume the partitioning algorithm builds a partition of data and n objects present in the database. Hence each section will be represented $ask \leq n$. This gives an idea that the classification of the data is in k groups, which can be shown below

Figure 1 shows original points in clustering



Figure 2 shows Partition clustering after applying an algorithm



This indicates that each group has at least one object, and every object, must belong to exactly one group.

- **Density-Based Method**

These algorithms produce clusters in a determined location based on the high density of data set participants. It aggregates some range notion for group members in clusters to a density standard level. Such processes can perform less in detecting the group's Surface areas.

- **Centroid-based Method**

A vector of values references almost every cluster in this type of grouping technique. Compared to other groups, each object is part of the group with a minimum difference in value. The number of groups should be predefined, which is the most significant algorithm problem of this type. This methodology is the closest to the subject of identification and is widely used for problems of optimization.

- **Hierarchical Method**

The method will create a hierarchical decomposition of a given set of data objects. Based on how the hierarchical decomposition is formed, we can classify hierarchical methods. This method is shown as follows

Agglomerative Approach is also known as Button-up Approach. Here we begin with every object that

constitutes a separate group. It continues to fuse items or groups close together

Divisive Approach is also known as the Top-Down Approach. We begin with all the things in the same cluster. This method is rigid, i.e., it can never be undone once a fusion or division is completed

Advantages of Hierarchical Clustering are as follows

- It applies to any attribute type.
- It provides flexibility related to the level of granularity.
- **Grid-Based Method**

Grid-based methods work in the object space instead of dividing the data into a grid. Grid is divided based on characteristics of the data. By using this method, non-numeric data is easy to manage. Data order does not affect the partitioning of the grid. An important advantage of a grid-based model it provides faster execution speed.

- **Model-Based Method**

This method uses a hypothesized model based on probability distribution. By clustering the density function, this method locates the clusters. It reflects the data points' spatial distribution.

(A) Difference between Hierarchical and Partitioning algorithms:

The most significant difference between hierarchical and partitional clustering is the running time. The partitional algorithms handle one piece of data, finding this piece's optimum point, and it's faster than the hierarchical method. (If you are working embedded system, this difference can be a life-changer.) However, the higher the value of k, the higher the transaction cost. Another

difference is input-output. This difference consists of different approaches of the hierarchical and partitional algorithms. The partitional algorithm finds many optimum points, and due to these optimum points, it generally gives us the local optimum, not the global optimum. However, sometimes if you are so lucky, this local optimum can be the global optimum. Or the global optimum can be found by deterministic annealing and genetic algorithms. But hierarchical methods handle all of the data, and it provides an optimum point that handles it. Also, for this reason, there is an assumption difference between the two algorithms, and due to handling all the data in one step, the result will equal the global optimum.

(B) Difference between Complete and Partial Clustering

Complete Clustering allocates each object to a cluster, whereas **partial Clustering** does not. The inspiration for a **Partial Clustering** is that a few objects in a data set may not belong to distinct groups. Most of the time, objects in the data set may produce outliers, noise, or "uninteresting background." For example, some news headlines stories may share a common subject, such that "Industrial production shrinks globally by 1.1 percent," While different stories are more frequent or one-of-a-kind. Consequently, to locate the significant topics in the last month's stories, we might need to search only for clusters of documents that are firmly related by a common subject. In other cases, a complete Clustering of objects is desired. For example, an application that utilizes Clustering to sort out documents for browsing needs to ensure that all documents can be browsed.

(C) Difference between Exclusive and Overlapping Clustering

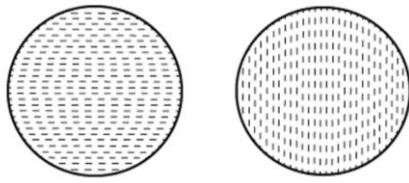
The Clustering that appeared in the figure is all **exclusive**, as they give the responsibility to each object to a single cluster. There are numerous circumstances in which a point could sensibly be set in more than one cluster, and these circumstances are better addressed by non-exclusive Clustering. In general terms, an **overlapping or non-exclusive Clustering** is used to reflect the fact that an object can together belong to more than one group (class). For example, a person at a company can be both a trainee student and an employee of the company. A non-exclusive Clustering is also usually used if an object is "between" two or more than two clusters and could sensibly be allocated to any of these clusters. Consider a point somewhere between two of the clusters rather than make an entirely random task of the object to a single cluster. it is put in all of the clusters to "equally good" clusters.

4. With suitable examples explain different types of clusters.

Ans:

Well-separated cluster:

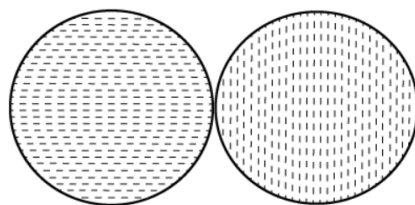
A cluster is a set of objects where each object is closer or more similar to every other object in the cluster. Sometimes a limit is used to indicate that all the objects in a cluster must be adequately close or similar to each other. The definition of a cluster is satisfied only when the data contains natural clusters that are quite far from one another. The figure illustrates an example of well-separated clusters that comprise of two points in a two-dimensional space. Well-separated clusters do not require to be spherical but can have any shape.



Well-separated clusters

Prototype-Based cluster:

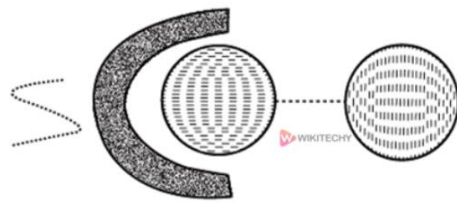
When the data has definite characteristics, the prototype is usually a medoid that is the most representative point of a cluster. Data with continuous characteristics, the prototype of a cluster is usually a centroid. For some sorts of data, the model can be viewed as the most central point, and in such examples, we commonly refer to prototype-based clusters as center-based clusters. As anyone might expect, such clusters tend to be spherical.



Center-based clusters

Graph-Based Cluster:

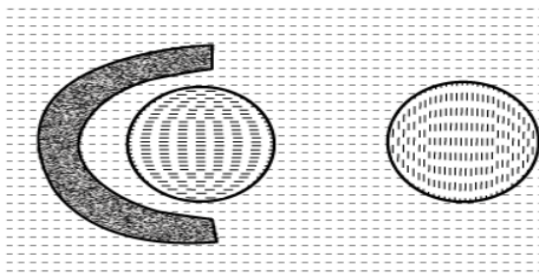
It is a group of objects that are associated with each other, but that has no association with objects that is outside the group. Other kinds of graph-based clusters also possible. One such way describes a cluster as a clique. It is like prototype-based clusters, and such clusters tend to be spherical. Clique is a set of nodes in a graph that is completely associated with each other.



Contiguity-based clusters

Density-Based Cluster:

A cluster is a compressed domain of objects that are surrounded by a region of low density. The two spherical clusters are not merged, as in the figure, because the bridge between them fades into the noise. Similarly, the curve that is present in the Figure disappears into the noise and does not form a cluster in Figure. It also disappears into the noise and does not form a cluster shown in the figure. A density-based definition of a cluster is usually occupied when the clusters are irregularly and intertwined, and when noise and outliers exist. The other hand contiguity-based definition of a cluster would not work properly for the data of Figure. Since the noise would tend to form a network between clusters.

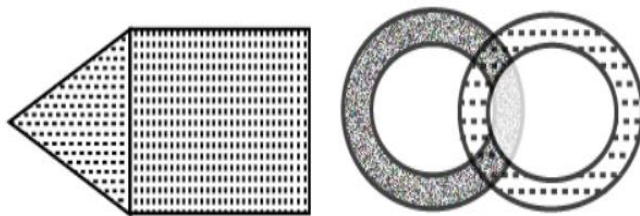


Density-based clusters

Shared-property or Conceptual Clusters:

A center-based cluster shares property that they closest to the similar centroid or medoid. the shared-property approach additionally incorporates new types of the cluster.

Consider the cluster given in the figure. A triangular area (cluster) is next to a rectangular one, and there are two intertwined circles (clusters). In both cases, a Clustering algorithm would require a specific concept of a cluster to recognize these clusters effectively. The way of discovering such clusters is called conceptual Clustering.



Conceptual clusters

5. What is K-Means Algorithm? Explain the same with examples. Also mention its strength and weakness.

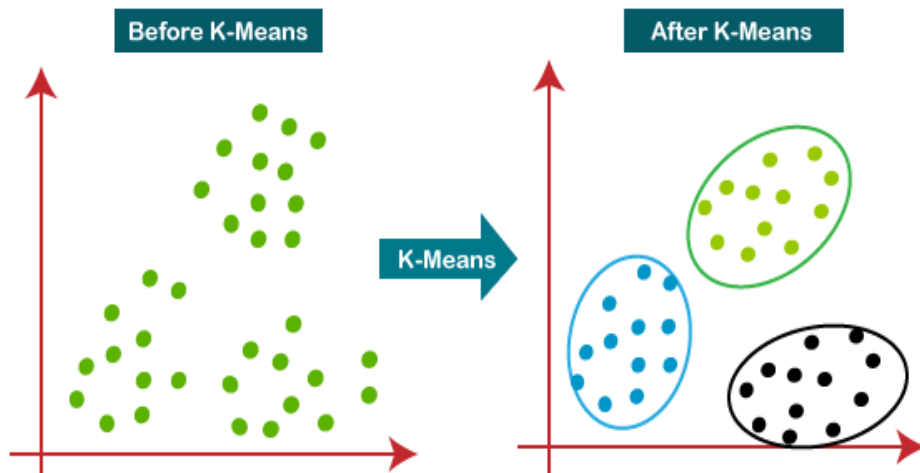
Ans:

K-Means Clustering is an Unsupervised Learning algorithm, which groups the unlabelled dataset into different clusters. Here K defines the number of pre-defined clusters that need to be created in the process, as if $K=2$, there will be two clusters, and for $K=3$, there will be three clusters, and so on.

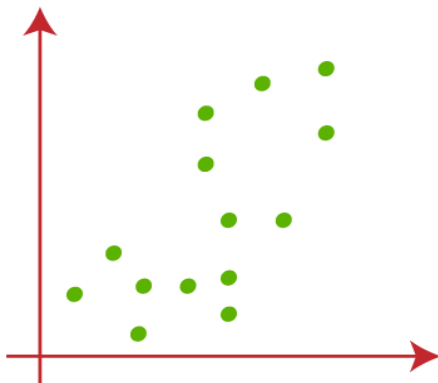
“It is an iterative algorithm that divides the unlabelled dataset into k different clusters in such a way that each dataset belongs only one group that has similar properties.”

It is a centroid-based algorithm, where each cluster is associated with a centroid. The main aim of this algorithm is to minimize the sum of distances between the data point and their corresponding clusters. The algorithm takes the unlabelled dataset as input, divides the dataset into k-number of clusters,

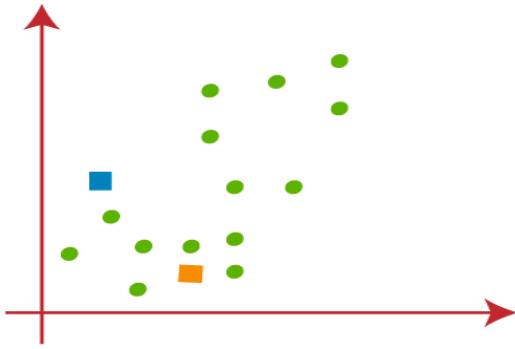
and repeats the process until it does not find the best clusters.
The value of k should be predetermined in this algorithm.



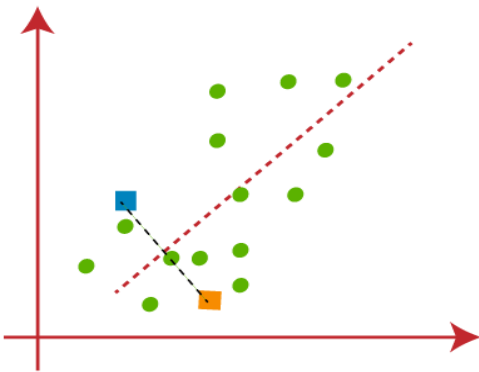
Let's understand the above steps by considering the visual plots:



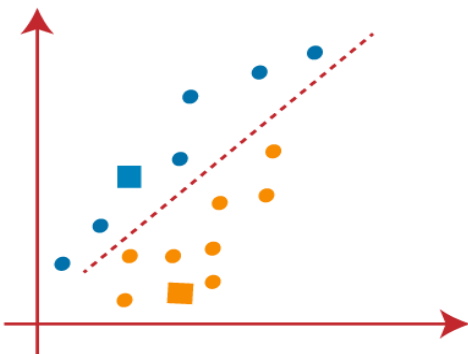
Let's take number k of clusters, i.e., $K=2$, to identify the dataset and to put them into different clusters. It means here we will try to group these datasets into two different clusters. We need to choose some random k points or centroid to form the cluster. These points can be either the points from the dataset or any other point. So, here we are selecting the below two points as k points, which are not the part of our dataset



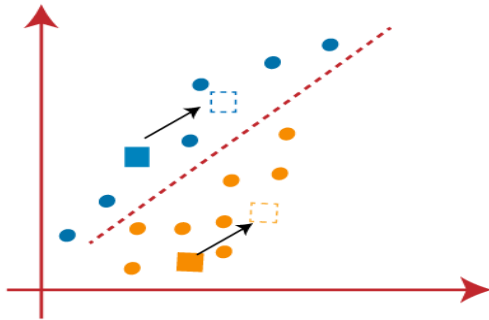
Now we will assign each data point of the scatter plot to its closest K-point or centroid. We will compute it by applying some mathematics that we have studied to calculate the distance between two points. So, we will draw a median between both the centroids. Consider the below image



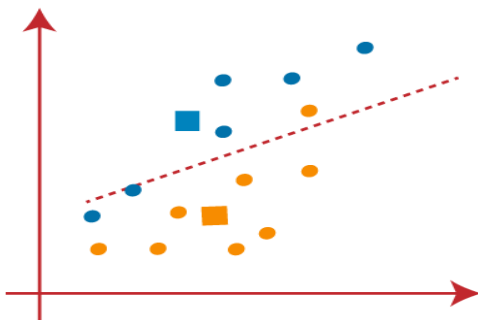
From the above image, it is clear that points left side of the line is near to the K1 or blue centroid, and points to the right of the line are close to the yellow centroid. Let's colour them as blue and yellow for clear visualization.



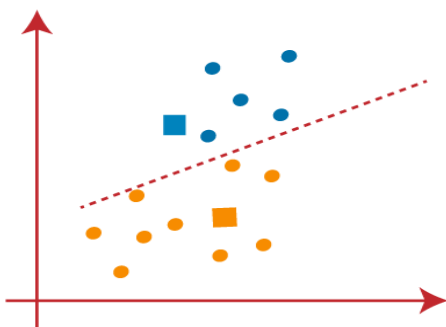
As we need to find the closest cluster, so we will repeat the process by choosing a new centroid. To choose the new centroids, we will compute the centre of gravity of these centroids, and will find new centroids as below



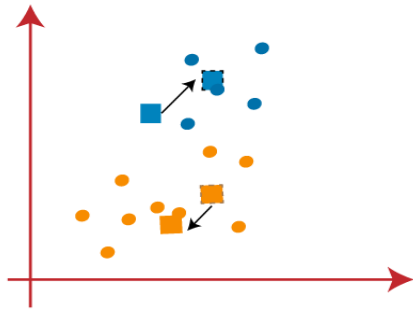
Next, we will reassign each datapoint to the new centroid. For this, we will repeat the same process of finding a median line. The median will be like below image:



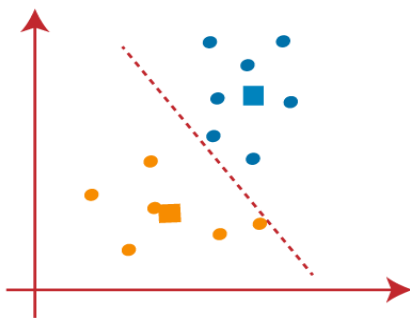
From the above image, we can see, one yellow point is on the left side of the line, and two blue points are right to the line. So, these three points will be assigned to new centroids.



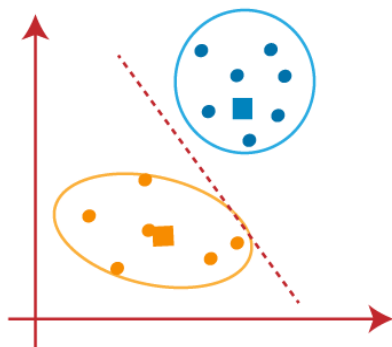
As reassignment has taken place, so we will again go to the step-4, which is finding new centroids or K-points. We will repeat the process by finding the centre of gravity of centroids, so the new centroids will be as shown in the below image:



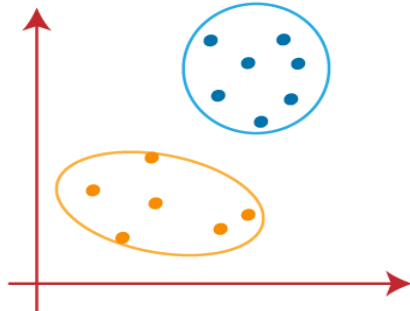
As we got the new centroids so again will draw the median line and reassign the data points. So, the image will be:



We can see in the above image; there are no dissimilar data points on either side of the line, which means our model is formed. Consider the below image:



As our model is ready, so we can now remove the assumed centroids, and the two final clusters will be as shown in the below image:



Algorithm 8.1 Basic K-means algorithm.

- 1: Select K points as initial centroids.
 - 2: **repeat**
 - 3: Form K clusters by assigning each point to its closest centroid.
 - 4: Recompute the centroid of each cluster.
 - 5: **until** Centroids do not change.
-

Advantages of k-means:

- Relatively simple to implement
- Wide variety of data types
- Easily adapts to new examples
- Generalizes to clusters of different shapes and sizes

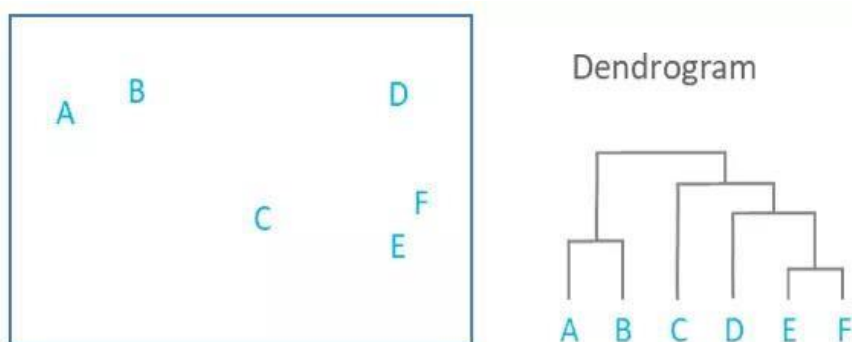
Disadvantages of k-means:

- Choosing k manually
- It is not suitable for all types of data
- Cannot handle large data sets Lacks consistency

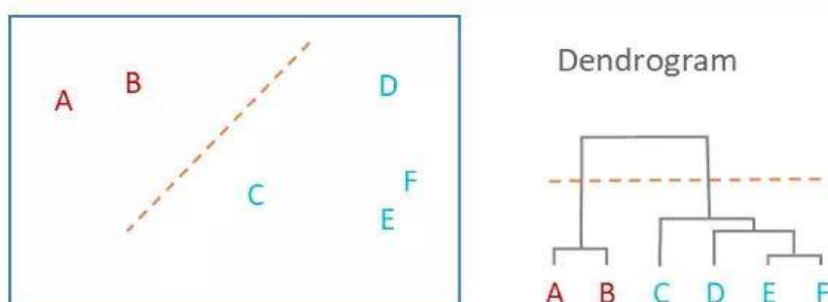
6. What is a Dendrogram? Give an example.

Ans:

A dendrogram is a diagram that shows the hierarchical relationship between objects. It is most commonly created as an output from hierarchical clustering. The main use of a dendrogram is to work out the best way to allocate objects to clusters.



In the example we can see that E and F are most similar, as the height of the link that joins them together is the smallest. The next two most similar objects are A and B. In the dendrogram, the height of the dendrogram indicates the order in which the clusters were joined. A more informative dendrogram can be created where the heights reflect the distance between the clusters as is shown below. In this case, the dendrogram shows us that the big difference between clusters is between the cluster of A and B versus that of C, D, E, and F.



7. Give Basic Agglomerative Hierarchical Clustering Algorithm

Ans:

Agglomerative hierarchical clustering

Agglomerative clustering is one of the most common types of hierarchical clustering used to group similar objects in clusters. Agglomerative clustering is also known as **AGNES** (Agglomerative Nesting). In agglomerative clustering, each data point act as an individual cluster and at each step, data objects are grouped in a bottom-up method. Initially, each data object is in its cluster. At each iteration, the clusters are combined with different clusters until one cluster is formed.

Step 1: Compute the proximity matrix, if necessary.

Step 2: repeat

Step 3: Merge the closest two clusters.

Step 4: Update the proximity matrix to reflect the proximity between the new cluster and the original cluster.

Step 5: Only one cluster remains.

Example:

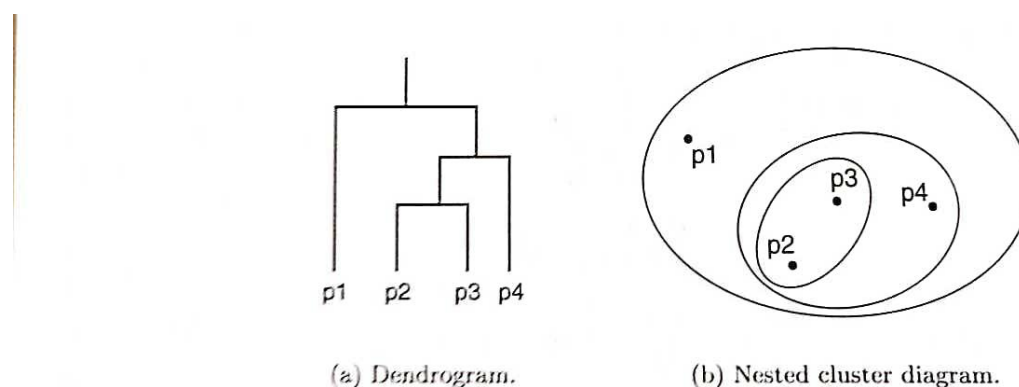
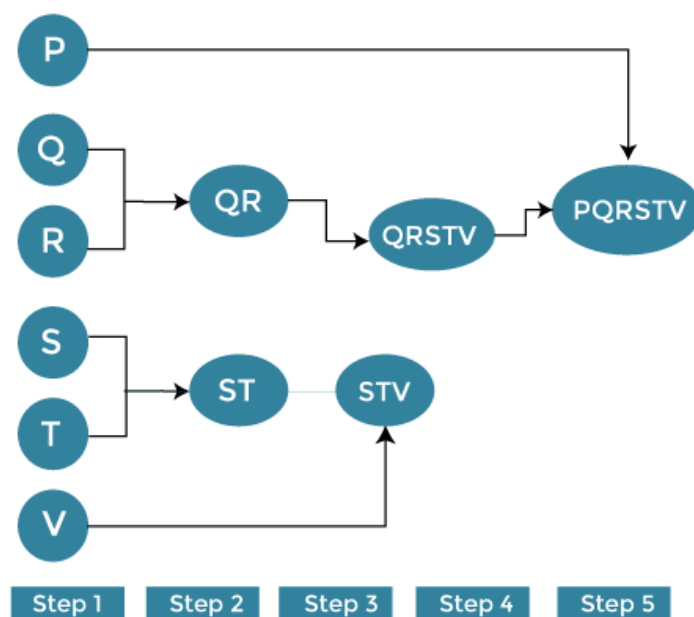


Figure 8.13. A hierarchical clustering of four points shown as a dendrogram and as nested clusters.

Let's understand this concept with the help of graphical representation using a dendrogram.

With the help of given demonstration, we can understand that how the actual algorithm work. Here no calculation has been done below all the proximity among the clusters are assumed. Let's suppose we have six different data points P, Q, R, S, T, V.



Step 1:

Consider each alphabet (P, Q, R, S, T, V) as an individual cluster and find the distance between the individual cluster from all other clusters.

Step 2:

Now, merge the comparable clusters in a single cluster. Let's say cluster Q and Cluster R are similar to each other so that we can merge them in the second step. Finally, we get the clusters [(P), (QR), (ST), (V)]

Step 3:

Here, we recalculate the proximity as per the algorithm and combine the two closest clusters [(ST), (V)] together to form new clusters as [(P), (QR), (STV)]

Step 4:

Repeat the same process. The clusters STV and PQ are comparable and combined together to form a new cluster. Now we have [(P), (QQRSTV)].

Step 5:

Finally, the remaining two clusters are merged together to form a single cluster [(PQRSTV)]

8. Explain the following methods of defining proximity between clusters.

a) Single Link b) Complete Link c) Group Average

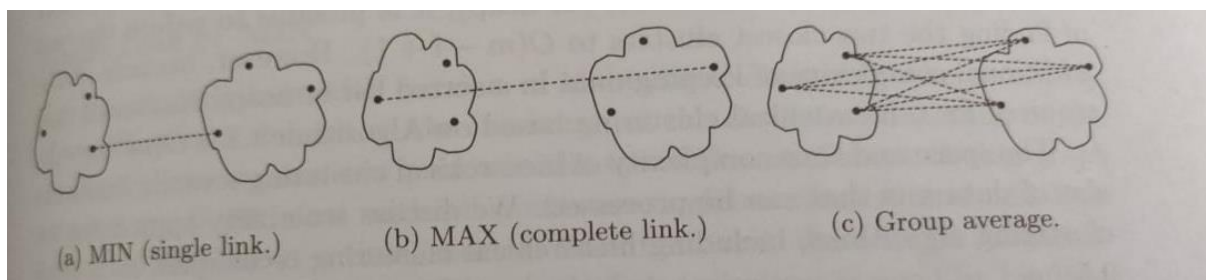
Ans:

Hierarchical clustering treats each data point as a singleton cluster, and then successively merges clusters until all points have been merged into a single remaining cluster. A hierarchical clustering is often represented as a dendrogram.

- In **single-link (MIN) hierarchical clustering**, we merge in each step the two clusters whose two closest members have the smallest distance (or: the two clusters with the smallest **minimum** pairwise distance). The time complexity of single-link clustering is $O(n^2)$
- In **complete-link (MAX) hierarchical clustering**, we merge in each step the two clusters whose merger has the

smallest diameter (or: the two clusters with the smallest **maximum** pairwise distance). The worst case time complexity of complete-link clustering is at most $O(n^2 \log n)$.

- **Average-link (Group Average) clustering** is a **compromise between the sensitivity of complete-link clustering to outliers and the tendency of single-link clustering to form long chains** that do not correspond to the intuitive notion of clusters as compact, spherical objects. **The time complexity of average-link clustering is $O(n^2 \log n)$.**



9. Explain the concept of core, border, and noise points with the help of suitable examples.

Ans:

The centre-based approach to density allows us to classify a point as being

- (1) **In the interior of a dense region (a core point)**
- (2) **On the edge of a dense region (a border point)**
- (3) **In a sparsely occupied region (a noise or background point)**

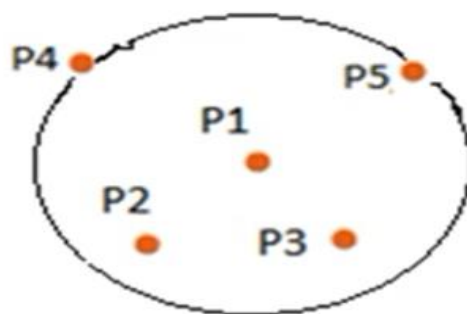
Core points: These points are in the interior of a density-based cluster. A point is a core point if the number of points within a given neighbourhood around the points as determined by the distance function and a user-specified distance parameter, Eps, exceeds a certain threshold, MinPts, which is also a user-specified parameter in figure, point P1 is a core point, for the indicated radius (Eps) if $\text{MinPts} \leq 4$.

P1: Centre point. Radius of the circle is 2 unit.

Min Pts:4 MinPts:5

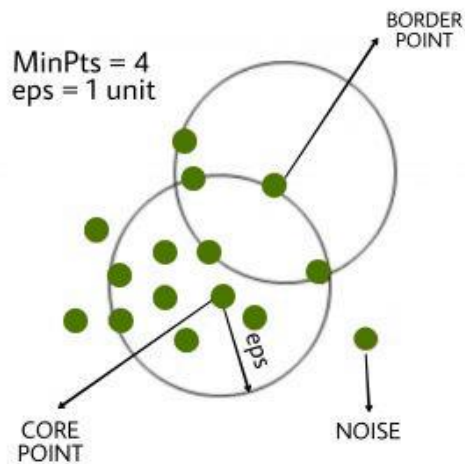
P1 can form a cluster. (P2, P3, P4, P5]

P1 is core point.



Border points: A border point is not a core point, but falls within the neighbourhood of a core point. A border point can fall within the neighbourhoods of several core points.

Noise points: A noise point is any point that is neither a core point nor a border point.



10. State and explain the DBSCAN Algorithm. (Write DBSCAN clustering algorithm and estimate time and space complexity).

Ans:

DBSCAN algorithm requires **two parameters**:

- **eps**: It defines the neighborhood around a data point i.e., if the distance between two points is lower or equal to 'eps' then they are considered neighbors. If the eps value is chosen too small then large part of the data will be considered as outliers. If it is chosen very large then the clusters will merge and the majority of the data points will be in the same clusters. One way to find the eps value is based on the **k-distance graph**.
- **MinPts**: Minimum number of neighbors (data points) within eps radius. Larger the dataset, the larger value of MinPts must be chosen. As a general rule, the minimum MinPts can be derived from the number of dimensions D in the dataset as, $\text{MinPts} \geq D+1$. The minimum value of MinPts must be chosen at least 3.

In this algorithm, we have 3 types of data points.

Core Point: A point is a core point if it has more than MinPts points within eps.

Border Point: A point which has fewer than MinPts within eps but it is in the neighborhood of a core point.

Noise or outlier: A point which is not a core point or border point.

DBSCAN ALGORITHM

- 1: Label all points as core, border, or noise points.
- 2: Eliminate noise points.
- 3: Put an edge between all core points that are within Eps of each other.
- 4: Make each group of connected core points into a separate cluster.
- 5: Assign each border points to one of the clusters of its associated core points.

Time and Space Complexity

The basic time complexity of the DBSCAN algorithm is $O(m * \text{time to find points in the Eps-neighbourhood})$, where m is the number of points. In the worst case, this complexity is $O(m^2)$. However, in the d -dimensional spaces, there are data structures, such as kd-trees, that allow efficient retrieval of all points within a given distance of a specified points, and the time complexity

can be as low as $O(m \log m)$. The space requirement of DBSCAN, even for high-dimensional data, is $O(m)$ because it is only necessary to keep a small amount of data for each point, i.e., the cluster label and the identification of each point as a core, border, or noise point.

The **DBSCAN algorithm** is based on this intuitive notion of “clusters” and “noise”. The key idea is that for each point of a cluster, the neighborhood of a given radius has to contain at least a minimum number of points.

DBSCAN algorithm can be abstracted in the following steps:

- Find all the neighbor points within ϵ and identify the core points or visited with more than MinPts neighbors.
- For each core point if it is not already assigned to a cluster, create a new cluster.
- Find recursively all its density connected points and assign them to the same cluster as the core point.
- A point a and b are said to be density connected if there exist a point c which has a sufficient number of points in its neighbors and both the points a and b are within the *eps distance*. This is a chaining process. So, if b is neighbor of c , c is neighbor of d , d is neighbor of e , which in turn is neighbor of a implies that b is neighbor of a .
- Iterate through the remaining unvisited points in the dataset. Those points that do not belong to any cluster are noise.

11. Distinguish Unsupervised and Supervised clustering evaluation methods

Ans:

Supervised Clustering:

- In supervised learning algorithms, the output for the given input is known.
- The algorithms learn from labelled set of data. This data helps in evaluating the accuracy on training data.
- It is Predictive Modelling technique which predicts the future outcomes accurately.
- It includes classification and regression algorithms.
- This type of learning is relatively complex as it requires labelled data.
- It is more accurate than unsupervised learning as input data and corresponding output is well known, and the machine only needs to give predictive.
- It is online process of data analysis and does.
- Supervised measures are often called external indices because they use information not present in the data set.
- Measures the goodness of a clustering structure without respect to external information.
- Some algorithms of supervised learning are Linear Regression, Naïve Bayes, and Neural Networks.

Unsupervised Clustering:

- In unsupervised learning algorithms, the output for the given input is unknown.

- The algorithm is provided with unlabelled data where it tries to find patterns and associations in between the data items
- It is Descriptive Modelling technique, which explains the real relationship between the elements and history of the elements.
- It includes clustering and association rules learning algorithms.
- It is less complex as there is no need to understand and label data.
- It has less accuracy as the input data is unlabelled. Thus, the machine has to first understand and label the data and then give predictions.
- This is real time analysis of data.
- Unsupervised measures are often called internal indices because they use only information present in the data set.
- Measures the extent to which the clustering structure discovered by a clustering algorithm matches some external structure.
- Some algorithms for unsupervised learning are k-means clustering, Apriori, etc.

12. Explain Classification-Oriented measures of Cluster Validity

Ans:

There are a number of measures- **entropy, purity, precision, recall and the F-measure**, that are commonly used to evaluate the performance of the classification model.

Why do we need these measures?

- To avoid finding patterns in noise.

- To compare clustering algorithms.
- To compare two sets of clusters.
- To compare two clusters.

Entropy:

The degree to which each cluster consists of objects of a single class. For each cluster we compute the probability that a member of cluster i belongs to class j.

i.e., $p_{ij} = m_{ij} / m_i$

where m_i is the number of objects in cluster i and m_{ij} is the number of objects of class j in cluster i.

Using this class distribution, entropy of each cluster i is calculated using the standard formula,

$$e_i = - \sum_{j=1}^L p_{ij} \log_2 p_{ij},$$

where L is the number of classes.

The total entropy for the set of clusters is calculated as

$$e = \sum_{i=1}^K \frac{m_i}{m} e_i,$$

where k is the number of clusters and m is the number of data points.

Purity:

Another measure of the extent to which a cluster contains objects of a single class.

Using the previous terminology, the purity of the cluster i is

$$p_i = \max_j p_{ij}$$

Overall purity of a clustering is

$$\sum_{i=1}^K \frac{m_i}{m} p_i$$

Precision:

The fraction of a cluster that consists of objects of a specified class.

The precision of cluster i with respect to class j is

$$\text{precision}(i, j) = p_{ij}$$

Recall:

The extent to which a cluster contains all objects of a specified class.

The Recall of cluster i with respect to class j is

$$\text{recall}(i, j) = m_{ij} / m_j$$

where m_j is the number of objects in class j .

F-Measure:

A combination of both precision and recall that measures the extent to which a cluster contains only objects of a particular class and all objects of that class.

The F-Measure of cluster i with respect to class j is

$$F(i,j) = (2 * \text{precision}(i,j) * \text{recall}(i,j)) / \text{precision}(i,j) + \text{recall}(i,j)$$

Consider the example to calculate the measures

Table 8.9. K-means clustering results for the *LA Times* document data set.

Cluster	Entertainment	Financial	Foreign	Metro	National	Sports	Entropy	Purity
1	3	5	40	506	96	27	1.2270	0.7474
2	4	7	280	29	39	2	1.1472	0.7756
3	1	1	1	7	4	671	0.1813	0.9796
4	10	162	3	119	73	2	1.7487	0.4390
5	331	22	5	70	13	23	1.3976	0.7134
6	5	358	12	212	48	13	1.5523	0.5525
Total	354	555	341	943	273	738	1.1450	0.7203

Precision, Recall and F-Measure can be calculated for each cluster.

Let's consider cluster 1 and metro class of the table.

Precision => $506/677 = 0.75$

Recall => $506/943 = 0.26$

From Precision and Recall we get,

F-Measure => $(2 * 0.75 * 0.26) / (0.75 + 0.26) = 0.39$

Cluster Validity for Hierarchical Clustering:

The key idea of this approach is to evaluate whether a hierarchical clustering contains, for each class, at least one cluster that is relatively pure and includes most of the objects of that class. To evaluate hierarchical clustering with respect to this goal, we compute, for each class, the F-Measure for each cluster

in the cluster hierarchy. For each class, we take the maximum F-Measure attained for any cluster and finally calculate the overall F-Measure for the hierarchical clustering by computing the weighted average of all per class measures, where the weights are based on the class sizes.

The hierarchical F-Measure is defined as follows

$$F = \sum_j \frac{m_j}{m} \max_i F(i, j)$$

where maximum is taken over all clusters i at all levels, m_j is the number of objects in class j and m is the total number of objects.

13. Explain Similarity Oriented Measure of Cluster Validity.

Ans:

The measures that we discuss in this section are all based on the premise that any two objects that are in the same cluster should be in the same class and vice versa. We can view this approach to cluster validity as involving the comparison of two matrices:

Ideal cluster similarity matrix:

This matrix is previously discussed, which has a 1 in the ij th entry if two objects, i and j are in the same cluster and 0, otherwise.

Ideal class similarity matrix:

This is defined with respect to class labels, which has a 1 in the ij th entry if two objects, i and j , belong to the same class, and a 0 otherwise. As before, we can take the correlation of these two

matrices as the measure of cluster validity. This measure is known as the I statistic in clustering validation literature.

Example for Correlation between Cluster and Class Matrices

Consider this involving five data points, P1, P2, P3, P4, P5, two clusters, $C1 = \{P1, P2, P3\}$ and $C2 = \{P4, P5\}$, and two classes, $L1 = \{P1, P2\}$ and $L2 = \{P3, P4, P5\}$. The ideal cluster and class similarity matrices are given in Tables, the correlation between the entries of these two matrices is 0.359.

Point	p1	p2	p3	p4	p5
p1	1	1	1	0	0
P2	1	1	1	0	0
P3	1	1	1	0	0
P4	0	0	0	1	1
p5	0	0	0	1	1

Ideal Cluster Similarity Matrix

Point	p1	p2	p3	p4	p5
p1	1	1	0	0	0
P2	1	1	0	0	0
P3	0	0	1	1	1
P4	0	0	1	1	1
p5	0	0	1	1	1

Ideal Class Similarity Matrix

More generally, we can use any of the measures for binary similarity that we saw in proximity measures (so here for example, we can convert these two matrices into binary vectors by appending the rows.) We repeat the definitions of the four quantities used to define those similarity measures, but modify

our descriptive text to fit the current context. Specifically, we need to compute the following four quantities for all pairs of distinct objects. (There is $m(m - 1) / 2$ such pairs, if m is the number of objects.)

f_{00} = number of pairs of objects having a different class and a different cluster

f_{01} = number of pairs of objects having a different class and the same cluster

f_{10} = number of pairs of objects having the same class and a different cluster

f_{11} = number of pairs of objects having the same class and the same cluster

In particular, the simple matching coefficient, which is known as the Rand statistic in this context, and the Jaccard coefficient are two of the most frequently used cluster validity measures.

$$\text{Rand statistic} = \frac{f_{00} + f_{11}}{f_{00} + f_{01} + f_{10} + f_{11}}$$

$$\text{Jaccard coefficient} = \frac{f_{11}}{f_{01} + f_{10} + f_{11}}$$

Rand and Jaccard Measures

Based on these formulas, we can readily compute the Rand statistic and Jaccard coefficient for the example based on earlier two tables. Noting that $f_{00} = 4$, $f_{01} = 2$, $f_{10} = 2$ and $f_{11} = 2$,

$$\text{Rand statistics} = (2 + 4) / 10 = 0.6$$

Jaccard coefficient = $2 / (2 + 2 + 2) = 0.33$.

Note: This is just a two-way contingency table for determining whether pairs of objects are in the same class and same cluster.

	Same Cluster	Different Cluster
Same Class	f ₁₁	f ₁₀
Different Class	f ₀₁	f ₀₀

14. What is Hierarchical clustering?

Ans:

Hierarchical clustering refers to an unsupervised learning procedure that determines successive clusters based on previously defined clusters. It works via grouping data into a tree of clusters. Hierarchical clustering starts by treating each data point as an individual cluster. The endpoint refers to a different set of clusters, where each cluster is different from the other cluster, and the objects within each cluster are the same as one another.

There are two types of hierarchical clustering

- Agglomerative Hierarchical Clustering
- Divisive Clustering

Advantages of Hierarchical clustering

- It is simple to implement and gives the best output in some cases.
- It is easy and results in a hierarchy, a structure that contains more information.
- It does not need us to pre-specify the number of clusters.

Disadvantages of Hierarchical clustering

- It breaks the large clusters.
- It is Difficult to handle different sized clusters and convex shapes.
- It is sensitive to noise and outliers.
- The algorithm can never be changed or deleted once it was done previously.